

The Elasticity of Taxable Income Across Countries

Claudio A. Agostini Zareh Asatryan Laurent Bach Govindadeva Bernier
Marinho Bertanha Katarzyna Bilicka Anne Brockmeyer Jaroslav Bukovina
Guillermo Falcone Pablo Garriga Yuxuan He Petr Jansky
Evangelos Koumanakos Tomas Lichard Tomás Martins Jan Palguta
Elena Patel João Pereira dos Santos Louis Perrault Nathan Seegert
Oliver Skultety Kristina Strohmaier Maximilian Todtenhaupt Guillermo Vuletin
Branislav Zudel*[†]

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Abstract

We produce the largest cross-country dataset of corporate elasticity of taxable income estimates to analyze why these elasticities differ across countries. We develop a harmonized method and apply it to administrative firm-level tax data from 19 countries. Exploiting bunching at zero taxable income, we estimate firm-level elasticities. Elasticities range from 0.075 in Ecuador to 1.91 in Canada, with an average of 0.6. To explain this variation, we pair our estimates with 95 country-level characteristics. A random forest model and Shapley decomposition show that tax policy, country fundamentals, and firm characteristics each explain about one-third of the cross-country variation in ETIs.

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*Respectively: Universidad Adolfo Ibañez, ZEW & CESifo, ESSEC Business School, Office of the Parliamentary Budget Officer Canada, University of Notre Dame, Utah State University and NBER, World Bank, Government Office of the Slovak Republic, World Bank, World Bank, Duke University, Charles University, Hellenic Open University, Prague University of Economics and Business, ISEG – University of Lisbon, CUNEF University, Brookings, ISEG – University of Lisbon, Office of the Parliamentary Budget Officer Canada, Northeastern University, Charles University, University Duisburg-Essen, Leibniz University Hannover & Norwegian School of Economics, World Bank, Ministry of Finance of the Slovak Republic

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1 Introduction

Corporate tax policy plays a central role in shaping firm behavior, but estimates of how firms respond to tax changes remain incomplete. The elasticity of taxable income (ETI) captures how reported income adjusts to changes in tax incentives and is a key parameter for evaluating both the efficiency costs and revenue implications of tax reforms. As policymakers frequently revisit optimal corporate tax policy, it is critical to measure how firms respond to corporate taxes and how these responses differ across countries. This is even more salient given the ongoing global tax reform negotiations.¹ While extensive work has estimated ETIs in specific settings, we still lack a systematic, cross-country understanding of what drives differences in these responses across contexts (e.g., Bachas and Soto, 2021; Coles et al., 2022; Devereux et al., 2014; Krapf and Staubli, 2020; Lediga et al., 2019). Why do firms in some countries respond strongly to tax incentives while others appear far less sensitive? Which institutional, structural, or policy features explain this variation?

In this paper, we provide the first globally comparable estimates of the corporate ETI using a unified empirical framework applied to administrative tax return data from 19 countries across a wide range of regions and income levels. Our analysis leverages administrative data on millions of firms and harmonized methods to ensure comparability across settings. Specifically, we provide estimates for: Armenia, Canada, Chile, China, Costa Rica, Czechia, Dominican Republic, Ecuador, France, Greece, Guatemala, Honduras, Montenegro, Norway, Portugal, Rwanda, Senegal, Slovakia, and South Africa. In most cases, our study provides the first estimate of the ETI for these countries. Notable exceptions include Costa Rica, Slovakia, and South Africa (Bachas and Soto, 2021; Bukovina et al., 2025; Lediga et al., 2019).

We document substantial heterogeneity in corporate ETIs across countries, ranging from near zero in some low-income, high-compliance environments to well above one in wealthier, more financially integrated economies. Although the heterogeneity we document is meaningful, the range is narrower than that suggested by previous estimates. Elasticities range from 0.075 in Ecuador to 1.91 in Canada, with an average of 0.6. In other words, firms in Canada are the most sensitive to changes in marginal tax rates — these firms respond to a ten percent increase in net-of-tax rate by increasing taxable income by 19.1 percent and

¹For example, the Organization for Economic Co-operation and Development (OECD) has led the largest multi-national corporate tax negotiation in recent history, referred to as the Global Tax Deal. This reform would, among other changes, harmonize the minimum corporate tax rate faced by corporations across tax jurisdictions (e.g., a 15% global minimum tax). The OECD estimates that mismatches in tax policy cost countries \$100 to \$240 billion in lost revenue every year. <https://www.oecd.org/tax/beps/about/>

Ecuadorian firms are the least sensitive to marginal tax rates — these firms respond to a ten percent increase in the net-of-tax rate by increasing taxable income by 7.5 percent. These estimates suggest that the efficiency costs of higher tax rates are larger in Canada than in Ecuador. We interpret these as real elasticities, capturing firms’ investment and production responses to tax changes rather than reporting responses alone.² Importantly, because we estimate elasticities at a common kink across countries, our results reflect differences in behavior among comparable firms rather than differences in methodological approaches or sample composition.

A key innovation of our approach is the use of a standardized estimation package, which we deploy within each country’s secure data environment, adhering to local data confidentiality constraints.³ This ensures that all elasticity estimates are generated using harmonized methods, rendering them directly comparable across countries. We estimate the ETI using data on the distribution of taxable income building on pioneering work by Saez (2010), Kleven and Waseem (2013), Bertanha et al. (2023), and Coles et al. (2022). Our estimation focuses on bunching at a universal kink in the tax schedule at zero taxable income, which allows us to identify responses consistently across countries. Unlike many prior studies that exclude firms with negative income, our framework explicitly models firms operating at a loss, capturing the full distribution of corporate outcomes observed in administrative data. While focusing on the kink at zero has many advantages because it captures the most significant density of firms, captures profitable and loss-making firms, and exists consistently across countries, we also address potential limitations, such as the influence of loss carryforwards and bunching at round numbers. We estimate our model using a censoring model following recent advances in this literature (Bertanha et al., 2024, 2023). We also provide nonparametric bounds and semi-parametric estimates of the elasticity (Bertanha et al., 2023).

We develop a model that incorporates heterogeneity in both firm characteristics and country-level production technologies. Within a standard neoclassical model, we allow firms to be heterogeneous in productivity and fixed costs and allow the curvature of the production function to be heterogeneous across countries. As a result, we estimate an ETI that varies at the firm level. This heterogeneity allows our model to match the dispersed distribution of income. In particular, our model allows for non-positive values, which represent a non-negligible fraction of the corporate income distribution in most countries (e.g., over 40% in the UK, Bilicka (2019)). In contrast, most papers estimating elasticities of taxable in-

²In the Appendix, we extend our model to incorporate potential reporting responses.

³The package and code to simulate data can be found at www.nathanseegert.com/code.

come focus on individuals where negative income is negligible, and therefore, those methods explicitly exclude the possibility of negative income (e.g., Saez, 2010).

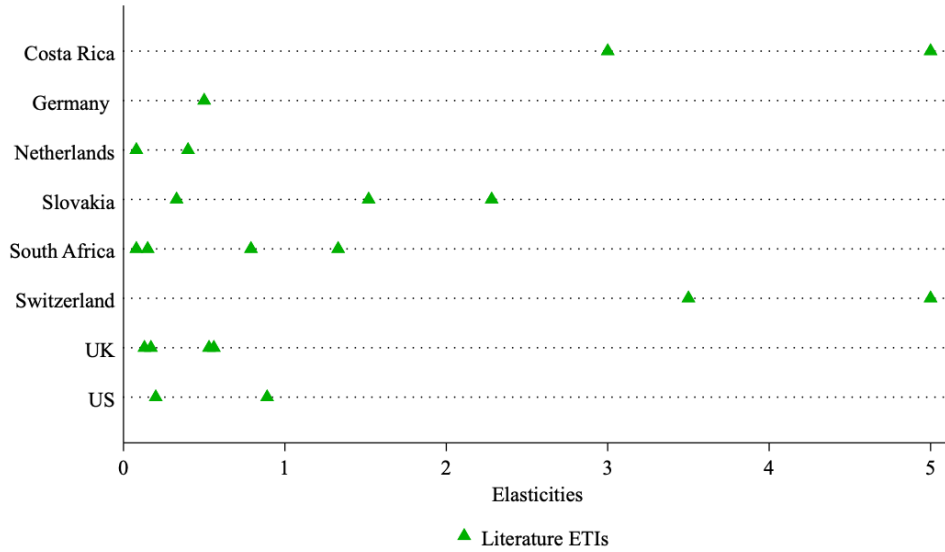
To understand the sources of elasticity variation, we pair our country-level estimates with 95 observable characteristics, spanning firm-level characteristics, tax system design, and country fundamentals. We show that our elasticity estimates are negatively correlated with statutory tax rates and positively related to GDP per capita and formality in the economy. Using a random forest model and Shapley decomposition, we find that three factors—statutory tax rates, total tax revenues, and the share of manufacturing in GDP—consistently explain most of the variation and have the minimum mean squared error. Our Shapley decomposition shows that each factor accounts for roughly one-third of the total, highlighting that firm characteristics, country fundamentals, and tax system design all contribute meaningfully to firms’ tax responsiveness. Our findings are robust to model extensions that include four and five predictors, with correlations of 0.89 to 0.99 between predicted and estimated elasticities. Using this framework, we generate out-of-sample predictions for 224 countries, producing a global elasticity map to inform both national and multilateral tax policy design.

Our contribution is threefold. First, we develop a flexible and scalable estimation method that enables consistent ETI measurement across countries. Second, we produce the largest cross-country dataset of corporate ETI estimates to date, creating a new empirical foundation for international tax research. Third, we demonstrate how machine learning methods can be used to decompose and extrapolate elasticity variation in the absence of administrative data access.

Our estimates contribute to a growing literature that views the elasticity of taxable income (ETI) as a central parameter for understanding both taxpayer behavior and the welfare implications of tax policy (Feldstein, 1995). Early research focused on individuals and produced widely varying estimates—ranging from -0.83 to 3 —due in part to differences in data sources and empirical strategies (Feldstein, 1995; Goolsbee, 1999). Subsequent work adopted a range of methods, including instrumental variables (Auten and Carroll, 1999; Gruber and Saez, 2002), panel-based identification (Gelber, 2014; Giertz, 2005; Kopczuk, 2005; Weber, 2014), and bunching estimators that exploit kinks and notches in the tax code (Kleven and Waseem, 2013; Saez, 2010).

More recently, attention has turned to the corporate ETI. Estimates for the United States range from 0.2 using IV strategies (Gruber and Saez, 2002) to 0.89 using control group methods (Coles et al., 2022), while bunching approaches yield estimates of 0.5 for

Figure 1: **Dispersion in existing estimates**



Notes: This figure graphs past estimates in the literature from different countries. This figure includes estimates from Costa Rica (Bachas and Soto, 2021), Germany (Dwenger and Steiner, 2012), Netherlands (Massenz, 2024), South Africa (Lediga et al., 2019), Switzerland (Krapf and Staubli, 2020), Slovakia (Bukovina et al., 2025), and the US (Coles et al., 2022; Gruber and Saez, 2002), the UK (Devereux et al., 2014).

the United Kingdom (Devereux et al., 2014). A growing number of studies have produced ETI estimates for other countries, including Costa Rica (Bachas and Soto, 2021), Germany (Dwenger and Steiner, 2012), the Netherlands (Massenz, 2024), South Africa (Lediga et al., 2019), Switzerland (Krapf and Staubli, 2020), and Slovakia (Bukovina et al., 2025). These estimates span a striking range—from 0 to 5—even among relatively similar economies. Figure 1 illustrates this dispersion.

There are two broad explanations for this heterogeneity. First, true differences in country fundamentals, tax system designs, and firm-level characteristics may generate genuine variation in elasticities across countries. This variation is important to understand for policy design. Second, as in the individual income literature, methodological differences across studies may themselves drive much of the observed dispersion. A key contribution of our paper is to minimize this second source of variation by applying a unified estimation framework to administrative data from a diverse set of countries, allowing us to isolate and study the drivers of corporate tax responsiveness.

The cross-country differences in elasticities that we find suggest that harmonizing tax

rates globally would result in varying costs and benefits for each country. For example, it has been long understood that global tax competition leads to a race-to-the-bottom in corporate tax rates as countries compete with each other for investment. Underpinning this dynamic is the notion that firms can easily relocate operations across taxing jurisdictions to take advantage of lower corporate tax rates. Consequently, harmonizing initiatives, such as, the OECD’s effort to introduce a global minimum tax, will likely affect the migration of firms across countries. The estimates and methods produced in this paper provide a foundation for future work that explores how firms will respond to these and other corporate tax policy proposals, especially in a global tax environment.

2 Neoclassical Two-Period Model

In this section, we develop a two-period neoclassical model of corporate behavior. With this model, we derive a relationship between bunching at a kink point in the marginal tax schedule and the elasticity of taxable income with respect to the net-of-tax rate. This model is robust to many additional features, though for ease of exposition, we present a parsimonious model.⁴

2.1 Model Fundamentals

Consider a firm, denoted i , in country k , that is owned by a single shareholder and begins period 1 with retained earnings, $X_{i,k}$. Firms are heterogeneous in their productivity, captured by $A_{i,k}$, and their fixed costs, captured by $C_{i,k}$. In period 1, firm i chooses its level of capital in period 2, $K_{i,k}$. Firms choose their level of capital in period 2 by determining the amount of retained earnings to distribute as a dividend payment ($D_{i,k} \geq 0$), or the amount of equity to issue ($E_{i,k} \geq 0$) but not both; $K_{i,k} = X_{i,k} + E_{i,k} - D_{i,k}$.⁵ In addition to equity, shareholders may hold government bonds with a tax-exempt rate of return, $r_k > 0$, which can vary by country k .

In period 2, capital generates income net-of-variable costs, such as depreciation costs,

⁴For example, the neoclassical model presented in this section is consistent with the more in depth model presented in Patel et al. (2014) that includes debt and dividend taxation.

⁵For ease of exposition, attention is restricted to equilibria where the firm does not pay out a dividend and issue equity concurrently. In the general model in Patel et al. (2014), the restriction that a firm does not pay out a dividend and issue equity concurrently is derived as equilibrium behavior with a dividend tax. The restriction does not change the following analysis.

according to a strictly concave production function

$$I_{i,k}(K_{i,k}) = \frac{1 + e_k}{e_k} A_{i,k}^{1/(1+e_k)} K_{i,k}^{\frac{e_k}{1+e_k}}. \quad (1)$$

Here, e_k determines the curvature of the production function and varies by country k . Finally, profit $F_{i,k}$ is income $I_{i,k}$ net of fixed costs $C_{i,k}$,

$$F_{i,k}(K_{i,k}) = I_{i,k}(K_{i,k}) - C_{i,k}. \quad (2)$$

A profit-maximizing firm may have negative profits in the presence of fixed costs, which we explicitly include here. This is a significant contribution to the literature, which often overlooks fixed costs and, therefore, cannot adequately explain negative profits in the data.⁶

In our baseline model, taxable income is equal to profits $Y_{i,k}(K_{i,k}) = F_{i,k}(K_{i,k})$. In Appendix A, we expand the model to allow firm i to also choose how much to avoid or evade taxation $\rho_{i,k}$, such that $Y_{i,k}(K_{i,k}) = F_{i,k}(K_{i,k}) - \rho_{i,k}$. Our estimation is robust to this addition, and we relegate it to the appendix as the real and reporting responses are not the focus of this paper.⁷

At the end of period 2, all firms liquidate, returning their principal and profits to their shareholders.

With this model in mind, firm i chooses $K_{i,k}$ to maximize its value to its shareholder:

$$\max_{K_{i,k}} V_{i,k} = X_{i,k} - K_{i,k} + \frac{(1 - t_{c,k})Y_{i,k}(K_{i,k}) + K_{i,k}}{1 + r_k}, \quad (3)$$

where $X_{i,k} - K_{i,k} = D_{i,k} - E_{i,k}$ are net distributions in period 1 valued by its shareholder.

The benefit of higher capital in period 2 is higher profit. Profit is taxed at the rate $t_{c,k}$ and discounted at the rate r_k .⁸ The cost of higher capital in period 2 is reduced distributions in period 1 (fewer dividends or more equity issuances).

Consider the case where there is a kink in the marginal tax rate schedule such that

⁶Firms often also have the ability to carry-forward losses to future periods. Loss carry-forwards have several implications. First, loss carry-forwards provide firms with losses in the past a stock of credits that they can use to lower their taxable income in future periods. As a result, we expect that tax systems with more generous loss carry-forwards to have higher elasticities. Second, loss carry-forwards have implications for the effective tax rates, especially around a kink at zero. We discuss losses and effective tax rates more in Section 4.

⁷For more information on the real and reporting trade-off, see Coles et al. (2022).

⁸The equilibrium rate of return r_k is assumed to be exogenous, abstracting from all general equilibrium effects.

$t_{c,k} = t_{0,k}$ for $Y_{i,k}(K_{i,k}) \leq \kappa$ and $t_{c,k} = t_{1,k}$ for $Y_{i,k}(K_{i,k}) > \kappa$, where $t_{0,k} < t_{1,k}$. Under this marginal rate schedule, the objective function faced by the firm is

$$\begin{aligned} \max_{K_{i,k}} \quad & V_{i,k} = X_{i,k} - \frac{1}{1+r_k} r_k K_{i,k} \\ & + \mathbb{I}(Y_{i,k}(K_{i,k}) \leq \kappa) \frac{(1-t_{0,k})Y_{i,k}(K_{i,k})}{1+r_k} \\ & + \mathbb{I}(Y_{i,k}(K_{i,k}) > \kappa) \frac{(1-t_{0,k})\kappa + (1-t_{1,k})(Y_{i,k}(K_{i,k}) - \kappa)}{1+r_k}, \end{aligned} \quad (4)$$

where $\mathbb{I}(Y_{i,k}(K_{i,k}) \leq \kappa)$ and $\mathbb{I}(Y_{i,k}(K_{i,k}) > \kappa)$ are indicator functions for taxable income being below or above the kink.

2.2 Model Solution

Firms choose their capital in period 2 to equalize the marginal benefit and marginal cost of additional capital. Traditionally, the Hall-Jorgenson formula (Hall and Jorgenson, 1967) sets the marginal product of capital equal to the opportunity-cost rate of return given by the risk- and tax-free rate r divided by one minus the corporate tax rate. The marginal product of capital decreases with capital because the income function is increasing and concave in capital. Firms subject to higher tax rates have higher opportunity-cost rates of return, and firms increase their marginal product of capital by decreasing their level of capital.

Our setting updates the Hall-Jorgenson formula for firms subject to a piecewise linear tax schedule with a kink. To provide intuition, it is helpful to consider firm capital choices for three representative firms that differ based on the curvature of their production function, seen in Figure 2. An easy way to understand these firms' optimal choices with a kinked tax schedule, depicted in panel (c), is to first consider their choices with kink-free schedules with high and low opportunity-cost rates, respectively, in panels (a) and (b).

In panel (a) of Figure 2, Firms 1, 2, and 3 are subject to a kink-free tax schedule with a low opportunity-cost rate (low tax rate, t_0). In this case, the Hall-Jorgenson formula applies, and all firms set their marginal product of capital equal to the opportunity-cost rate of return $\partial Y_{i,k}(K_{i,k})/\partial K_{i,k} \equiv Y'_{i,k}(K_{i,k}) = r_k/(1-t_{0,k})$. Firm 3 sets its capital such that its taxable income is below the kink. Firms 1 and 2, by comparison, set their capital such that their taxable income is above the kink.

In Panel (b) of Figure 2, Firms 1, 2, and 3 are subject to a kink-free tax schedule with a high opportunity-cost rate (high tax rate, $t_{1,k}$). The optimal choices for Firms 1 and 3 change relative to panel (a) according to the Hall-Jorgenson formula applied to the higher opportunity-cost rate. However, their taxable income values remain above the kink for Firm 1 and below the kink for Firm 3, as in panel (a). As a result, with a kinked tax schedule, Firm 1 will be subject to the high opportunity-cost rate and will choose its taxable income as depicted in panel (b). Firm 3, on the other hand, under a kinked tax schedule, will be subject to the low opportunity-cost rate and will choose its taxable income as depicted in panel (a).

In contrast, Firm 2's taxable income values change from above the kink in panel (a) to below the kink in panel (b). As a result, when Firm 2 is faced with a kinked schedule, it sets its taxable income exactly at the kink point, depicted in panel c. At this point, its marginal product of capital is greater than the opportunity-cost rate of return with the lower tax rate and less than the opportunity-cost rate of return with the higher tax rate. We depict the optimal choices of the three firms with the kinked tax system in panel c of Figure 2.

Formally, we derive the equilibrium capital and taxable income by taking the derivative of firm value with respect to capital in two regions where the derivative exists.

$$\frac{\partial V_{i,k}}{\partial K_{i,k}} = \begin{cases} \frac{1}{1+r_k} \left(-r_k + (1 - t_{0,k}) \frac{\partial Y_{i,k}(K_{i,k})}{\partial K_{i,k}} \right), & Y_{i,k}(K_{i,k}) < \kappa \\ \frac{1}{1+r_k} \left(-r_k + (1 - t_{1,k}) \frac{\partial Y_{i,k}(K_{i,k})}{\partial K_{i,k}} \right), & Y_{i,k}(K_{i,k}) > \kappa. \end{cases} \quad (5)$$

The solution for taxable income $Y_i(K_i)$ has a similar form to solutions derived in different contexts in this literature (Bertanha et al., 2023; Coles et al., 2022; Saez, 2010):

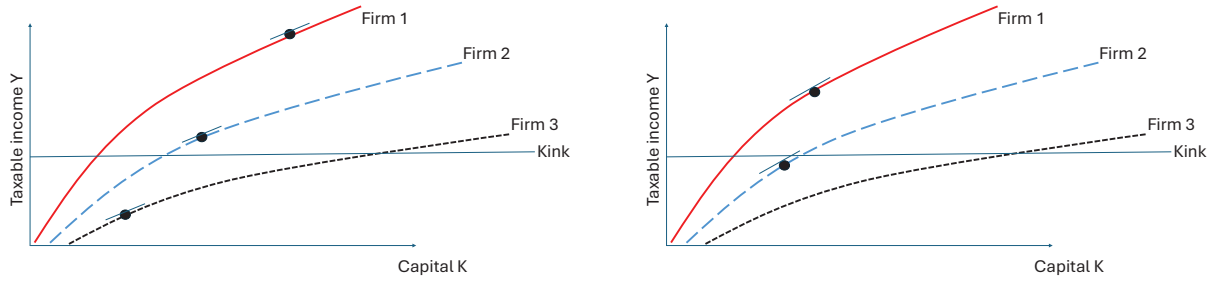
$$Y_{i,k}(K_{i,k}) = \begin{cases} \frac{1+e_k}{e_k} r_k^{-e_k} (1 - t_{0,k})^{e_k} A_{i,k} - C_{i,k}, & A_{i,k} \leq \underline{A}_{i,k} \\ \kappa, & \underline{A}_{i,k} < A_{i,k} < \bar{A}_{i,k} \\ \frac{1+e_k}{e_k} r_k^{-e_k} (1 - t_{1,k})^{e_k} A_{i,k} - C_{i,k}, & A_{i,k} \geq \bar{A}_{i,k}. \end{cases} \quad (6)$$

The thresholds are found by setting the optimal taxable income equal to the kink κ with both tax rates:

$$\underline{A}_{i,k} = (\kappa + C_{i,k})/\theta_{0,k}, \quad \text{and} \quad \bar{A}_{i,k} = (\kappa + C_{i,k})/\theta_{1,k}, \quad (7)$$

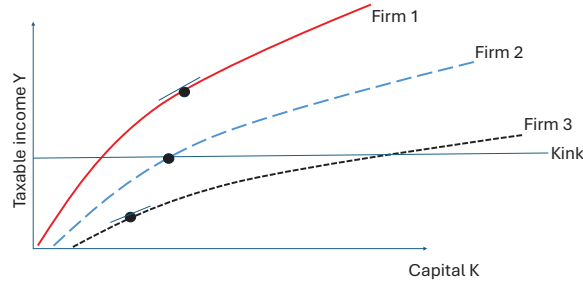
where $\theta_{c,k} = \frac{1+e_k}{e_k} r_k^{-e_k} (1 - t_{c,k})^e$, $c = 0, 1$.

Figure 2: **Firm optimization**



a Low opportunity-cost rate of return $r_k/(1 - t_{0,k})$ kink-free schedule

b High opportunity-cost rate of return $r_k/(1 - t_{1,k})$ kink-free schedule



c Kinked tax schedule with low and high opportunity-cost rates of return

Notes: This figure depicts the optimal choices of three firms in three different tax regimes in panels a, b, and c. Panel a depicts choices with a kink-free tax schedule with a low opportunity-cost rate of return. Panel b does the same with a high opportunity-cost rate of return. Finally, Panel c depicts the optimal capital taxable income choices with the kinked tax schedule with low and high opportunity-cost rates of return.

2.3 Elasticities

The key parameter in our model is e_k , which determines the curvature of the profit function with $A_{i,k}$ and varies by country k . In addition, e_k determines how much firms adjust their capital in response to a change in tax rate and, therefore, characterizes the elasticity of income with respect to the net-of-tax rate. To see this, we write equilibrium income, net of depreciation costs, as

$$I_{i,k}^* = \begin{cases} \frac{1+e_k}{e_k} r_k^{-e_k} (1-t_{0,k})^{e_k} A_{i,k}, & A_{i,k} \leq \underline{A}_{i,k} \\ \kappa + C_{i,k}, & \underline{A}_{i,k} < A_{i,k} < \bar{A}_{i,k} \\ \frac{1+e_k}{e_k} r_k^{-e_k} (1-t_{1,k})^{e_k} A_{i,k}, & A_{i,k} \geq \bar{A}_{i,k}. \end{cases} \quad (8)$$

This representation of income is simply the profit function before taking into account fixed costs $C_{i,k}$.

The elasticity of income with respect to the net-of-tax rate is given by

$$\frac{\partial I_{i,k}^*}{\partial(1-t_{0,k})} \frac{(1-t_{0,k})}{I_{i,k}^*} = e_k \frac{1+e_k}{e_k} r_k^{-e_k} (1-t_{0,k})^{e_k-1} A_{i,k} (1-t_{0,k}) \frac{1}{I_{i,k}^*} = e_k \quad \text{if} \quad A_{i,k} \leq \underline{A}_{i,k} \quad (9)$$

$$\frac{\partial I_{i,k}^*}{\partial(1-t_{1,k})} \frac{(1-t_{1,k})}{I_{i,k}^*} = e_k \frac{1+e_k}{e_k} r_k^{-e_k} (1-t_{1,k})^{e_k-1} A_{i,k} (1-t_{1,k}) \frac{1}{I_{i,k}^*} = e_k \quad \text{if} \quad A_{i,k} \geq \bar{A}_{i,k} \quad (10)$$

The elasticity of *taxable income* with respect to the net-of-tax rate, $\varepsilon_{i,k}$ varies by firm and country and depends on the elasticity of income,

$$\begin{aligned} \varepsilon_{i,k} &= \frac{\partial Y_{i,k}(K_{i,k})}{\partial(1-t_{0,k})} \frac{(1-t_{0,k})}{Y_{i,k}(K_{i,k})} = \frac{\partial I_{i,k}^*}{\partial(1-t_{0,k})} \frac{(1-t_{0,k})}{Y_{i,k}(K_{i,k})} = e_k \frac{I_{i,k}^*}{Y_{i,k}(K_{i,k})} \\ &= e_k \left(1 + \frac{C_{i,k}}{Y_{i,k}} \right) \end{aligned} \quad (11)$$

In our model, the elasticity of taxable income, ETI, is heterogeneous by firm and country and depends on the shape of the production function, captured by e_k , $A_{i,k}$, and $C_{i,k}$. For example, firms in countries with greater income elasticities will have greater taxable income elasticities, all else equal. We also show that firms with greater fixed costs are more elastic, potentially because they have lower marginal costs once their fixed infrastructure is in place. In a special case that we examine in this paper, i.e., at zero taxable income kink, the average

elasticity across firms within a country may differ from the elasticity at different points in the distribution. This is because the elasticity varies by firm and our model explicitly allows for that. We further discuss the heterogeneity across firms within a country in Section 4.4, and for expositional ease, otherwise focus on a single elasticity by country.

Coles et al. (2022) shows that the elasticity of taxable income is a combination of the real and reporting responses. In the paper, we focus on the real response. In Appendix A, we show how this equation is updated when we allow for reporting responses.

Equation (6) maps the unobserved variables $A_{i,k}$ and $C_{i,k}$ to the observed variable $Y_{i,k}$. This mapping depends on the kink point κ , the value of production amenities to the left of the kink $\theta_{0,k} = \frac{1+e_k}{e_k} r_k^{-e_k} (1 - t_{0,k})^e$ and on the right $\theta_1 = \frac{1+e_k}{e_k} r_k^{-e_k} (1 - t_{1,k})^e$, and the income elasticity e_k . We can use this mapping to write the mixed continuous-discrete distribution of $Y_{i,k}(K_{i,k})$, which is observed by the researcher, as a function of the continuous distribution of $(A_{i,k}, C_{i,k})$, which is unobserved. The identification problem consists of using this mapping to back out e_k using the observed distribution of $Y_{i,k}$, which requires assumptions on the unobserved distribution of $(A_{i,k}, C_{i,k})$. We discuss solutions to this problem in the next section.

3 Datasets and Empirical Methods

In this section, we summarize the datasets and discuss the empirical methodology we use to estimate elasticities across countries. Each estimate was performed separately and is based on administrative tax data, which reflects average firm sensitivity to tax rates within the context of the tax system in which they operate. Each country has information on taxable income that is common across all settings and a set of covariates that differ across countries, but always include at least a measure of revenues and deductions to allow us to estimate fixed costs.

3.1 Datasets and Comparison of Countries

In this paper, we use corporate tax returns datasets from the following 19 countries: Armenia, Canada, Chile, China, Costa Rica, Czechia, Dominican Republic, Ecuador, France, Greece, Guatemala, Honduras, Montenegro, Norway, Rwanda, Senegal, Slovakia, South Africa, and Uruguay. We provide detailed information on each dataset used in Appendix D. That information includes details on the data, how it was accessed, which time period was

used to estimate the elasticities, and a brief description of the corporate tax system in each country.

The countries in our sample represent places with different tax systems, firm characteristics, and country fundamentals. In Table 1, we provide details of a country's tax system using data from the Center for Business Taxation tax database. Countries in our sample have tax rates that range from 10% in Montenegro to 33% in France (column 2). We expect a country's elasticity may depend on its tax rate, though this is an open question. In Columns 3 - 7, we report rules about loss carryforwards and carryback, loss consolidation, minimum taxes, and tax holidays. Some countries allow indefinite loss carryforwards (e.g., Chile, France), some allow only 3 years (Senegal) or none (Guatemala). Most countries have no loss carrybacks, but three allow some (Canada, France, and Norway). Six countries in our sample allow loss consolidation across multiple affiliates that belong to the same firm within the country (Dominican Republic, France, Montenegro, Norway, Senegal, and South Africa). Many countries also have a minimum tax, which can limit deductions or tax holidays, which may facilitate reporting responses. We anticipate countries with more flexible tax codes, such as those that allow more carryforwards or tax holidays, to have higher elasticity.

Firms also differ across countries in our sample. In Table 2, we provide details of firm distribution within each country using Orbis data reported in US dollars. In columns 2-4, we report measures of average firm size: assets, revenues, and number of employees, respectively. Slovakia and Greece have relatively small average firms, and South Africa and Canada have relatively large average firms. In columns 5-7, we report the 10th, 90th, and 99th percentiles of asset distribution. We expect that firm size is related to firm elasticity. For example, larger firms may be more tax sophisticated and adjust their employment and investment in response to higher taxes. Alternatively, smaller firms may have fewer fixed costs and be able to adjust production faster.

Finally, countries also differ in ways other than their tax system and firm distribution, which we call country fundamentals. In Table 3, we report a few of these characteristics using data from the World Bank World Development Indicators to demonstrate the wide span of these characteristics across countries in our sample. We consider characteristics of the economy (column 2, log GDP per capita), the organization of economic activity (column 3, indicator of formality), and the quality of government (column 4, governance quality index). The Dominican Republic has a low index of formality, while Canada and Norway have a high level of formality. Canada and Chile have a higher governance quality index, while Ecuador and Honduras have lower indexes. We complement these by reporting the percentage of

Table 1: Tax System Characteristics

Country	Tax rate	Loss carry forward	Loss carry back	Loss cons.	Min tax	Tax holiday
(1)	(2)	(3)	(4)	(5)	(6)	(7)
Armenia	0.2	5	No	No	Yes	Yes
Canada	0.17	20	3	No	Yes	No
Chile	0.20	indefinitely	No	No	No	Yes
China	0.26	5	No	No	No	Yes
Costa Rica	0.30	5	No	No	No	Yes
Czechia	0.19	5	2	No	No	No
Dominican Republic	0.27	5	No	Yes	No	No
Ecuador	0.24	5	No	No	No	No
France	0.33	indefinitely	3	Yes	Yes	No
Greece	0.27	5	No	No	No	No
Guatemala	0.29	No	No	No	Yes	No
Honduras	0.25	3	No	No	Yes	Yes
Montenegro	0.10	5	No	Yes	No	Yes
Norway	0.27	indefinitely	2	Yes	No	No
Portugal	0.21	5	No	Yes	No	No
Rwanda	0.31	5	No	No	No	No
Senegal	0.28	3	No	Yes	Yes	No
Slovakia	0.23	7	No	No	No	Yes
South Africa	0.32	indefinitely	No	Yes	No	Yes

Note: We present statutory corporate tax rates in the years when each country estimates are available. These numbers come from CBT Tax Database. The remainder of characteristics were provided by each country's co-authors. All countries have VAT registration (over some threshold) and payment advances (monthly or quarterly).

GDP that comes from natural resources (column 5), the Gini coefficient of income (column 6), which measures the distribution of resources in an economy, and population (column 7). In principle, each of these characteristics may influence how elastic firms are in each country and countries we have in our sample range from small (Costa Rica) to large (China) economies.

In Section 4.5, we use these differences in characteristics across countries in our sample to provide evidence on how these characteristics influence how elastic firms are.

Table 2: Firm Characteristics

Country	Assets	Revenue	Employees	p10 Assets	p90 Assets	p99 Assets
Armenia	22,000,000	9,482,662	172	5,588,624	39,100,000	194,000,000
Canada	3,690,000,000	312,000,000	473	18,300,000	11,700,000,000	48,400,000,000
Chile	620,000,000	71,600,000	3,382	707,220	946,000,000	9,510,000,000
China	171,000,000	147,000,000	412	386,456	118,000,000	2,870,000,000
Costa Rica	630,000,000	90,700,000	519	7,967,706	1,420,000,000	9,450,000,000
Czechia	2,666,367	2,406,674	26	6,538	2,264,301	33,000,000
Dominican Republic	357,000,000	31,100,000	1,399	2,613,614	427,000,000	11,500,000,000
Ecuador	375,000,000	64,800,000	679	12,100,000	1,170,000,000	4,890,000,000
France	16,900,000	4,546,513	26	37,703	3,157,918	59,000,000
Greece	5,153,476	3,679,307	22	174,401	8,967,624	56,800,000
Guatemala	570,000,000	70,000,000	—	9,269,739	1,620,000,000	7,570,000,000
Honduras	799,000,000	194,000,000	873	335,000,000	1,820,000,000	3,520,000,000
Montenegro	22,900,000	8,937,527	151	182,616	53,100,000	211,000,000
Norway	12,400,000	2,768,294	9	14,821	3,783,857	73,100,000
Portugal	2,700,000,000	95,600,000	462	40,500,000	3,580,000,000	58,200,000,000
Rwanda	19,400,000	11,000,000	4	794,872	58,500,000	58,500,000
Senegal	463,000,000	33,600,000	365	103,000,000	1,160,000,000	1,380,000,000
Slovakia	2,937,793	1,648,581	23	6,755	1,949,026	27,500,000
South Africa	4,100,000,000	1,110,000,000	1,133	1,973,567	5,850,000,000	69,200,000,000

Note: Firm-level data comes from Orbis and represents averages for 2004 - 2019. All amounts are reported in US Dollars. p10 is the 10th percentile of the distribution of assets, p90 is the 90th percentile and p99 is the 99th percentile.

Table 3: Country fundamentals.

Country	log GDP pc	% Formally Reg.	Governance	Total Taxes	Gini	Population	Manuf. VA (% GDP)	Manuf. Imports
Armenia	8.22	97.23	-0.27	3.44	32.14	2.98	10.03	56.95
Canada	10.78	—	1.62	3.38	32.99	34.69	10.18	77.02
Chile	9.59	96.90	1.14	3.32	45.80	17.48	10.98	70.31
China	8.90	95.80	-0.51	4.21	38.83	1354.64	30.40	60.23
Costa Rica	9.26	87.90	0.59	4.06	48.49	4.66	12.56	74.44
Czechia	9.91	98.23	0.91	3.84	25.56	10.46	23.03	81.06
Dominican Republic	8.81	79.50	-0.32	3.72	44.39	9.68	14.65	61.85
Ecuador	8.68	88.20	-0.67	3.53	45.53	15.51	13.75	74.57
France	10.61	99.20	1.19	4.21	32.09	63.24	10.24	73.86
Greece	9.98	98.10	0.43	3.88	34.76	10.95	8.54	52.14
Guatemala	8.22	89.10	-0.60	3.67	48.30	15.30	18.55	66.06
Honduras	7.71	85.40	-0.61	3.77	49.34	8.49	17.16	60.53
Montenegro	8.89	97.03	0.05	3.26	37.81	0.62	4.13	60.90
Norway	11.38	—	1.74	3.69	27.36	5.02	6.64	77.40
Portugal	9.98	93.10	1.04	3.74	34.66	10.45	11.83	65.57
Rwanda	6.56	85.15	-0.32	3.56	44.40	10.52	2.26	73.74
Senegal	7.21	83.35	-0.19	3.82	38.30	13.82	17.22	46.43
Slovakia	9.78	99.07	0.75	3.89	25.33	5.40	18.56	78.99
South Africa	8.80	93.40	0.27	3.45	63.00	53.06	12.03	64.10

Note: Country-level fundamentals data comes from World Bank World Development Indicators. We provide details of data sources and definitions for each variable in Table C1.

3.2 Empirical Methods

This section discusses the identification of the parameter e_k using data with bunching at zero profits. Two key features of the corporate income tax data are that many firms (roughly half in some countries) report negative income in a given year, and the density of firms is highest around zero. Because of these two features and to make estimates of e_k most comparable across countries, we estimate ETIs at the zero kink. In contrast, the original bunching identification strategy was developed for personal income tax data (Saez, 2010), where all values are positive. Recent bunching methods (Bertanha et al., 2023; Goff, 2022; Pollinger, 2024) apply the logarithmic transformation to the data and therefore cannot be applied immediately here.

We start with the mapping between observables and unobservables in Equation (6). We say the parameter e_k is identified if we are able to solve for e_k as a function of the distribution of the data $Y_{i,k}$ while imposing minimal assumptions on the distribution of the unobserved variables $(A_{i,k}, C_{i,k})$. In order to make progress on this identification problem and propose an estimator, we draw a parallel with Bertanha et al. (2023). They study a related problem and seek to identify the elasticity of labor supply with respect to income taxes. Their mapping between observables and unobservables is simpler than ours in at least three dimensions (compare our Equation (6) to their Equation (3)).

First, they have one unobserved variable, while we have two, productivity A_i and fixed costs C_i . Second, their observed income variable and kink point are always positive, allowing for the logarithmic transformation, which is not possible in our case because of negative taxable income and a kink that is zero in most contexts. Third, the threshold values on the unobserved variable $A_{i,k}$ that dictates the three regimes in Equation (6) are firm-specific in our case while constant across individuals in their case. In light of these differences, we develop a novel bunching identification strategy by transforming our problem into the problem studied by Bertanha et al. (2023); we then rely on their arguments to identify our parameter of interest e_k .

Assumptions on fixed costs The first step is to reduce the dimension of the problem from two unobserved variables, $A_{i,k}$ and $C_{i,k}$, to only one unobserved variable, $A_{i,k}$. We use firm-level data to identify a functional form relating revenue and deductions, where deductions serve as a proxy for total costs (Coles et al., 2022). The firm fixed cost equals the deductions amount predicted by the functional form when revenues are set to zero.

For a unknown vector of parameters $\beta \in \mathbb{R}^n$, assume there exist known functions $f(\cdot)$

and $g(\cdot; \beta)$ such that

$$f_k(\text{Deductions}_{i,k}) = g_k(\text{Revenues}_{i,k}; \beta) + \varepsilon_{i,k}, \quad (12)$$

where $\varepsilon_{i,k}$ is an idiosyncratic shock independent of revenues, $f_k(\cdot)$ is invertible, and there is sufficient variation in revenues to identify β_k . Once we have β , we retrieve $\varepsilon_{i,k} = f_k(\text{Deductions}_{i,k}) - g_k(\text{Revenues}_{i,k}; \beta_k)$ and the firm fixed costs is

$$C_{i,k} = f_k^{-1}(g_k(0; \beta_k) + \varepsilon_{i,k}). \quad (13)$$

The choice of deduction variables and functional form varies according to the data availability in each country. Table B4 in the Appendix provides the details. For example, one possible choice is $f_k(y_{i,k}) = \log(y_{i,k})$ and $g_k(x_{i,k}; \beta_{i,k}) = \beta_0 + \beta_{1,k}x_{i,k}$, in which case $C_{i,k} = \exp(\beta_{0,k} + \varepsilon_{i,k})$. In cases where panel data are available, $\beta_{0,k} + \varepsilon_{i,k}$ is replaced by a firm fixed effect. Estimation of $C_{i,k}$ follows naturally from readily available estimators for β_k in each context.

Operating leverage The second step in our identification argument is to create a new variable $W_{i,k}$ as a function of $Y_{i,k}$ and $C_{i,k}$, which is a measure of operating leverage. We define operating leverage W_i as profits $Y_{i,k}$ plus fixed cost $C_{i,k}$ divided by fixed cost $C_{i,k}$,

$$W_{i,k} = \frac{Y_{i,k} + C_{i,k}}{C_{i,k}}. \quad (14)$$

Recall that $Y_{i,k}$ equals revenue minus variable and fixed costs, so that $Y_{i,k} + C_{i,k}$ equals revenue minus variable costs. The $W_{i,k}$ ratio equals 1 when profit equals zero and the firm's revenue simply covers their costs, that is, they are at their break-even point. The ratio is greater than 1 when the firm is generating profits. The ratio is less than 1 but above zero when revenue covers the variable cost but not the fixed cost, so the firm is above their shutdown point. As such, this ratio is similar to the degree of operating leverage (Ferri and Jones, 1979). The researcher is able to construct $W_{i,k}$ because $Y_{i,k}$ is directly observed and $C_{i,k}$ is constructed as explained above via Equation (13).

Using Equation (6) in Equation (14) and assuming the most common case of $\kappa = 0$, we

find W_i to have three regimes:

$$W_{i,k} = \begin{cases} \theta_{0,k} A_{i,k}/C_{i,k}, & \text{if } A_{i,k}/C_{i,k} \leq 1/\theta_{0,k}, \\ 1, & \text{if } 1/\theta_{0,k} < A_{i,k}/C_{i,k} < 1/\theta_{1,k}, \\ \theta_{1,k} A_{i,k}/C_{i,k}, & \text{if } 1/\theta_{1,k} \leq A_{i,k}/C_{i,k}. \end{cases} \quad (15)$$

For a continuous distribution of $A_{i,k}/C_{i,k}$, operating leverage $W_{i,k}$ has a mixed continuous-discrete distribution with a mass point at $W_{i,k} = 1$. Below the mass point, $W_{i,k}$ increases with the ratio $A_{i,k}/C_{i,k}$, with a slope of $\theta_{0,k} = \frac{1+e_k}{e_k} r_k^{-e_k} (1 - t_{0,k})^{e_k}$. Similarly, above the kink, W_i increases with the ratio $A_{i,k}/C_{i,k}$, with a slope of $\theta_{1,k} = \frac{1+e_k}{e_k} r_k^{-e_k} (1 - t_{1,k})^{e_k}$. The three regimes summarized above have important characteristics. First, the mapping presented in Equation (15) relates one unobserved variable $A_{i,k}/C_{i,k}$ to the observed variable $W_{i,k}$. Second, the variable $W_{i,k}$ is always positive, allowing for a logarithmic transformation.⁹ Third, the threshold values on the variable $A_{i,k}/C_{i,k}$ that dictate the three regimes are constant across firms.

Next, we rewrite the mapping in Equation (15) using the logarithmic transformation. Define $w_{i,k} = \log(W_{i,k})$, $s_{c,k} = \log(1 - t_{c,k})$, $c = 0, 1$, $\mu_k = \log\left[\left(\frac{1+e_k}{e_k}\right) r^{-e_k}\right]$, $a_{i,k} = \log(A_{i,k}/C_{i,k})$. It follows that

$$w_{i,k} = \begin{cases} \mu_k + e_k s_{0,k} + a_{i,k}, & \text{if } a_{i,k} \leq -\mu_k - e_k s_{0,k}, \\ 0, & \text{if } -\mu_k - e_k s_{0,k} < a_{i,k} < -\mu_k - e_k s_{1,k}, \\ \mu_k + e_k s_{1,k} + a_{i,k}, & \text{if } -\mu_k - e_k s_{1,k} \leq a_{i,k}. \end{cases} \quad (16)$$

Define $n_{i,k}^* = \mu_k + a_{i,k}$. We finally arrive at

$$w_{i,k} = \begin{cases} n_{i,k}^* + e_k s_{0,k}, & \text{if } n_{i,k}^* \leq -e_k s_{0,k}, \\ 0, & \text{if } -e_k s_{0,k} < n_{i,k}^* < -e_k s_{1,k}, \\ n_{i,k}^* + e_k s_{1,k}, & \text{if } -e_k s_{1,k} \leq n_{i,k}^*, \end{cases} \quad (17)$$

which is the same mapping studied by Bertanha et al. (2023) with a kink value of 0 (i.e., see their Equation (4)).

⁹Empirically, we do find negative values of $W_{i,k}$ when firms have very high losses and their fixed costs do not explain these losses. Our bunching identification strategy relies on the distribution of $W_{i,k}$ local to its mass point at 1 and thus negative values of $W_{i,k}$ do not harm the identification of e_k .

Bunching methods The final step of our identification argument is to apply the identification strategies of Bertanha et al. (2023) to data on $w_{i,k}$. These strategies seek to identify the elasticity parameter e_k under various assumptions on the distribution of the unobserved variable $n_{i,k}^*$. It is known that identification is impossible when the distribution of $n_{i,k}^*$ belongs to the nonparametric class of all continuous distributions (Blomquist et al. (2021), Bertanha et al. (2018)). Thus, bunching is only informative of e_k with assumptions stronger than continuity. In contrast, parametric assumptions on the distribution of $n_{i,k}^*$ yield identification of the elasticity, but are often deemed too strong. In the next paragraphs, we revisit two identification methods of Bertanha et al. (2023) that balance the two extremes.

The first method relies on the weakest type of assumption and yields partial identification of the elasticity parameter e_k . The assumption is that the probability density function of $n_{i,k}^*$ is Lipschitz continuous with slope bounded by a constant $M_k \in (0, \infty)$. Under this nonparametric class of distributions, Theorem 2 of Bertanha et al. (2023) gives analytical expressions for upper and lower bounds on e_k . The expressions depend on M_k and other quantities that can be estimated from the data. We refer to this first method as nonparametric bounds.

The nonparametric bounds provide insights into the heterogeneity across countries without relying on strong assumptions. These bounds nest the original estimators that either assume $M_k = 0$ or the so-called trapezoidal approximation (Chetty, 2009; Saez, 2010). Given a reasonable range of values for the slope restriction, M_k , we plot bounds as functions of M_k for each country. As we increase M_k , we loosen the shape restriction on the distribution of $n_{i,k}^*$ and can, therefore, examine the sensitivity of elasticity bounds to assumptions on the heterogeneity distribution.

The second method point-identifies the elasticity parameter e_k by relying on additional firm-level data $X_{i,k}$ and a semi-parametric distributional assumption on $n_{i,k}^*$. Bertanha et al. (2023) show how to identify e_k using a Tobit model with a censoring point in the interior of the distribution. The Tobit model typically assumes the normality of $n_{i,k}^*$ conditional on $X_{i,k}$. Lemma 1 by Bertanha et al. (2023) shows that such a normality condition is not necessary for the identification of the elasticity. In fact, a sufficient condition requires that the distribution of n_i^* equal an average of normal distributions, where the average is taken over the distribution of X_i . Tobit models are extremely easy to estimate with statistical software that does so is readily available. A simple way to assess the distributional assumptions of this second method is to compare the best-fit distributions produced by the model with the raw histogram of the data. Note that this second method relies on stronger assumptions than the first method above. In particular, in the absence of additional data $X_{i,k}$, this method

simply assumes the normality of $n_{i,k}^*$.

We can further loosen the distributional assumptions by relying on a truncated Tobit model, where the dependent variable $w_{i,k}$ is truncated at an interval centered at the mass point at zero. Intuitively, the distributional assumption that a mixture of normals approximates the distribution of $n_{i,k}^*$ only needs to hold over a smaller region rather than the full support. Further, we can plot the estimated value of the elasticity as a function of decreasing truncation intervals to examine the sensitivity of estimates to the distributional assumptions.

We follow the advice of Bertanha et al. (2023) and use both methods to assess the robustness of elasticity values to different types of assumptions, from the weaker assumption of the nonparametric bounds to the stronger distributional assumption of the Tobit regressions. We further conduct sensitivity analyses by varying the strength of the assumption and examining how elasticity values change. In the first method, we vary the maximum slope value M_k around values that are consistent with slope magnitudes observed in the distribution of $w_{i,k}$. In the second method, we vary the truncation window around values for which the model best-fit distributions are similar to raw histograms of the data. Bunching is most informative on the elasticity parameter whenever elasticity values are robust to these sensitivity analyses across methods. Finally, estimators that follow the two identification strategies described in this section are implementable in Stata through the package named **bunching**. See Bertanha et al. (2022) for full details on the package.

3.3 Limitations

We finish this section by outlining a set of limitations that our empirical approach has. First, there is a fundamental question about whether the elasticity of taxable income is a structural parameter. In our model, it is not. The ETI we estimate varies by country and firm, which we explore more in section 4.4. The elasticity of taxable income may also depend on real and reporting responses. Based on our model, we interpret the bunching response as a real response—coming only from the curvature in the production function that we estimate. However, some of this response could be a reporting response. To disentangle the real and reporting responses, we would need additional assumptions or data, which we explore in Appendix A. Our approach of not treating the ETI as a structural parameter follows the literature that describes many potential reasons why the elasticity of taxable income may be context-specific, for example, because of the rules around loss carry forwards (Doerrenberg et al., 2015; Saez et al., 2012).

Second, we focus on the kink at zero because of its many advantages, but it also has

some limitations. The advantages include comparability since all countries have a kink there and the ability to generalize the estimates since a large number of firms are at or have been around zero. The limitation is that firms may use net operating losses to get their income to be zero exhibiting bunching at that point. Since this is a behavioral response to the structure of the tax system, we are explicitly interested in this response. However, policymakers may be interested in other responses that are separate from those, and we further explore those in Appendix B. Specifically, we consider a subset of firms without net operating losses and provide ETI estimates only for those firms.

Third, an extension of our model could consider heterogeneous elasticities based on different firm characteristics. For example, an extensive literature discusses differences between loss- and profit-making firms (Joos and Plesko, 2005). One reason these firms may have different elasticities of taxable incomes is that many countries allow firms to carry forward (or back) the losses they make to offset profits in different years. Therefore, firms' effective marginal tax rate is positive even when in a loss position (see an extensive literature that estimates effective tax rates (Blouin et al., 2010; Graham, 1996)). The rules around how firms can use these losses make them more or less elastic. For example, firms will be more elastic in countries that allow firms to use loss carry-forwards with few restrictions. Another example of firm heterogeneity could be that firms of different sizes may respond differently to changes in tax rates, as discussed above. Unfortunately, in most countries, we do not have data on firm characteristics beyond taxable income that would allow us to explicitly model any of this heterogeneity. Instead, we explore differences in ETIs that these loss offset provisions and firm-size differences will generate using country-level data in Section 4.

Another set of extensions to our model would be to make it dynamic, allowing us to think about potential mechanisms that result in firms bunching at zero. In that, we could consider the short-run intertemporal shifting of income¹⁰, firms taking advantage of tax incentives or adjusting employment or investment and the related adjustment costs, or firms using net operating losses and related tax provisions. Since the short and long-run elasticities may differ, a dynamic model would help explore these differences and allow us to estimate how much elasticity variation is due to intertemporal shifts, adjustment costs, or net operating losses. Unfortunately, our datasets do not systematically cover many firm-level characteristics, which means we cannot consider which firms bunch at zero.

¹⁰Le Maire and Schjering (2013); Miller et al. (2024) discuss income shifting across time in the context of small businesses, either self-employed or owner-managers.

4 Results

We start by outlining our baseline results in which we provide a set of semi-parametric point estimates of ETI for all 19 countries in our sample. We then discuss the role of truncation and nonparametric bounds for each of our estimates. We follow with analysis of factors that determine the differences in ETIs across countries, both descriptive and machine learning based. We finish with an extension in which we propose how to use our estimates to predict elasticities in the remaining countries not covered by our sample.

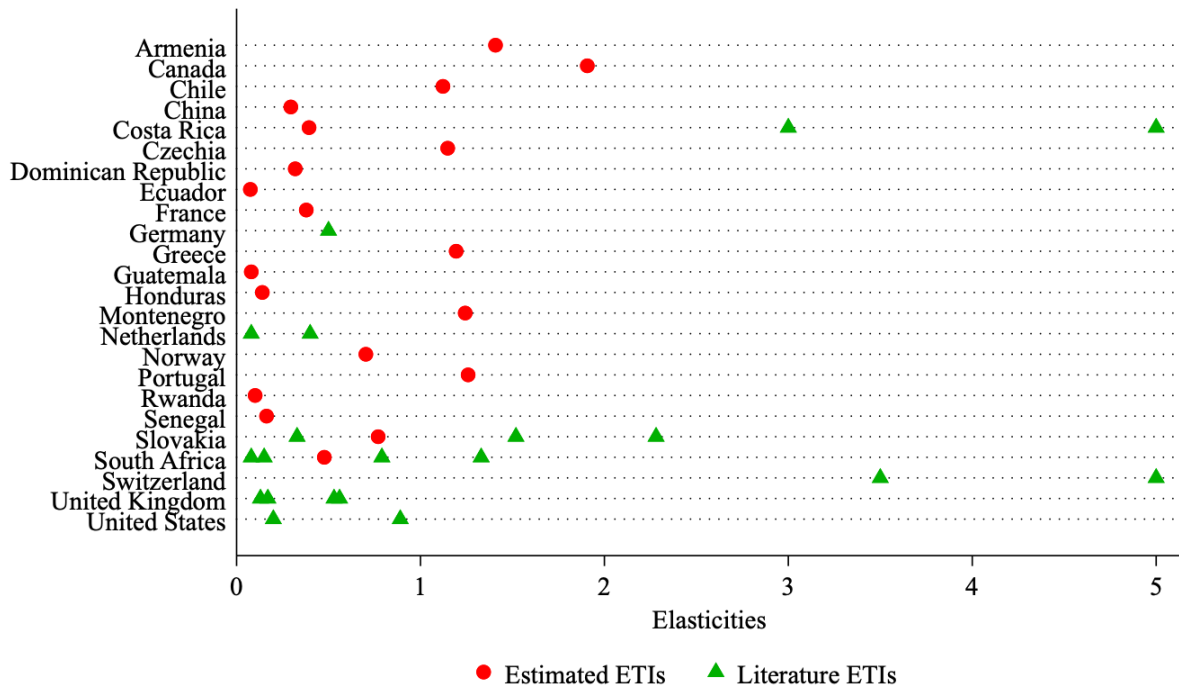
4.1 Cross country comparisons of the elasticity of taxable income

To ease comparison across countries, we start by reporting a single income elasticity estimate, e_k , for each country with standard errors. Note that this income elasticity is equivalent to taxable income elasticity, ϵ_k when the fixed costs are zero. We summarize our estimates in Figure 3 and in Table 4 focusing on our preferred specifications based on 60% of the data. In the Table, in column (1), we present elasticity estimates themselves, in column (2) we include standard errors, in column (3) we show the number of firms (or observations) that are being used to estimate elasticities, and in column (4) the sample duration.

Our elasticity estimates range from 0.05 in Czechia to 1.9 in Canada, which implies that a 10% increase in the net-of-tax rate results in between 0.5% and 19% of an increase in taxable income. Twelve countries in our sample have ETIs below 0.5, five above one, and the remaining two have ETIs between 0.5 and 1. As such, most of our estimates are considerably lower than the prevailing estimate of the responsiveness of U.S. corporations, which are predicted to respond to a 10% increase in the net-of-tax rate with an increase in taxable income of 8.9%.

Czechia, Ecuador, and Guatemala have the lowest ETIs in our sample. They all have mid-range corporate tax rates, with relatively low loss offset provisions, but are very different in terms in their country fundamentals. Czechia has relatively high GDP per capita, quite good governance and relatively higher formality of their economy with low reliance of natural resources. In turn, Ecuador relies heavily on natural resources, has very poor governance and lower formality and GDP per capita. On the other end of the spectrum, economies like Armenia, Canada, Greece, Chile and Montenegro all have ETIs above one, yet they are all very different both in terms of corporate tax rates and country fundamentals. For example, Montenegro has a 10% tax rate and Greece has 27% tax rate. Canada has very high GDP per capita, very formalized economy, exceptionally high governance quality and

Figure 3: Estimated Elasticities



Notes: This figure plots the estimated and previous literature ETIs. The estimated ETIs are in filled red circles and the previous literature ETIs in filled green triangles.

Table 4: Corporate Elasticity of Taxable Income: cross-country comparison.

Country	Elasticity of income (1)	Std. error (2)	Firms (3)	Years used (4)
Armenia	1.408	0.024	80,839	2007-2017
Canada	1.907	0.068	10,056	2012-2019
Chile	1.122	0.057	14,148	2018
China	0.295	0.009	269,225	2009
Costa Rica	0.394	0.001	542,880	2006-2020
Czechia	1.148	0.018	244,870	2018-2022
Dominican Republic	0.319	0.005	118,033	2007-2010
Ecuador	0.075	0.001	98,742	2014-2018
France	0.379	0.001	716,950	2010-2016
Greece	1.194	0.063	25,712	2002-2004
Guatemala	0.080	0.001	82,238	2006-2020
Honduras	0.140	0.002	44,065	2014-2020
Montenegro	1.243	0.019	45,898	2011-2020
Norway	0.703	0.002	284,336	2006-2013
Portugal	1.259	0.008	449,185	2016-2019
Rwanda	0.101	0.005	76,896	2010-2020
Senegal	0.163	0.004	14,415	2010-2020
Slovakia	0.770	0.067	48,922	2013
South Africa	0.477			2010-2018

Notes: This table presents estimates of the corporate elasticity of income across countries, outlining the fraction of the data used. Note that the elasticity of income reported here is equal to the Elasticity of Taxable Income (ETI) evaluated for a firm with zero fixed costs.

low reliance on natural resources. Chile relies heavily on natural resources, while Greece and Montenegro have relatively lower governance quality. Some of those countries, such as, for example, Greece, have historically struggled to raise revenue to fund the provision of public goods, consistent with the hypothesis that Greek taxpayers are very sensitive to tax rates and/or the Greek tax authority struggles with tax administration and enforcement. In all, no definite patterns emerge in which high tax rates, reliance on natural resources, or different firm distribution appear to be characteristics of countries with higher ETIs. We return to this analysis in depth in Section 4.4 of this paper, where we systematically evaluate all of the potential country-level observable determinants of differences in ETIs.

The novelty of this paper lies in the fact that we can compare our ETI estimates across countries and the differences must be driven by country, firm or tax system fundamentals, for example, differences in the risk preferences of the firms and/or the detection and enforcement mechanisms of the tax authority, as highlighted by canonical models of optimal tax evasion (Allingham and Sandmo, 1972). Recall that past empirical estimates of the corporate elas-

ticity of taxable income have varied from close to zero to as high as five. Here, we show that across a wide range of countries, the elasticity of taxable income is much less variable than was previously suggested, with our top estimates being less than half of what the top ETI estimates in the literature are. This reduction in variability has important consequences for policymakers considering changes in business tax policy, an area of active policy debate. Our new estimates suggest that tax receipts may be considerably less affected by changes in business tax rates than previous literature suggests.

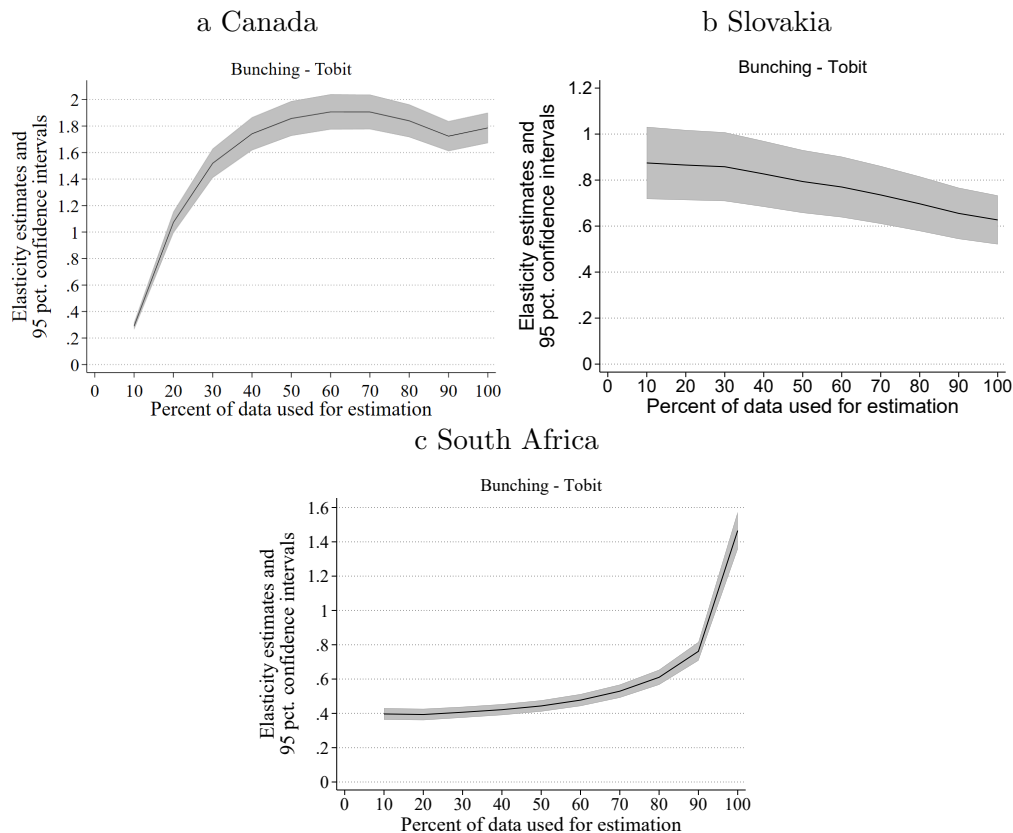
4.2 Truncation results

Our preferred estimates rely on a truncated sample rather than the full dataset because the assumption of normality is often too restrictive when applied to the entire distribution of firms. As illustrated in panels (d) of Figures D4–D14 in Appendix B, the normality assumption yields a poor fit to the data when all observations are included. Consequently, estimates based on this assumption are likely to be biased or inconsistent. By contrast, restricting the sample to exclude the extremes of the distribution typically enhances the plausibility of the identifying assumptions. This improvement is evident in panels (e) of Figures D4–D14, which present results using our preferred truncation—60 percent of the data. In these panels, the density provides a substantially better fit. We adopt this specification as our baseline because it is the one most likely to satisfy the identifying assumptions.¹¹

In Figure 4, we focus on three illustrative cases—Canada, Slovakia, and South Africa to highlight differences in how the estimated ETI varies with the degree of truncation. These countries were selected because the shape of the elasticity curve with respect to sample inclusion exhibits distinct patterns. In Canada, the estimated elasticity rises steeply when using between 10 and 30 percent of the data, after which it stabilizes. In contrast, estimates for Slovakia decline gradually as more data are included, while those for South Africa remain relatively flat until a sharp increase occurs at the 90th and 100th percentiles. Across all three cases, the interior portion of the distribution yields the most stable estimates, suggesting that moderate truncation delivers more robust inference than relying on either tail of the data.

¹¹Panel (a) of Figures D4–D14 displays the raw data for each country. In each case, we observe significant bunching at zero profits—the point at which the marginal corporate tax rate discontinuously increases. For example, the tax rate jumps from 0 percent to 25 percent in China, from 0 percent to 28 percent in Norway, and from 0 percent to 23 percent in Slovakia.

Figure 4: Corporate Elasticity of Taxable Income: cross-country comparison of elasticity estimates across the fraction of data used.

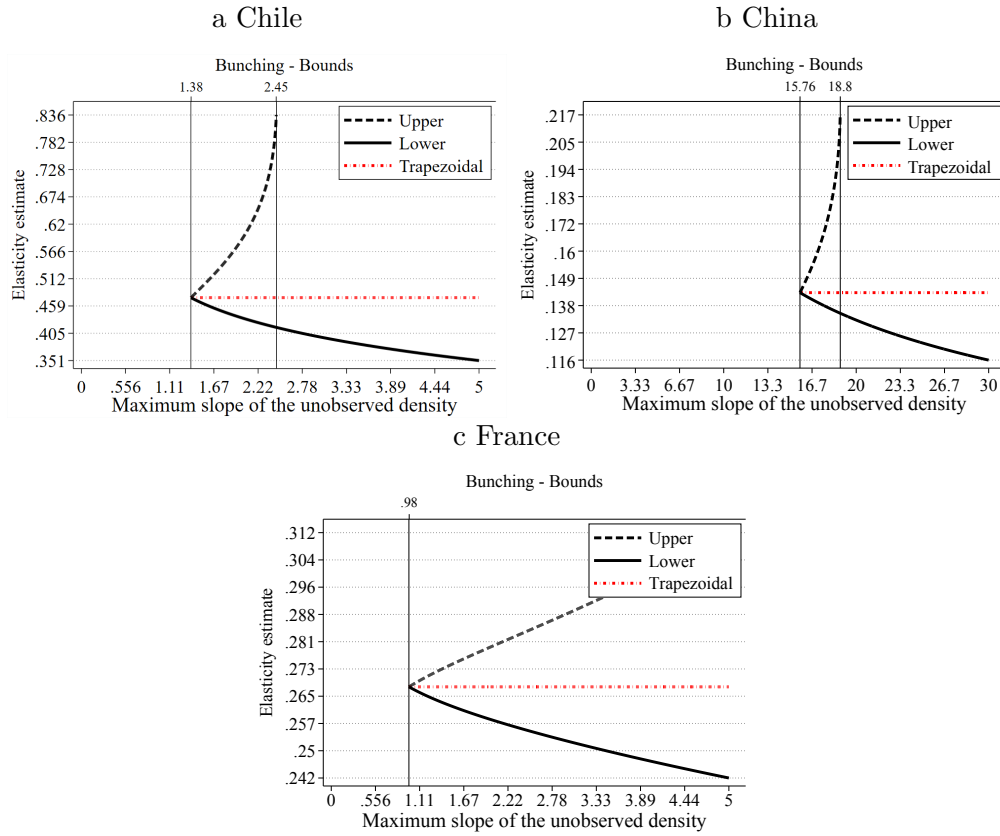


Notes: This figure plots the variation in the estimated elasticity of taxable income based on the amount of data used in the estimation. This graph is reported for all countries in the appendix.

4.3 Nonparametric bounds

As an additional robustness check, we construct nonparametric bounds on the elasticity estimates. These bounds shed light on the identifying power of the slope restrictions imposed on the counterfactual distribution. Figure 5 presents results for three representative countries: Chile, China, and France. The horizontal axis denotes the maximum slope permitted for the counterfactual distribution. The dashed line plots the resulting upper bound on the elasticity, while the solid line shows the corresponding lower bound. For reference, the dashed red line indicates the estimate obtained under the trapezoidal assumption proposed by Saez (2010). As the maximum slope increases, the bounds widen, reflecting the increasing flexibility of the counterfactual distribution. This divergence is particularly pronounced in Chile,

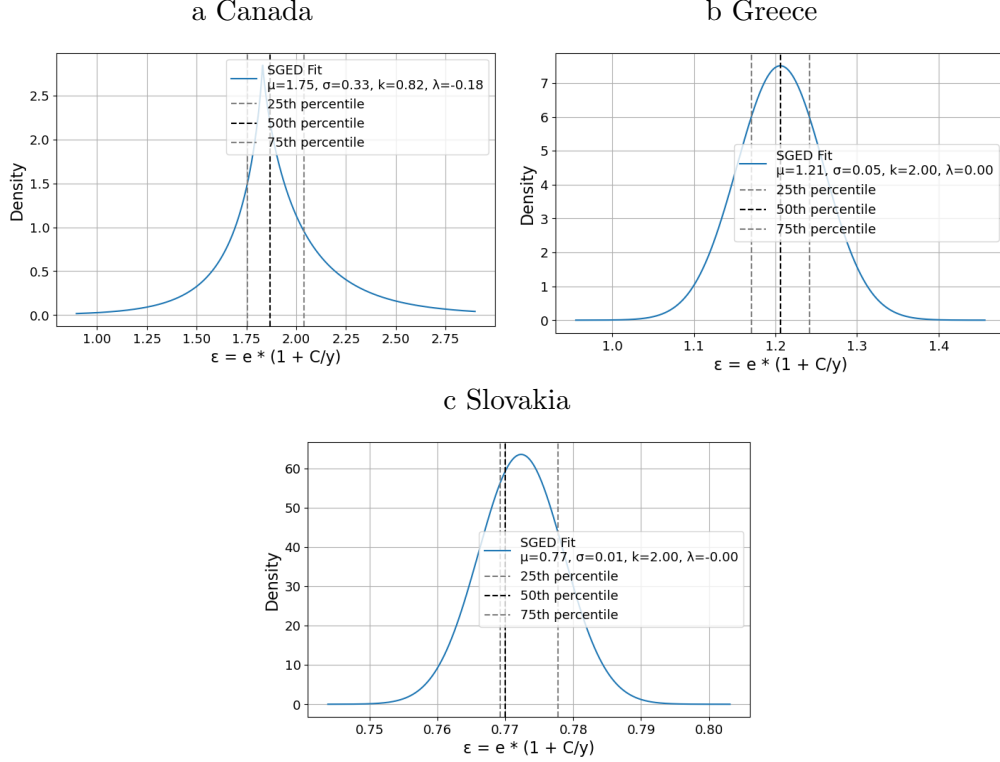
Figure 5: Corporate Elasticity of Taxable Income: cross-country comparison of bunching bounds.



Notes: This figure plots diagnostic tools for the traditional bunching estimate. Specifically, each figure plots the resulting upper and lower bounds on e using the nonparametric bounds method. This graph is reported for all countries in the appendix.

where the bounds expand rapidly, suggesting greater sensitivity of the elasticity estimate to the identifying assumptions. In contrast, the bounds in China and France remain relatively tight, indicating more robust identification. We report the full set of nonparametric bounds for each country in panel (b) of Figures D4–D14 in Appendix B.

Figure 6: Distribution of elasticities



Notes: This figure plots the distribution of elasticities for Canada, Greece, and Slovakia. The estimation procedure is outlined in Appendix B.3. This graph is reported for all available countries in the appendix.

4.4 Distribution of ETIs

We then use the estimates of e_k to estimate the *distribution* of taxable income elasticities across firms within each country in our sample. The model emphasizes that firm-level elasticities vary systematically with production amenities and fixed costs, implying that firms with different cost structures respond differentially to changes in tax rates. We illustrate this heterogeneity in two steps. First, Table B5 reports the distribution of the fixed-cost-to-output ratio, $\frac{C_{i,k}}{Y_{i,k}}$, across all countries in our sample. This distribution enables conversion from e_k to $\epsilon_{i,k}$. Second, Figure 6 plots the resulting distribution of elasticities for three representative countries—Canada, Greece, and Slovakia, based on the empirical distribution of $\frac{C_{i,k}}{Y_{i,k}}$.¹²

The estimated distributions exhibit substantial cross-country heterogeneity. Canada's distribution displays a pronounced mode, suggesting strong central tendency, whereas the

¹²Appendix B.3 provides details on the estimation of the elasticity distributions.

distributions in Greece and Slovakia are smoother and more dispersed. The shape of the distribution also varies: Greece exhibits an approximately symmetric distribution, with evenly spaced quartiles, while the distributions in Canada and Slovakia are right-skewed, indicating a longer upper tail of high-elasticity firms. Finally, the interquartile range differs meaningfully across countries. In particular, the elasticity distribution in Slovakia is notably more compressed than in Greece and Canada, suggesting a lower degree of within-country heterogeneity.

4.5 Why are elasticities different across countries?

We begin by documenting simple correlations between our elasticity estimates and key country-level characteristics to shed light on the cross-country variation in estimated elasticities. Figure 7 plots our elasticity estimates (hollow blue diamonds) against a range of country attributes, with a fitted line superimposed. In each panel, the elasticity is on the vertical axis, and the country characteristic is on the horizontal axis.

We find that our elasticity estimates positively correlate with the log GDP per capita and the share of firms formally registered and negatively correlate with the logarithm of total tax revenues. As such, in our sample, elasticities are larger for larger, high-income, more formal economies but not necessarily those with high tax revenues.

Elasticities are also negatively correlated with statutory corporate tax rates and average tariffs, and positively correlated with the presence of loss carryforward provisions. This pattern suggests that while higher statutory tax rates are associated with lower elasticities, more generous loss offset provisions appear to increase them.¹³

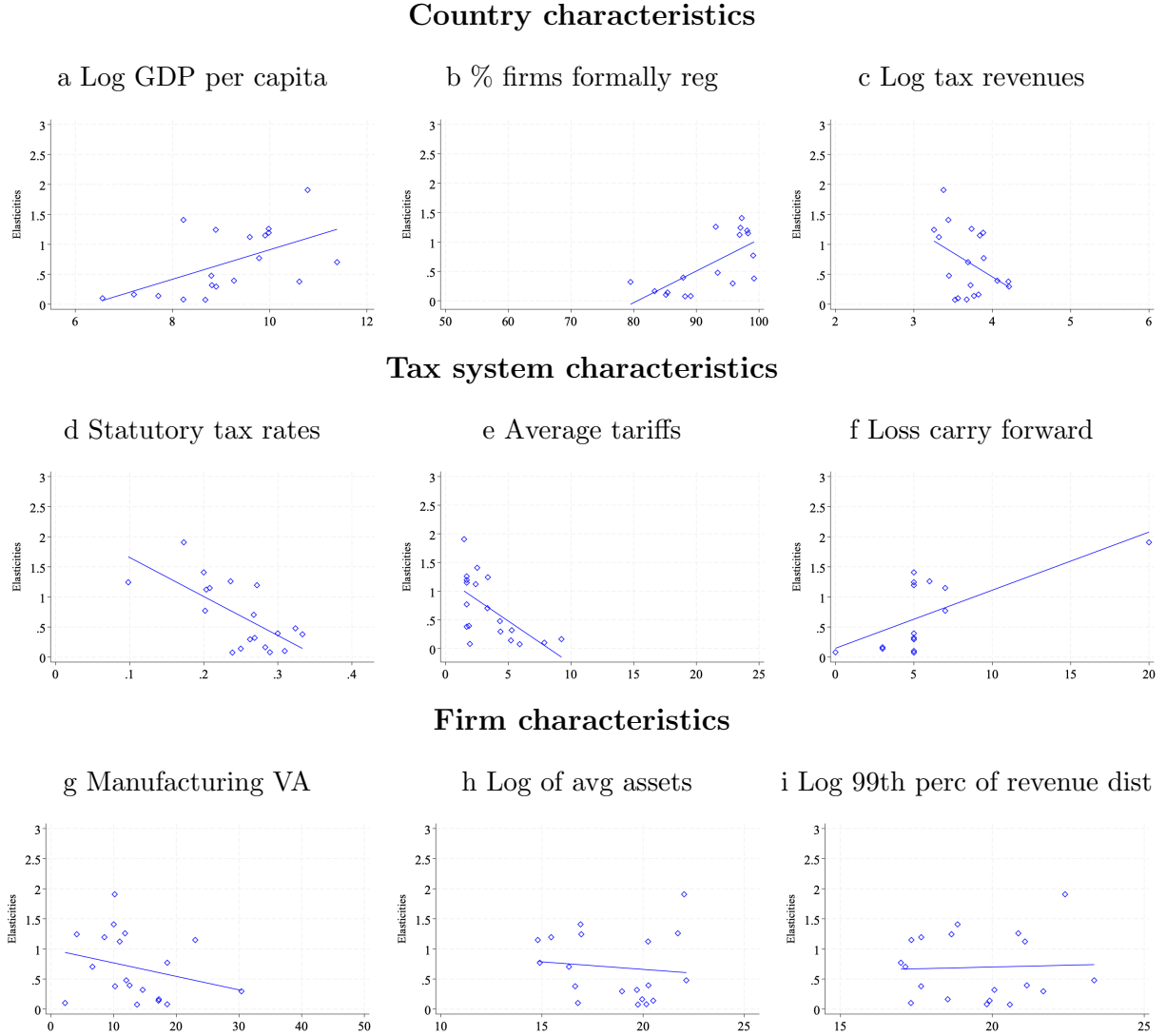
We also observe a negative correlation between elasticities and the share of manufacturing in GDP, and little correlation with the average size of firm assets or the 99th percentile of firm revenue (both in logs). This implies that countries with a larger manufacturing base tend to exhibit lower elasticities, whereas within-country firm size distributions do not meaningfully predict elasticity estimates.¹⁴

Taken together, these patterns suggest that macroeconomic fundamentals and tax system design are systematically associated with elasticity estimates, while firm-level characteristics such as firm size may play a more limited role.

¹³Panel g of Figure 7 excludes countries with unlimited provisions: Chile, France, Norway, and South Africa.

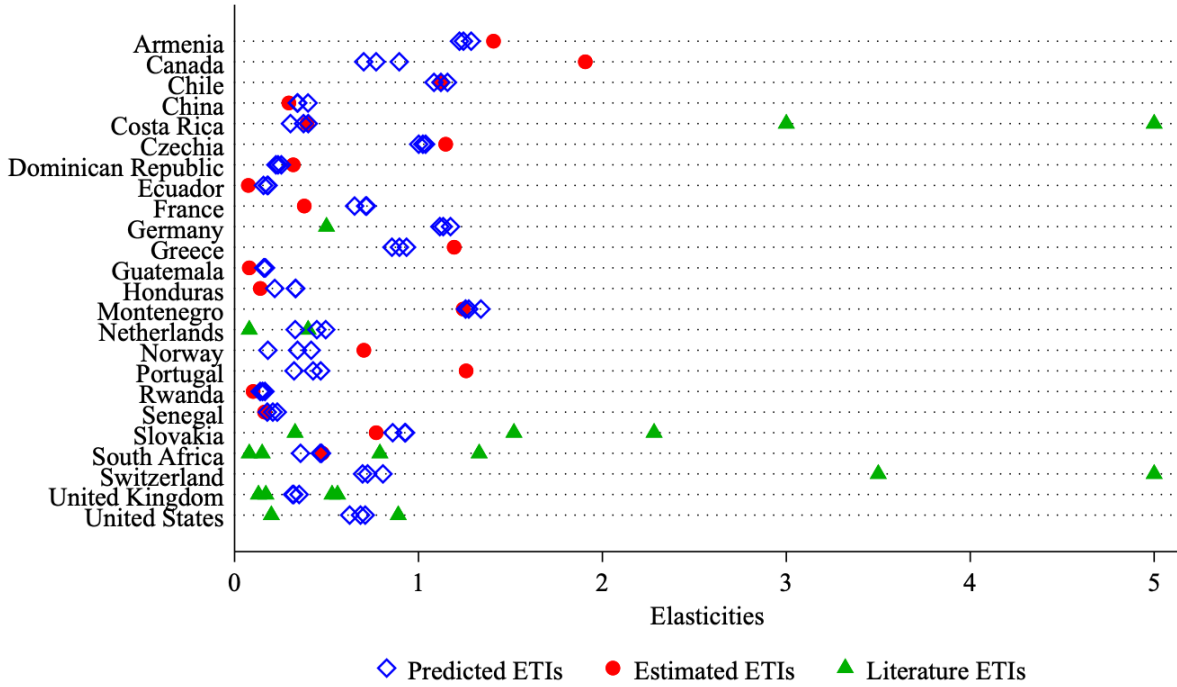
¹⁴Firm-level characteristics are based on Orbis data, which may have limited coverage in low-income countries.

Figure 7: Elasticity correlates across observables.



Notes: This figure plots the correlations between our estimated elasticities and observable country level characteristics. The blue line represents the linear estimation of best fit. Each correlate is listed in the figure title. Note that in panel i, we remove observations with unlimited loss carry forward, which are Chile, France, Norway and South Africa. We code them as 9999 in the dataset.

Figure 8: Predicted Elasticities



Notes: This figure plots the predicted ETIs using the top 3 selected combinations of variables using a Random Forest. These specifications include models with 3, 4 and 5 features. We plot the predicted ETIs in blue hollow diamonds, the estimated ETIs in filled red circles, and the previous literature ETIs in filled green triangles.

We next formalize this simple approach to understand which factors contribute the *most* to explaining the differences in ETIs across countries. To do so, we use a random forest prediction model, our estimated elasticities, and 95 country-level observable characteristics that we list in Table C1 in the Appendix. We organize the country-level observable characteristics into three broad categories (i) tax system features, (ii) firm-level characteristics, and (iii) country fundamentals such as size, population, GDP, exports, and foreign direct investment. We use data from the Center for Business Taxation, Orbis, and the World Bank, respectively.¹⁵ These sources provide broad cross-country coverage, enabling us in later sections to extrapolate elasticity predictions beyond our main sample. Nonetheless, some countries have missing data for select variables. To address this, we use a nearest neighbor imputation method following Troyanskaya et al. (2001).

The random forest prediction model selects three variables as the most predictive of ETI variation based on both mean-squared error and R-squared; see Figure C1. These characteristics are the logarithm of tax revenue, the statutory corporate tax rate, and manufacturing value-added as a share of GDP. We categorize these characteristics as country fundamentals, tax system design, and firm characteristics, respectively.

Figure 8 compares our predicted elasticities (blue diamonds) from the random forest model to both the estimates reported in the literature (green triangles) and our own empirical estimates (red circles). The close alignment between predicted and observed elasticities indicates that the model performs well in capturing cross-country variation. In addition to the baseline predictions from the three-factor model, we also report predictions from extended models that include a fourth and fifth factor—specifically, the share of formally registered firms and manufacturing imports as a percentage of GDP, respectively. Figure 8 includes 24 countries: for three, we observe all three types of estimates (Costa Rica, Slovakia, and South Africa); for five, only literature estimates and our predicted estimates are available (Germany, Netherlands, Switzerland, UK, US); and for the remaining countries we have estimates only from our paper and the predicted estimates.

We do not observe a systematic ranking among the predicted, estimated, and literature-based elasticities. For example, in Costa Rica, the elasticity estimate from the literature exceeds both our estimated and predicted values, likely reflecting differences in data sources, methodology, and sample composition. In contrast, for South Africa and Slovakia, our predicted and estimated elasticities lie between the literature estimates. Notably, our predicted elasticities closely approximate those from the literature in countries such as the United

¹⁵See: <https://oxfordtax.sbs.ox.ac.uk/cbt-tax-database>.

States and the United Kingdom, even though these countries are excluded from the model’s training sample. Overall, the predictive model generates elasticities that are broadly consistent with both our own estimates and those in the existing literature, reinforcing the model’s credibility and supporting its use in decomposing the drivers of elasticity variation across countries.

Table 5: **Percentage contribution of factors to deviations from average ETI**

Country	Tax system	Country characteristics	Firm characteristics
Armenia	0.36	0.32	0.31
Canada	0.34	0.40	0.26
Chile	0.35	0.42	0.23
China	0.29	0.16	0.55
Costa Rica	0.43	0.22	0.35
Czechia	0.00	0.45	0.55
Dominican Republic	0.26	0.39	0.35
Ecuador	0.34	0.36	0.30
France	0.33	0.23	0.44
Greece	0.07	0.23	0.69
Guatemala	0.25	0.36	0.39
Honduras	0.27	0.34	0.39
Montenegro	0.43	0.38	0.19
Norway	0.15	0.33	0.53
Rwanda	0.36	0.39	0.25
Senegal	0.22	0.36	0.42
Slovakia	0.65	0.04	0.31
South Africa	0.39	0.39	0.21
Uruguay	0.37	0.25	0.39
Median	0.34	0.36	0.35
Mean	0.31	0.32	0.37

Notes: This table quantifies the role of different factors in explaining deviations from average ETI across countries. We use the top 3 factors that explain the differences in ETIs across countries: the log of total tax revenue, the statutory corporate tax rate, and manufacturing sector value added as a share of GDP. We call these country characteristics, tax system, and firm characteristics, respectively. We quantify the contribution of each factor in percent terms using absolute values.

We next quantify the contribution of each of the three key factors identified by the random forest model to cross-country variation in ETIs. To do so, we apply the Shapley Additive Explanations (SHAP) framework, which decomposes each country’s predicted elasticity into marginal contributions from each factor by averaging over all possible orderings of inclusion.

A key advantage of the SHAP approach is that it captures nonlinearities and interactions among predictors. For each country, the predicted elasticity begins from a common baseline of 0.56, and the contribution of each factor may be positive or negative depending on the country’s characteristics. We present the full country-level decomposition in Figure C2 and report the corresponding level deviations in Table C2. For ease of interpretation, we summarize percentage contributions by factor in Table 5.

The results indicate that all three factors contribute meaningfully and relatively evenly to cross-country differences in estimated elasticities. While the average marginal effect of each factor is close to zero by construction, the average absolute contribution is largest for firm characteristics, which shift a country’s predicted elasticity by 0.15 from the baseline, followed by country fundamentals (0.14) and tax system characteristics (0.13). In proportional terms, firm characteristics account for 37% of the variation, with tax and country characteristics contributing 31% and 32%, respectively. Median contributions reflect a similar distribution. Taken together, these findings suggest that a single factor does not drive differences in ETIs but rather reflects the joint influence of firm-level characteristics, country fundamentals, and tax system design. While this analysis provides a first step toward unpacking the drivers of cross-country elasticity variation, it also highlights the need for further research to understand the underlying mechanisms.

4.6 Extending our estimates to other countries

We conclude the paper by leveraging the random forest model to predict ETIs for 224 countries. A central contribution of our analysis is the estimation of 19 country-specific elasticity estimates using administrative tax data and a unified empirical framework. This represents a substantial expansion relative to the prior literature, which reported estimates with different methods for only eight countries. Nevertheless, many countries still lack direct elasticity estimates.

To address this gap, we propose two strategies. First, researchers with access to administrative tax records may directly implement our empirical approach using our code to generate elasticity estimates for their respective countries. Second, in settings where administrative data are unavailable, researchers may rely on the predicted elasticities presented in Table A1. These predictions are derived from observable characteristics of countries, firms, and tax systems and demonstrate strong predictive performance, as documented in Appendix C.

We report estimates from models incorporating three, four, and five covariates. The resulting ETI estimates are highly correlated across specifications. The average predicted

elasticity is 0.66 in the three-variable model and 0.59 in the four- and five-variable models.

5 Conclusion

This paper provides us with a better understanding of the corporate elasticity of taxable income across the world. We generate the most comprehensive collection of comparable estimates across countries to date, including both developed and developing countries. The fact that all elasticities are estimated using the same methodology allows a meaningful comparison among them.

We find substantial heterogeneity in our ETI estimates across countries. These differences suggest there is scope for differences in tax regulation and enforcement to have a large effect on this elasticity. More specifically, we show that a large portion of the difference in elasticities is due to country-level fundamentals, tax systems, and firm characteristics that make it easier for firms to respond to changes in tax rates.

A second important result is that we find the differences across countries to be substantially smaller than those found and reported in the literature. Our estimates range between 0.05 and 1.9, while estimates in the literature range from 0 to 5. The substantially smaller range suggests that differences in methods across studies explain a large portion of the differences found across countries. This finding highlights the need to use methods with reasonable and explicit identifying assumptions suited for the context that can be broadly applied across countries for comparative purposes.

The predicted estimates provide a sandbox for future research. For example, we may have thought that if capital moves freely across countries, the elasticity would be the same everywhere. The question arises then to what extent does the variation across countries we observe indicate capital frictions, and the answer to this question may change how we interpret elasticity estimates. Further, is a low elasticity an indicator of a functional tax system or many frictions? Specifically, the high elasticity in Switzerland (both the one we predict and the one in the literature) might imply a porous tax system, but it could also be that capital is more elastic in Switzerland. In addition, important questions remain as to what constitutes a friction. For example, to what extent is uniqueness a friction, such as capital employed in Chile for copper mining? Similarly, to what extent do infrastructure and human capital that are complementary to capital create frictions and a lower elasticity? We hope the estimates in this paper provide a starting point for future research to investigate these and other important questions about capital and its responsiveness to tax policy.

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Table A1: Predicted ETIs across the world.

Beginning of Table				
	Country	3 predictors	4 predictors	5 predictors
		(1)	(2)	(3)
1	Afghanistan	0.73	0.54	0.55
2	Albania	1.01	0.89	0.88
3	Algeria	0.28	0.25	0.24
4	American Samoa	0.40	0.58	0.59
5	Andorra	1.07	0.82	0.80
6	Angola	0.69	0.61	0.60
7	Anguilla	1.03	0.64	0.65
8	Antarctica	0.54	0.58	0.57
9	Argentina	0.39	0.36	0.36
10	Armenia	1.24	1.07	1.07
11	Aruba	0.70	0.61	0.61
12	Australia	0.76	0.61	0.59
13	Austria	0.37	0.36	0.36
14	Azerbaijan	0.65	0.72	0.68
15	Bahamas	0.95	0.89	0.88
16	Bahrain	1.03	0.64	0.66
17	Bangladesh	0.18	0.22	0.23
18	Barbados	0.57	0.56	0.57
19	Belarus	0.34	0.41	0.44
20	Belgium	0.41	0.39	0.39
21	Belize	0.69	0.61	0.62
22	Benin	0.40	0.31	0.33
23	Bermuda	1.39	0.96	0.96
24	Bhutan	0.57	0.72	0.72
25	Bolivia	0.62	0.56	0.55
26	Bosnia and Herzegovina	1.33	0.88	0.89
27	Botswana	0.87	0.83	0.80
28	Brazil	0.54	0.67	0.66
29	Bulgaria	1.09	0.73	0.76

Continuation of Table A1				
	Country	3 predictors	4 predictors	5 predictors
30	Burkina Faso	0.59	0.58	0.58
31	Burundi	0.65	0.61	0.62
32	Cabo Verde	0.56	0.56	0.57
33	Cambodia	0.93	0.64	0.64
34	Cameroon	0.47	0.28	0.30
35	Canada	1.53	1.01	1.00
36	Cayman Islands	1.05	0.82	0.82
37	Central African Republic	0.35	0.27	0.26
38	Chad	0.50	0.61	0.62
39	Chile	1.17	1.02	1.02
40	China	0.30	0.33	0.36
41	Colombia	0.39	0.38	0.37
42	Comoros	0.57	0.61	0.62
43	Congo	0.58	0.61	0.59
44	Congo, DR	0.36	0.27	0.29
45	Cook Islands	0.66	0.47	0.47
46	Costa Rica	0.40	0.31	0.30
47	Croatia	0.97	0.74	0.75
48	Cuba	0.29	0.32	0.35
49	Curaçao	0.64	0.56	0.57
50	Cyprus	1.30	1.04	1.03
51	Czechia	0.22	0.38	0.38
52	Cote d'Ivoire	0.43	0.28	0.29
53	Denmark	0.61	0.44	0.44
54	Djibouti	0.52	0.70	0.70
55	Dominica	0.48	0.71	0.72
56	Dominican Republic	0.27	0.23	0.25
57	Ecuador	0.16	0.24	0.24
58	Egypt	0.14	0.28	0.31
59	El Salvador	0.20	0.22	0.23
60	Equatorial Guinea	0.38	0.27	0.29
61	Eritrea	0.60	0.75	0.75

Continuation of Table A1				
	Country	3 predictors	4 predictors	5 predictors
62	Estonia	0.44	0.43	0.43
63	Eswatini	0.15	0.30	0.32
64	Ethiopia	0.50	0.70	0.68
65	Faroe Islands	1.01	0.80	0.80
66	Fiji	0.56	0.60	0.62
67	Finland	0.16	0.31	0.34
68	France	0.57	0.71	0.69
69	French Guiana	0.58	0.58	0.57
70	French Polynesia	0.27	0.23	0.25
71	Gabon	0.24	0.23	0.23
72	Gambia	0.57	0.61	0.62
73	Georgia	1.33	1.07	1.05
74	Germany	0.75	0.58	0.59
75	Ghana	0.67	0.58	0.56
76	Gibraltar	1.17	1.04	1.03
77	Greece	0.92	0.74	0.75
78	Greenland	0.48	0.58	0.59
79	Grenada	0.45	0.58	0.59
80	Guadeloupe	0.58	0.58	0.57
81	Guam	0.49	0.58	0.59
82	Guatemala	0.14	0.23	0.23
83	Guernsey	1.06	0.89	0.87
84	Guinea	0.54	0.70	0.73
85	Guinea-Bissau	0.51	0.48	0.51
86	Guyana	0.46	0.58	0.59
87	Haiti	0.50	0.58	0.59
88	Holy See	0.54	0.58	0.57
89	Honduras	0.14	0.20	0.24
90	Hong Kong	1.28	1.04	1.01
91	Hungary	0.78	0.60	0.60
92	Iceland	1.36	0.90	0.90
93	India	0.41	0.28	0.30

Continuation of Table A1				
	Country	3 predictors	4 predictors	5 predictors
94	Indonesia	0.52	0.39	0.40
95	Iran	0.22	0.25	0.24
96	Iraq	1.45	0.96	0.94
97	Ireland	1.03	0.63	0.64
98	Isle of Man	0.94	0.80	0.80
99	Israel	0.72	0.47	0.48
100	Italy	0.36	0.36	0.37
101	Jamaica	0.60	0.61	0.61
102	Japan	0.43	0.34	0.38
103	Jersey	0.75	0.52	0.51
104	Jordan	1.08	0.69	0.72
105	Kazakhstan	0.56	0.67	0.66
106	Kenya	0.57	0.68	0.68
107	Kiribati	0.61	0.67	0.67
108	Korea, Republic of	0.13	0.24	0.27
109	Kuwait	0.86	0.72	0.71
110	Kyrgyzstan	0.68	0.54	0.55
111	Lao PDR	1.03	0.74	0.72
112	Latvia	1.06	0.88	0.89
113	Lebanon	1.24	0.87	0.87
114	Lesotho	0.61	0.40	0.43
115	Liberia	0.39	0.58	0.58
116	Libya	0.58	0.61	0.59
117	Liechtenstein	1.45	0.96	0.96
118	Lithuania	0.65	0.57	0.60
119	Luxembourg	0.92	0.86	0.84
120	Macao	0.95	0.80	0.78
121	Madagascar	0.66	0.58	0.59
122	Malawi	0.71	0.65	0.62
123	Malaysia	0.16	0.20	0.21
124	Maldives	1.27	0.96	0.96
125	Mali	0.81	0.61	0.62

Continuation of Table A1				
	Country	3 predictors	4 predictors	5 predictors
126	Malta	0.58	0.68	0.68
127	Marshall Islands	0.98	0.82	0.82
128	Martinique	0.58	0.58	0.57
129	Mauritania	0.72	0.59	0.60
130	Mauritius	1.06	0.65	0.66
131	Mayotte	0.54	0.58	0.57
132	Mexico	0.42	0.27	0.26
133	Moldova, Republic of	1.04	0.86	0.86
134	Monaco	0.62	0.62	0.62
135	Mongolia	1.04	0.81	0.80
136	Montenegro	1.29	1.07	1.07
137	Montserrat	0.44	0.27	0.29
138	Morocco	0.46	0.36	0.39
139	Mozambique	0.55	0.61	0.61
140	Myanmar	0.16	0.22	0.22
141	Namibia	0.86	0.68	0.68
142	Nauru	0.67	0.48	0.51
143	Nepal	0.69	0.65	0.66
144	Netherlands	0.64	0.62	0.62
145	Netherlands Antilles	0.50	0.53	0.55
146	New Caledonia	0.52	0.54	0.56
147	New Zealand	0.51	0.54	0.53
148	Nicaragua	0.37	0.28	0.27
149	Niger	0.51	0.58	0.57
150	Nigeria	0.55	0.58	0.59
151	Niue	0.55	0.58	0.57
152	Norfolk Island	0.54	0.58	0.57
153	North Macedonia	1.35	1.00	1.02
154	Norway	0.61	0.58	0.56
155	Oman	1.03	0.64	0.64
156	Pakistan	0.35	0.29	0.31
157	Palau	0.95	0.83	0.83

Continuation of Table A1				
	Country	3 predictors	4 predictors	5 predictors
158	Palestine, State of	0.75	0.52	0.53
159	Panama	0.60	0.71	0.68
160	Papua New Guinea	0.40	0.68	0.65
161	Paraguay	0.67	0.54	0.54
162	Peru	0.25	0.27	0.26
163	Philippines	0.21	0.31	0.32
164	Pitcairn	0.54	0.58	0.57
165	Poland	0.62	0.57	0.58
166	Portugal	0.27	0.38	0.38
167	Puerto Rico	0.33	0.26	0.29
168	Qatar	1.23	0.97	0.96
169	Romania	0.65	0.57	0.58
170	Russian Federation	0.45	0.43	0.42
171	Rwanda	0.35	0.58	0.55
172	Saint Kitts and Nevis	0.72	0.61	0.58
173	Saint Lucia	0.39	0.72	0.72
174	Samoa	0.92	0.73	0.74
175	San Marino	1.03	0.80	0.80
176	Sao Tome and Principe	0.57	0.57	0.58
177	Saudi Arabia	0.90	0.77	0.77
178	Senegal	0.20	0.22	0.25
179	Serbia	0.78	0.59	0.62
180	Seychelles	0.49	0.58	0.59
181	Sierra Leone	0.50	0.61	0.62
182	Singapore	1.02	0.73	0.73
183	Slovakia	0.75	0.60	0.60
184	Slovenia	0.67	0.57	0.58
185	Solomon Islands	0.89	0.74	0.75
186	Somalia	0.71	0.64	0.65
187	South Africa	0.50	0.46	0.46
188	South Sudan	0.93	0.74	0.72
189	Spain	0.67	0.69	0.69

Continuation of Table A1				
	Country	3 predictors	4 predictors	5 predictors
190	Sri Lanka	0.34	0.27	0.26
191	Sudan	0.49	0.72	0.70
192	Suriname	0.64	0.46	0.49
193	Sweden	0.37	0.35	0.34
194	Switzerland	1.19	0.69	0.70
195	Syrian Arab Republic	0.29	0.37	0.40
196	Tajikistan	1.06	0.91	0.88
197	Tanzania	0.62	0.58	0.59
198	Thailand	0.20	0.23	0.23
199	Timor-Leste	1.28	1.04	1.03
200	Togo	0.72	0.61	0.60
201	Tonga	0.86	0.81	0.82
202	Trinidad and Tobago	0.18	0.21	0.24
203	Tunisia	0.37	0.35	0.36
204	Turkey	0.23	0.34	0.35
205	Turkmenistan	0.75	0.49	0.50
206	Turks and Caicos Islands	1.27	0.96	0.96
207	Tuvalu	0.54	0.60	0.61
208	Uganda	0.55	0.58	0.57
209	Ukraine	0.53	0.46	0.47
210	United Arab Emirates	1.32	1.04	1.02
211	United Kingdom	0.66	0.67	0.64
212	United States of America	0.40	0.53	0.52
213	Uruguay	0.38	0.37	0.37
214	Uzbekistan	0.78	0.63	0.62
215	Vanuatu	1.29	0.96	0.96
216	Venezuela	0.44	0.47	0.45
217	Viet Nam	0.17	0.32	0.31
218	Virgin Islands (British)	1.30	0.96	0.96
219	Virgin Islands (U.S.)	0.49	0.58	0.59
220	Wallis and Futuna	0.54	0.58	0.57
221	Western Sahara	0.54	0.58	0.57

Continuation of Table A1				
	Country	3 predictors	4 predictors	5 predictors
222	Yemen	0.82	0.58	0.59
223	Zambia	0.83	0.74	0.72
224	Zimbabwe	0.54	0.61	0.64
	Average	0.66	0.59	0.59
	Median	0.58	0.58	0.59
	Max	1.53	1.07	1.07
	Min	0.13	0.20	0.21
	Correlation		0.89	
				0.99
End of Table				

ONLINE APPENDIX—NOT FOR PUBLICATION, October 20, 2025
“The Elasticity of Taxable Income Across Countries”

A	Real and reporting responses	2
A.1	Model Fundamentals	2
A.2	Model Solution	3
A.3	Elasticities	4
B	Additional results	5
B.1	Additional descriptive statistics	5
B.1.1	Transition probabilities	5
B.1.2	Comparison of firm revenues across taxable income distribution . . .	6
B.2	Additional empirical model details	8
B.2.1	Fixed costs estimation methods	8
B.3	Methodology for estimating the distribution of elasticities	10
C	Prediction appendix	11
D	Detailed Data Description	18
D.1	Armenia	18
D.2	Canada	20
D.3	Chile	23
D.4	China	26
D.5	Costa Rica	28
D.6	Czechia	30
D.7	Dominican Republic	32
D.8	Ecuador	34
D.9	France	36
D.10	Greece	38
D.11	Montenegro	40
D.12	Norway	42
D.13	Portugal	44
D.14	Slovakia	46
D.15	South Africa	48

A Real and reporting responses

In this appendix, we present a model with the addition of reporting responses. The key takeaway is that our equilibrium and methods for estimating the model primitive e are robust to this addition. We also show how to update the calculation of the elasticity of taxable income with reporting responses.

A.1 Model Fundamentals

Consider a firm, denoted Firm i , that is owned by a single shareholder and begins period 1 with retained earnings, X_i . Firms are heterogeneous in their productivity, captured by A_i , and their fixed costs, captured by C_i . In period 1, Firm i chooses its level of capital in period 2, K_i , and the amount of taxable income that will avoid taxes, ρ_i . Firms choose their level of capital in period 2 by determining the amount of retained earnings to distribute as a dividend payment ($D_i \geq 0$), and the amount of equity to issue ($E_i \geq 0$); $K_i = X_i + E_i - D_i$.¹⁶ In addition to equity, shareholders may hold government bonds with a tax-exempt rate of return, $r > 0$. The cost of avoiding taxes, including the probability of audit and cost of penalty, is given by $c(\rho_i)$, which is increasing and convex.

In period 2, capital generates income net-of-depreciation costs according to a strictly concave production function

$$I_i(K_i) = \frac{1+e}{e} A_i^{1/(1+e)} K_i^{\frac{e}{1+e}}. \quad (\text{A.1})$$

Here, e determines the curvature of the production function. More importantly, e determines the elasticity of income with respect to the net-of-tax rate, which is an input to the elasticity of taxable income with respect to net-of-tax rate. Finally, profit is income net of fixed costs. Modeling fixed costs is important to match the data since a large proportion of firms report negative profits.

$$F_i(K_i) = I_i(K_i) - C_i. \quad (\text{A.2})$$

Taxable income is defined as profit net of the amount of income that the firm decides to avoid or evade ρ_i : $Y_i(K_i) = F_i(K_i) - \rho_i$.

¹⁶For ease of exposition, attention is restricted to equilibria where the firm does not payout a dividend and issue equity concurrently. In the general model in Patel et al. (2014) the restriction that a firm does not pay out a dividend and issue equity concurrently is derived as equilibrium behavior with a dividend tax. The restriction does not change the following analysis.

At the end of period 2, all firms liquidate, returning their principal and profits to their shareholders.

Firm i maximizes its value to its shareholder:

$$\max_{K_i, \rho_i} V = X_i - K_i + \frac{(1 - t_c)Y_i(K_i, \rho_i) + \rho_i + K_i - c(\rho_i)}{1 + r}, \quad (\text{A.3})$$

where $X_i - K_i = D_i - E_i$ are net distributions in period 1 valued by its shareholder.

The benefit of higher capital in period 2 is higher profit. Profit is taxed at the rate t_c and discounted at the rate r .¹⁷ The cost of higher capital in period 2 is lower distributions in period 1 (fewer dividends or more equity issuances).

Consider the case where there is a kink in the marginal tax rate schedule such that $t_c = t_0$ for $Y_i(K_i, \rho_i) \leq \kappa$ and $t_c = t_1$ for $Y_i(K_i, \rho_i) > \kappa$, where $t_0 < t_1$. Under this marginal rate schedule, the objective function faced by the firm is

$$\begin{aligned} \max_{K_i} V_i = & X_i - \frac{1}{1 + r}(rK_i - c(\rho_i)) \\ & + \mathbb{I}(Y_i(K_i, \rho_i) \leq \kappa) \frac{(1 - t_0)Y_i(K_i, \rho_i) + \rho_i}{1 + r} \\ & + \mathbb{I}(Y_i(K_i, \rho_i) > \kappa) \frac{(1 - t_0)\kappa + (1 - t_1)(Y_i(K_i, \rho_i) - \kappa) + \rho_i}{1 + r}, \end{aligned} \quad (\text{A.4})$$

where $\mathbb{I}(Y_i(K_i, \rho_i) \leq \kappa)$ and $\mathbb{I}(Y_i(K_i, \rho_i) > \kappa)$ are indicator functions for taxable income being below or above the kink.

A.2 Model Solution

Formally, we derive the equilibrium capital, tax avoidance, and taxable income by taking the derivative of firm value with respect to capital and avoidance in two regions where the derivative exists.

$$\frac{\partial V_i}{\partial K_i} = \begin{cases} \frac{1}{1+r} \left(-r + (1 - t_0) \frac{\partial Y_i(K_i, \rho_i)}{\partial K_i} \right), & Y_i(K_i, \rho_i) < \kappa \\ \frac{1}{1+r} \left(-r + (1 - t_1) \frac{\partial Y_i(K_i, \rho_i)}{\partial K_i} \right), & Y_i(K_i, \rho_i) > \kappa. \end{cases} \quad (\text{A.5})$$

¹⁷The equilibrium rate of return r is assumed to be exogenous, abstracting from all general equilibrium effects.

$$\frac{\partial V_i}{\partial \rho_i} = \begin{cases} \frac{1}{1+r} (-c'(\rho_i) + t_0), & Y_i(K_i, \rho_i) < \kappa \\ \frac{1}{1+r} (-c'(\rho_i) + t_1), & Y_i(K_i, \rho_i) > \kappa. \end{cases} \quad (\text{A.6})$$

The solution for taxable income $Y_i(K_i)$ has a similar form to solutions derived in different contexts in this literature (Bertanha et al., 2023; Coles et al., 2022; Saez, 2010):

$$Y_i(K_i) = \begin{cases} \frac{1+e}{e} r^{-e} (1 - t_0)^e A_i - C_i - \rho_i, & A_i \leq \underline{A}_i \\ \kappa, & \underline{A}_i < A_i < \bar{A}_i \\ \frac{1+e}{e} r^{-e} (1 - t_1)^e A_i - C_i - \rho_i, & A_i \geq \bar{A}_i. \end{cases} \quad (\text{A.7})$$

The thresholds are found by setting the optimal taxable income equal to the kink κ with both tax rates;

$$\underline{A}_i = (\kappa + C_i + \rho_i)/\theta_0, \quad \text{and} \quad \bar{A}_i = (\kappa + C_i + \rho_i)/\theta_1. \quad (\text{A.8})$$

A.3 Elasticities

Here, we update the calculation of the elasticity of taxable income to account for both real and reporting responses. Critically, we note that the elasticity of income that we estimate remains a key piece to this calculation.

$$\begin{aligned} \varepsilon &= \frac{\partial Y_i(K_i, \rho_i)}{\partial (1-t)} \frac{(1-t)}{Y_i(K_i, \rho_i)} \\ &= \left(\frac{\partial F_i(K_i)}{\partial (1-t)} - \frac{\partial \rho_i}{\partial (1-t)} \right) \frac{(1-t)}{Y_i(K_i, \rho_i)} \end{aligned} \quad (\text{A.9})$$

$$= \frac{1}{Y_i(K_i, \rho_i)} (eF(K_i) + e_\rho \rho_i) \quad (\text{A.10})$$

Note to use the elasticity of taxable income for welfare analysis, it is also important to consider whether the reporting response (or to what extent the reporting response) is a resource cost or a transfer. For a more detailed discussion of the reporting response and its implications for welfare see Coles et al. (2022).

B Additional results

B.1 Additional descriptive statistics

B.1.1 Transition probabilities

Table B1: Transition probabilities across bins of income distribution.

Armenia				
	bin -1	bin 0	bin 1	bin 2
bin -1	84.82	6.19	1.40	7.60
bin 0	33.14	54.44	3.63	8.80
bin 1	40.59	16.99	22.63	19.78
bin 2	23.63	4.76	1.92	69.69
Canada				
	bin -1	bin 0	bin 1	bin 2
bin -1	0.70	0.01	0.20	0.09
bin 0	0.21	0.53	0.17	0.09
bin 1	0.18	0.01	0.75	0.07
bin 2	0.13	0.01	0.11	0.75
Costa Rica				
	bin -1	bin 0	bin 1	bin 2
bin -1	0.79	0.14		0.07
bin 0	0.16	0.81		0.02
bin 2	0.21	0.11		0.68
Czechia				
	bin -1	bin 0	bin 1	bin 2
bin -1	75.36	3.66	11.34	9.63
bin 0	7.99	86.35	4.72	0.93
bin 1	32.22	7.93	50.54	9.32
bin 2	24.57	1.02	9.33	65.08

Ecuador				
	bin -1	bin 0	bin 1	bin 2
bin -1	0.76	0.06	0.09	0.09
bin 0	0.17	0.73	0.07	0.03
bin 1	0.38	0.17	0.36	0.09
bin 2	0.26	0.04	0.07	0.63
Greece				
	bin -1	bin 0	bin 1	bin 2
bin -1	76.48	0.17	16.04	7.32
bin 0	14.51	55.78	22.58	7.13
bin 1	14.49	0.35	78.21	6.95
bin 2	13.90	0.16	8.93	77.01
Norway				
	bin -1	bin 0	bin 1	bin 2
bin -1	69.03	3.69	16.97	10.31
bin 0	20.91	43.14	25.93	10.03
bin 1	15.25	20.39	56.84	7.52
bin 2	14.92	14.14	10.87	60.07
Portugal				
	bin -1	bin 0	bin 1	bin 2
bin -1	72.73	0.74	14.26	12.27
bin 0	3.51	86.55	8.44	1.50
bin 1	28.56	4.67	55.28	11.49
bin 2	28.72	1.07	15.98	54.23
Slovakia				
	bin -1	bin 0	bin 1	bin 2
bin -1	74.28	0.08	17.62	8.02
bin 0	42.67	12.89	27.11	17.33
bin 1	37.48	0.26	53.34	8.91
bin 2	20.85	0.11	11.23	67.80

Notes: This table shows the set of transition probabilities for firms to move between 4 bins of taxable profits, bin -1: negative profits, bin 0: zero taxable profits, bin 1: Q1-Q4 of the distribution of non-zero taxable profits and bin 2: Q5 of the distribution of non-zero taxable profits. Note that for countries which only have 1 year of data, we cannot produce the transition matrix. These are Chile, China, Czechia.

B.1.2 Comparison of firm revenues across taxable income distribution

Table B2: Average firm revenues and variable costs for zero and non-zero profit reporters.

	(1) mean at zero taxable profits	(2) mean away from zero taxable profits	(3) difference	t-stat
Armenia revenues	58155746.886	1.481e+08	-8.992e+07***	-24.381
Armenia var costs	56163823.722	1.355e+08	-7.932e+07***	-21.483
Armenia obs	8283	59363	67646	
Canada revenues	54759171.06	23792000.49	30967170.569***	4.638
Canada var costs	49404845.23	22566981.3	26837863.931***	4.579
Canada obs	945	30000	30945	
Chile revenues	2.391e+08	8.497e+08	-6.106e+08***	-17.503
Chile var costs	19506086.357	1.343e+08	-1.148e+08***	-24.080
Chile obs	2803	9076	11879	
Costa Rica revenues	22964883.061	82959176.909	-5.999e+07***	-164.222
Costa Rica var costs	27663001.104	80938353.886	-5.328e+07***	-146.894
Costa Rica obs	552210	885849	1438059	
Czechia revenues	3302921.296	7327190.686	-4024269.390***	-54.677
Czechia var costs	111279.100	459884.634	-348605.534***	-37.263
Czechia obs	104884	1480035	1584919	
Dominican Republic revenues	2690981.635	6764691.108	-4073709.473***	-37.012
Dominican Republic var costs	2380720.395	5944736.106	-3564015.711***	-36.506
Dominican Republic obs	82973	127459	210432	
Ecuador revenues	6780.104	413859.878	-407079.774***	-159.133
Ecuador var costs	5927.222	321639.911	-315712.690***	-139.959
Ecuador obs	99938	220680	320618	
Greece revenues	961067.257	1036162.997	-75095.740	-0.949
Greece var costs	962286.968	1024718.976	-62432.008	-0.789
Greece obs	411	50573	50984	
Montenegro revenues	58045.597	113577.022	-55531.426***	-29.289
Montenegro var costs	84128.178	115774.549	-31646.371***	-13.986
Montenegro obs	22251	120768	143019	
Norway revenues	11170799.289	4874725.996	6296073.293***	80.788
Norway var costs	9775212.073	4371239.255	5403972.817***	76.987
Norway obs	209041	940519	1149560	
Portugal revenues	1309.091	165084.282	-163775.191***	-479.003
Portugal var costs	1195.747	151425.441	-150229.694***	-467.398
Portugal obs	108865	1033129	1141994	
Slovakia revenues	720813.999	383557.586	337256.413*	1.885
Slovakia var costs	33714.470	17139.223	16575.246**	2.132
Slovakia obs	132	65125	65257	

Notes: This table presents the average of revenues for firms that locate at zero taxable profits (column 1) and those that do not (column 2). We define firms that do not locate at zero taxable profits as those in the first 4 quantiles of the distribution of firm revenues (omitting Q5). We present descriptives across all 5 quantiles of firm revenues in Table B3 in the Online Appendix.

Table B3: Summary of firm revenues at different points of taxable income distribution.

	(1) zero Y_i	(2) Q1	(3) Q2	(4) Q3	(5) Q4	(6) Q5
Armenia revenues	58.16	215.90	51.36	101.30	197.30	1,248.00
Armenia obs	8,283	14,842	14,840	14,841	14,840	14,826
Canada revenues	54.76	106.00	9.07	14.63	23.69	275.00
Canada obs	945	7,500	7,500	7,500	7,500	7,500
Chile revenues	239.10	2,929.00	170.90	526.40	1,260.00	6,088.00
Chile obs	2,803	2,269	2,269	2,269	2,269	2,269
Costa Rica revenues	22.96	61.24	39.67	63.56	105.90	561.00
Costa Rica obs	552,210	221,463	221,462	221,463	221,462	221,462
Czechia revenues	3.30	11.58	2.05	3.81	9.38	70.35
Czechia obs	104,884	381,168	284,614	403,001	406,656	410,764
Dominican Republic revenues	2.69	14.99	1.27	2.61	8.99	69.70
Dominican Republic obs	82,973	31,865	31,865	31,865	31,865	31,864
Ecuador revenues	0.01	0.59	0.06	0.21	0.56	4.04
Ecuador obs	99,938	55,171	55,170	55,170	55,170	55,170
Greece revenues	0.96	1.10	0.42	0.75	1.48	2.88
Greece obs	411	12,644	12,643	12,643	12,643	12,643
Montenegro revenues	0.06	0.15	0.03	0.07	0.16	0.70
Montenegro obs	22,251	30,192	30,196	30,197	30,183	30,191
Norway revenues	11.17	10.57	1.46	5.38	5.44	27.33
Norway obs	209,041	235,002	235,127	235,051	235,165	234,870
Portugal revenues	1.31	127.49	42.60	128.81	244.40	556.79
Portugal obs	108,865	258,284	258,282	258,283	258,281	258,282
Slovakia revenues	0.72	0.78	0.14	0.24	0.50	2.88
Slovakia obs	132	16,282	16,281	16,281	16,281	16,281

Notes: This table presents the average of revenues for firms across 5 quantiles of taxable income distribution (columns 2-6) in millions of local currency and at zero taxable profits (column 1).

B.2 Additional empirical model details

B.2.1 Fixed costs estimation methods

Table B4: Summary of fixed costs estimation data and methods.

country	(1) Functional form assumption	(2) variable costs proxy	(3) panel/ cross-section
Armenia	Cubic	Tax deductions	Panel with firm fixed effects
Canada	log-log	Total expenses (Income Statement)	panel with firms fixed effects
Chile	quintic	wages	cross section
China	cubic	interest deductions	cross section
Costa Rica	cubic	Material, and Operational Costs	repeated cross section
Czechia	cubic	Material and energy costs	cross section
Dominican Republic			
Ecuador	cubic	Labor and Material Costs	repeated cross section
France	cubic		repeated cross section
Greece	log-log	tax deductions	repeated cross section
Guatemala	cubic		repeated cross section
Honduras	cubic		repeated cross section
Montenegro	quintic	labor input + material input + operating input	repeated cross section
Norway	quintic	operational costs and financial costs	panel, with firm fixed effects
Portugal	cubic	Costs of goods sold and material consumed + Supplies and external services + Employee expenses.	repeated cross section
Rwanda	cubic		repeated cross section
Senegal	cubic		repeated cross section
Slovakia	log-log	depreciation of long-term tangible and intangible assets	cross section
South Africa	cubic	operational costs, financial costs, and other expenses	repeated cross section
Uruguay	cubic		repeated cross section

Notes: This table presents the variables choices that were used for the fixed costs estimations across countries.

Table B5: Summary of fixed costs scaling parameter.

	25th pct	Median	75th pct
Armenia	-0.62	0.00	1.10
Canada	-0.08	-0.02	0.07
Chile	-0.34	0.03	1.05
Costa Rica	-0.04	1.34	6.94
Czechia	0.00	0.02	0.08
Dominican Republic	-1.22	-0.21	1.24
Ecuador	-5.99	-1.00	1.02
Greece	-0.02	0.01	0.04
Montenegro	-2.37	0.18	32.07
Norway	-1.66	-0.81	0.29
Portugal	-1.24	-0.89	-0.27
Slovakia	-0.00	0.00	0.01

Notes: This table presents the distribution of fixed costs scalar, $(\frac{C_i}{Y_i})$, across countries. C_i is the estimated firm-level fixed cost parameter, while Y_i is the taxable profits. Note that Y_i can be negative.

B.3 Methodology for estimating the distribution of elasticities

We estimate and graph the distribution of elasticities within a country in Figure 6. We employ a flexible parametric approach to recover the shape of the distribution of elasticities within a country. Specifically, we use the Skewed Generalized Error Distribution (SGED), which is defined by four parameters—mean, standard deviation, shape (k), and skewness (λ). The advantage of this distribution is that it allows for asymmetric, heavy-tailed distributions. To recover the distributional parameters, we match three quantiles—the 25th, 50th, and 75th percentiles. Identification proceeds by minimizing a non-linear least squares objective function. The optimization is constrained to ensure economically meaningful parameters (e.g., positive standard deviation, bounded skewness) and is implemented with a quasi-Newton method (L-BFGS-B) for computational efficiency.

C Prediction appendix

This appendix provides additional details and results for our random forest analysis to predict elasticities. Table C1 provides a list of the characteristics we use in our random forest analysis. We include variables in three categories: tax system characteristics, country characteristics, and firm characteristics. The random forest procedure chooses the variables and the number of variables to include in our out-of-sample prediction. Figure C1 graphs the mean squared error and R-squared of the model as the number of factors increases from 1 to 20. The lowest MSE and the highest R-squared occur with three factors. We use three factors, specifically logarithm of tax revenues, statutory corporate tax rates, and manufacturing value added as % of GDP as our baseline. We also provide estimates using a fourth and fifth factors, percent of formally registered firms and manufacturing imports as a percent of GDP. The estimates are similar across using these three different models.

In Figure C2 we graph the contribution of of the three focal factors (logarithm of tax revenues, statutory corporate tax rates, and manufacturing value added as % of GDP) in explaining differences across elasticities based on the Shapley additive explanation (SHAP). For each country, the estimate starts with a base level of 0.56. Then, if a factor contributes positively to the difference from the base, it is shown above the base and in blue. If, instead, a factor contributes negatively to the difference from the base, it is shown below the base and in red. For example, the log of tax revenues adds to the elasticity for Canada while it adds negatively to Costa Rica. This comparison highlights that the same factor can cause estimates in some countries to be greater or less than the base estimate. Table C2 summarizes these waterfall graphs in a tabular format, while Table C3 and Table C4 show the percentage contributions of the four and five focal factors respectively.

Figure C1: Mean squared error and R-squared as number of factors increase

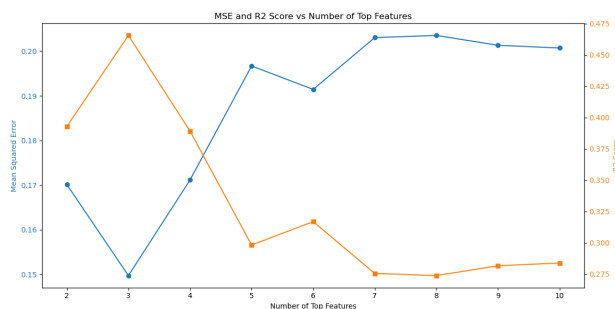
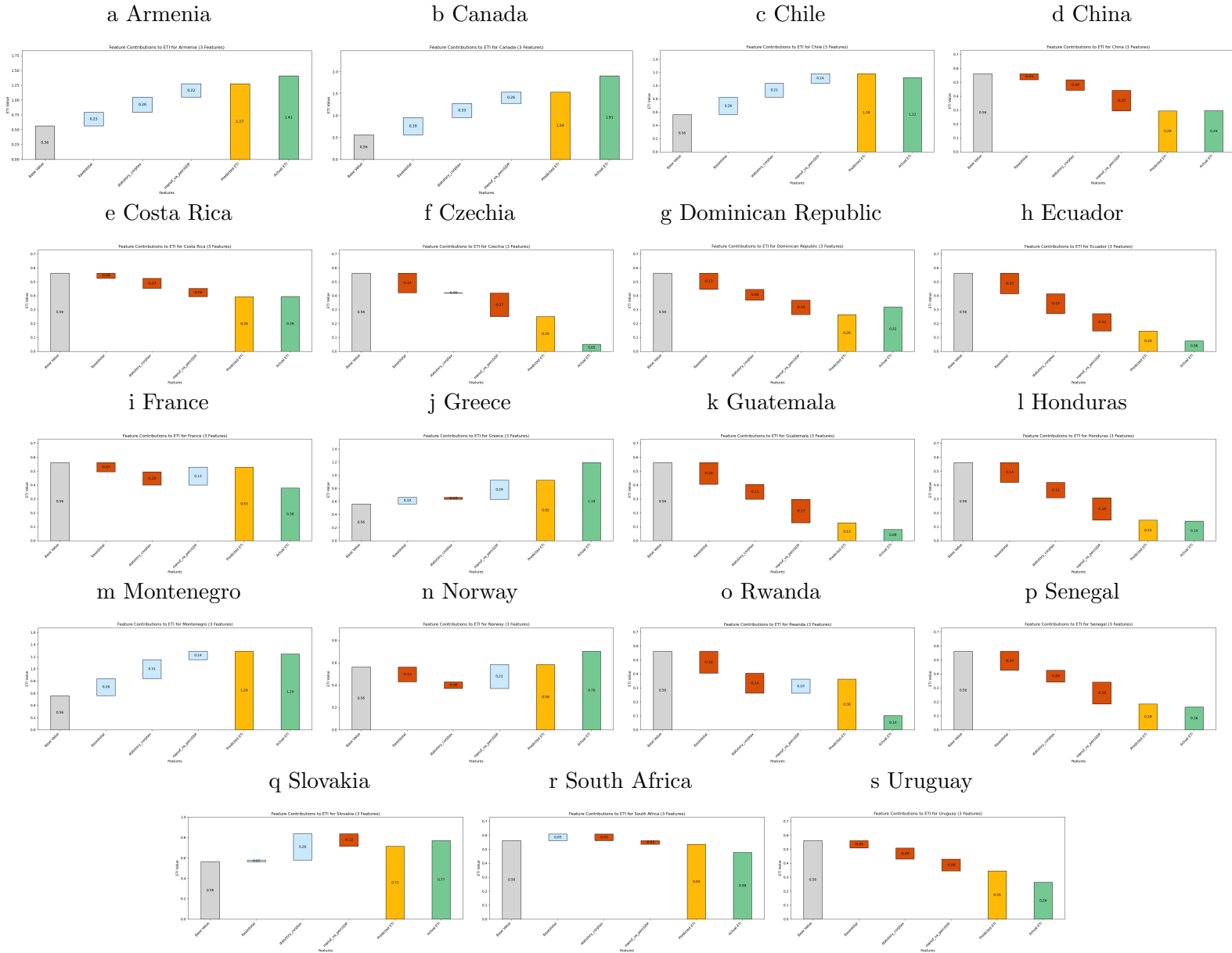


Table C1: List of all variables used in the random forest procedure.

variable name	variable description	source
Tax system characteristics		
statutory_corptax	statutory corporate tax rate	CBT tax database
loss_carry_forward	loss carryforward allowed years	CBT tax database
loss_carry_back	loss carryback allowed years	CBT tax database
national_loss_consolidation	dummy if loss consolidation allowed	CBT tax database
minimumtax	dummy if minimum ltax	CBT tax database
taxholiday	dummy if tax holiday	CBT tax database
SEZs	dummy if special economic zone	CBT tax database
EATR_oecd	effective average tax rate	OECD + CBT tax database
EMTR_oecd	effective marginal tax rate	OECD + CBT tax database
ltaxestotal	log total tax revenues	OECD + CBT tax database
taxesTTRprofit	total tax revenues	OECD + CBT tax database
ltaxesTTRprofit	log total tax revenues	OECD + CBT tax database
taxesTTRLabor	total tax revenues	OECD + CBT tax database
ltaxesTTRLabor	log total tax revenues	OECD + CBT tax database
taxesTTROther	total tax revenues	OECD + CBT tax database
ltaxesTTROther	log total tax revenues	OECD + CBT tax database
expenditure_percGDP	total tax revenues	OECD + CBT tax database
Country characteristics		
gdp_nom	nominal GDP (USD)	WDI
lgdp_pc	logarithm of GDP per capita (USD)	WDI
gdp_gr	GDP growth	WDI
gini	Inequality: Gini	WDI
govtexpperc	Government expenditures as perc of GDP	WEO
govtrevperc	Government revenues as perc of GDP	WEO
FDI_bopUSD	Foreign direct investment, net (BoP, current US\$)	WDI
FDI_inflows_percGDP	Foreign direct investment, net inflows (% of GDP)	WDI
FDI_outflows_percGDP	Foreign direct investment, net outflows (% of GDP)	WDI
CABperc	Current account balance as perc of GDP	WEO
exports_percGDP	Exports of goods and services (% of GDP)	WDI
improts_percGDP	Imports of goods and services (% of GDP)	WDI
manuf_exp	Manufactures exports (% of merchandise exports)	WDI
manuf_imp	Manufactures imports (% of merchandise imports)	WDI
natresourcesrents_percGDP	Total natural resources rents (% of GDP)	WDI
tariffatemean	Tariff rate, applied, weighted mean, all products (%)	WDI
formal	Index built based on the quantiles of the distribution of the percent of firms that are formally registered	WB Informal Economy Database
epayments	percentage of payments received electronically	ISORA
taxadminexp_GDPshare	Tax administration expenditures (% of GDP)	ISORA
taxadminstaff	logarithm of Tax administrationstaff (total FTE)	ISORA
lPopulation	logarithm of population	WEO
governancequality	governance quality: average of voice and accountability, political stability, government effectiveness, regulatory quality, rule of law and control of corruption	WGI
exchrte	exchange rate USD	
Characteristics of firms within each country		
ageofirm	average age of firms	WDI
percfirmwith10percown	Percent of firms with at least 10% of foreign ownership	WDI
manuf_va_percGDP	Manufacturing, value added (% of GDP)	WDI
petformallyregist	% of firms formally registered when they started operations	WB Informal Economy Database
lassets	logairithm of the average firm total assets	Orbis BvD
lp90assets	logairithm of the p90 of firm total assets	Orbis BvD
lp10assets	logairithm of the p10 of firm total assets	Orbis BvD
lp99assets	logairithm of the p99 of firm total assets	Orbis BvD
lrevenue	logarithm of average firm revenues	Orbis BvD
lp90revenue	logarithm of the p90 of firm revenues	Orbis BvD
lp10revenue	logarithm of the p10 of firm revenues	Orbis BvD
lp99revenue	logarithm of the p99 of firm revenues	Orbis BvD
lnbempl	logarithm of the average firm number of employees	Orbis BvD
lp90nbempl	logarithm of the p90 firm number of employees	Orbis BvD
lp10nbempl	logarithm of the p10 firm number of employees	Orbis BvD
lp99nbempl	logarithm of the p99 firm number of employees	Orbis BvD
lnbfirms	logarithm of the number of firms	Orbis BvD
llarge	logarithm of the number of above median size firms	Orbis BvD
lsmall	logarithm of the number of below median size firms	Orbis BvD
asstratio90_10	ratio of total assets of the p90 to p10 firm	Orbis BvD
share_small	share of below median firms in all firms	Orbis BvD

Figure C2: Contribution of factors to deviations from average ETI estimates



Notes: This figure quantifies the role of different factors in explaining differences in the corporate elasticity across countries. The factors that increase a country's elasticity are shown in blue and factors that decrease a country's elasticity are shown in red.

Table C2: **Level contribution of factors to deviations from average ETI, 3 factors**

	Base Value	Predicted ETI	Actual ETI	Country	Tax system	Firm
Armenia	0.562	1.273	1.408	0.230	0.259	0.221
Canada	0.562	1.537	1.907	0.392	0.327	0.256
Chile	0.562	1.182	1.122	0.262	0.214	0.143
China	0.562	0.294	0.295	-0.044	-0.077	-0.148
Costa Rica	0.562	0.392	0.394	-0.037	-0.073	-0.060
Czechia	0.562	0.249	0.051	-0.141	-0.002	-0.171
Dominican Republic	0.562	0.265	0.319	-0.116	-0.077	-0.104
Ecuador	0.562	0.146	0.075	-0.148	-0.143	-0.125
France	0.562	0.528	0.379	-0.067	-0.095	0.128
Greece	0.562	0.924	1.194	0.099	-0.031	0.294
Guatemala	0.562	0.128	0.080	-0.157	-0.108	-0.169
Honduras	0.562	0.147	0.140	-0.143	-0.112	-0.160
Montenegro	0.562	1.290	1.243	0.277	0.314	0.136
Norway	0.562	0.584	0.703	-0.133	-0.059	0.214
Rwanda	0.562	0.363	0.101	-0.158	-0.143	0.102
Senegal	0.562	0.185	0.163	-0.136	-0.084	-0.157
Slovakia	0.562	0.715	0.770	0.016	0.259	-0.123
South Africa	0.562	0.536	0.477	0.047	-0.048	-0.026
Uruguay	0.562	0.345	0.264	-0.053	-0.079	-0.084
Mean				0.000	0.013	0.009

Notes: This table quantifies the role of different factors in explaining deviations from average ETI across countries in absolute terms. We use the top 3 factors that explain the differences in ETIs across countries: $\ln(\text{total taxes})$, statutory corporate tax rate, manufacturing sector value added as a % of GDP. We call these country characteristics, tax system, firm characteristics respectively. We quantify the contribution of each factor in absolute values.

Table C3: **Percentage contribution of factors to deviations from average ETI, 4 factors**

	Tax system	Country-taxes	Firm	country-formality
Armenia	0.27	0.24	0.24	0.24
Canada	0.27	0.36	0.20	0.18
Chile	0.19	0.41	0.17	0.23
China	0.25	0.13	0.40	0.21
Costa Rica	0.26	0.15	0.20	0.39
Czechia	0.17	0.24	0.34	0.25
Dominican Republic	0.25	0.31	0.25	0.18
Ecuador	0.26	0.27	0.21	0.25
France	0.24	0.27	0.21	0.28
Greece	0.06	0.11	0.47	0.36
Guatemala	0.18	0.29	0.27	0.26
Honduras	0.24	0.27	0.27	0.22
Montenegro	0.33	0.31	0.13	0.22
Norway	0.19	0.23	0.29	0.30
Rwanda	0.23	0.32	0.20	0.26
Senegal	0.19	0.27	0.30	0.24
Slovakia	0.36	0.02	0.25	0.37
South Africa	0.09	0.26	0.18	0.47
Uruguay	0.33	0.09	0.24	0.34
Median	0.24	0.27	0.24	0.25
Mean	0.23	0.24	0.25	0.28

Notes: This table quantifies the role of different factors in explaining deviations from average ETI across countries. We use the top 4 factors that explain the differences in ETIs across countries: $\ln(\text{total taxes})$, statutory corporate tax rate, manufacturing sector value added as a % of GDP, % of firms formally registered. We call these country-taxes characteristics, tax system, firm characteristics, country-formality characteristics respectively. We quantify the contribution of each factor in absolute values.

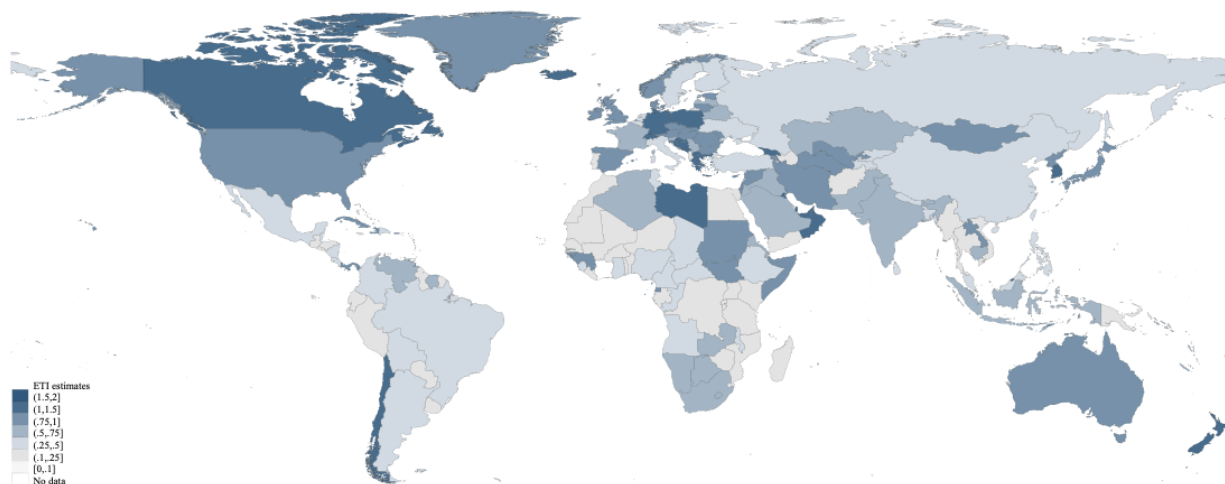
Table C4: **Percentage contribution of factors to deviations from average ETI, 5 factors**

	Tax system	Country taxes	Firm manufact VA	country formality	firm manufact imports
Armenia	0.34	0.23	0.22	0.12	0.08
Canada	0.36	0.34	0.23	0.01	0.06
Chile	0.32	0.40	0.17	0.07	0.04
China	0.30	0.14	0.40	0.13	0.02
Costa Rica	0.33	0.19	0.21	0.14	0.13
Czechia	0.16	0.29	0.35	0.17	0.03
Dominican Republic	0.31	0.29	0.29	0.10	0.00
Ecuador	0.30	0.27	0.22	0.11	0.10
France	0.19	0.23	0.23	0.20	0.15
Greece	0.14	0.01	0.39	0.20	0.26
Guatemala	0.23	0.27	0.31	0.11	0.07
Honduras	0.29	0.25	0.30	0.14	0.02
Montenegro	0.40	0.31	0.19	0.10	0.00
Norway	0.21	0.26	0.43	0.03	0.07
Rwanda	0.23	0.28	0.29	0.12	0.09
Senegal	0.20	0.20	0.25	0.12	0.23
Slovakia	0.49	0.08	0.21	0.18	0.03
South Africa	0.15	0.39	0.22	0.10	0.13
Uruguay	0.33	0.20	0.23	0.13	0.11
Median	0.30	0.26	0.23	0.12	0.07
Mean	0.28	0.24	0.27	0.12	0.09

Notes: This table quantifies the role of different factors in explaining deviations from average ETI across countries. We use the top 5 factors that explain the differences in ETIs across countries: $\ln(\text{total taxes})$, statutory corporate tax rate, manufacturing sector value added as a % of GDP, % of firms formally registered, manufacturing imports as a share of GDP. We call these country-taxes characteristics, tax system, firm-manufact VA characteristics, country-formality characteristics, firm-manufact imports characteristics respectively. We quantify the contribution of each factor in absolute values.

As we discuss in Section 4.5, we extend our prediction of elasticity estimates to 224 countries across the world—the near universe. In Figure C3, we report predicted elasticities for 224 countries, with dark blue indicating larger elasticities and light blue indicating smaller elasticities. We find substantial variation in the elasticities across and within each continent, with some countries within each continent having very high predicted elasticities while others have very low ones. For example, in North America, Canada has a large predicted elasticity, Mexico a small predicted elasticity, and the United States an intermediate value. In Asia, South Korea and Japan have large predicted elasticities, Thailand has small predicted elasticities, and China and Turkey have intermediate values.

Figure C3: **Cross-Country elasticity estimates**



Notes: This figure maps the predicted elasticities using our three-factor random forest model. Estimates for each country using our three-, four-, and five-factor models are provided in Table A1.

D Detailed Data Description

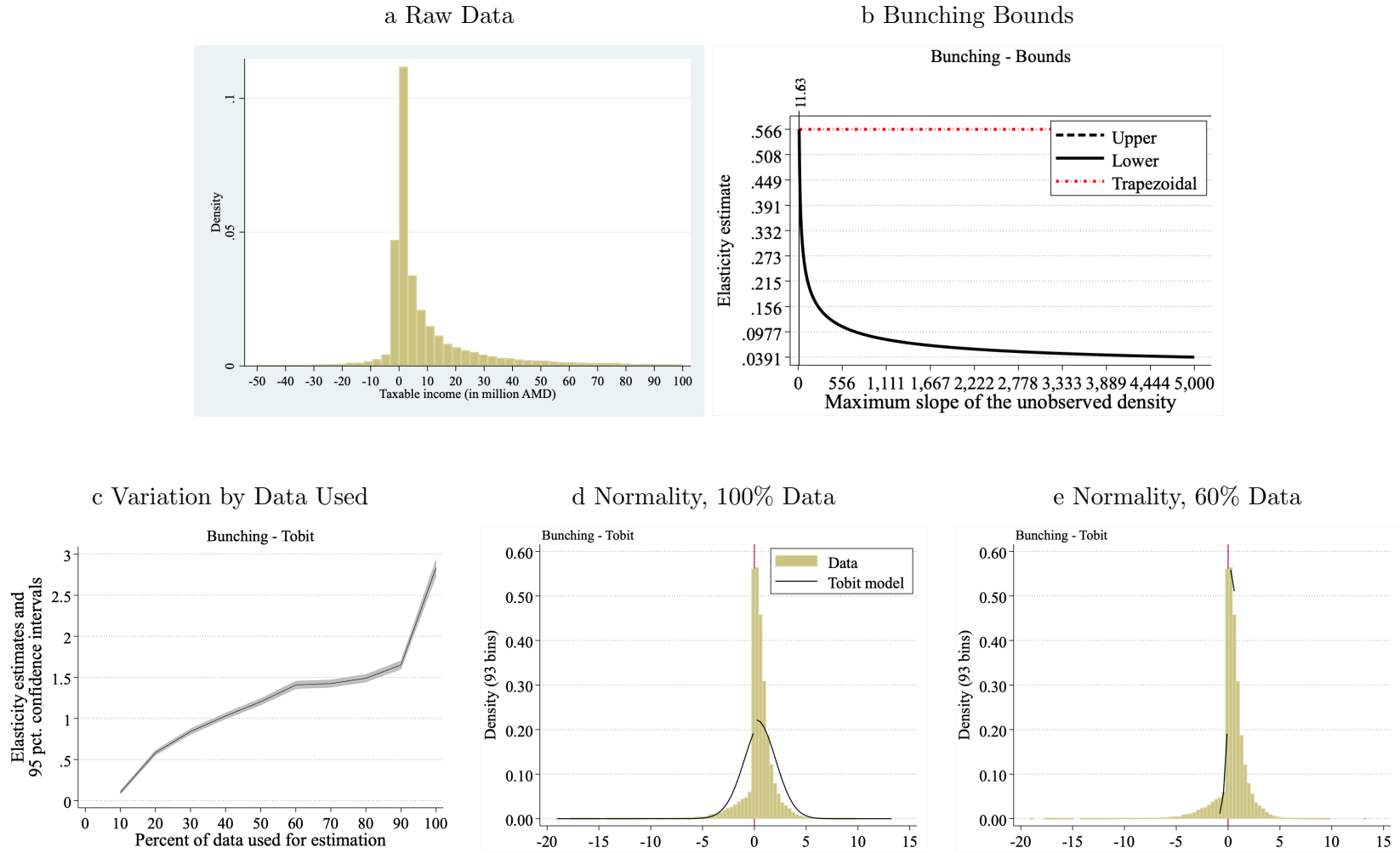
In this Appendix, we present detailed data description and information on elasticity estimates for each country. In all graphs and figures we report the elasticity of taxable income evaluated for a firm with zero fixed costs, which is the elasticity of W with respect to the net-of-tax rate. All in-line references to the elasticity of taxable income reflect e_W . As shown in equation (14), the elasticity of Y with respect to the net-of-tax rate is related to the elasticity of W with respect to the net-of-tax rate using a scale factor, $(1 + \frac{F}{Y})$. We provide additional information about the magnitude of this scale factor within each context to allow the reader to convert e_W to e_Y .

D.1 Armenia

Tax Data The estimations are based on administrative data provided by the State Revenue Committee of the Republic of Armenia. We use the universe of corporate income tax returns of firms filing taxes over the period 2007-2017. The data contains balance sheets, economic sector, tax regime, tax deductions and credits, depreciation, taxable income, and corporate income taxes paid. This data is also used in Asatryan and Joulfaian (2022).

Corporate Tax Context During the observation period 2007-2017, Armenian corporate firms were subject to a flat tax rate of 20% on their corporate profits. Firms report gross income which primarily consists of turnover or net receipts. This is reduced by cost of goods sold, employee compensation, interest expenses, and depreciation allowances among others costs. Consolidated returns are not permitted, each company must file a separate return. Tax losses may be carried forward and set off against tax profits for the following five years. The carry back of tax losses is not permitted. Companies are required to pay quarterly advance payments. The tax year is the calendar year.

Figure D1: Corporate Elasticity of Taxable Income: Armenia, 2007-2018



Notes: This figure plots diagnostic graphs related to the estimation of the corporate elasticity of taxable income using administrative data from Armenia. Panel (a) plots the raw distribution of accounting profit and highlights bunching behavior. Panel (b) plots diagnostic tools for the traditional bunching estimate. Panel (c) plots variation in the estimated elasticity of taxable income based on the amount of data used in the estimation. Panel (d) plots the fit of the normal distribution in the tobit model for 100% of the data. Panel (e) plots the fit of the normal distribution in the tobit model using 60% of the data, and reflects our preferred specification.

D.2 Canada

Tax data The estimations are based on administrative data from the Canada Revenue Agency accessed through Statistics Canada’s Microdata Access Division. The data consists of the T2 Corporation Income Tax Return and all its accompanying schedules (forms T2 SCH1 to T2 SCH200). These data contain hundreds of variables such as total revenue, expenses, assets and liabilities, as well as accounting profit, net income (or loss) for tax purposes, some tax deductions such as tax depreciation and losses carried forward, scientific research and experimental development expenditures, charitable donations, etc.

Our sample contains only public corporations (a corporation resident in Canada with a class of shares listed on a Canadian stock exchange) and corporations controlled by a public corporation, for the years 2012 to 2019. We exclude corporations with total expenses below \$100,000 CAD or total assets below \$500,000 CAD as well as those with dividend income representing more than 50% of their total revenue. We further restrict the sample by excluding corporations in the Finance and insurance sector (NAICS 52) as they face different tax rules. Our variable of interest is taxable income before the carryforward of previous years’ losses (that is net income (or loss) for income tax purposes minus charitable donations and gifts minus deductible dividends).

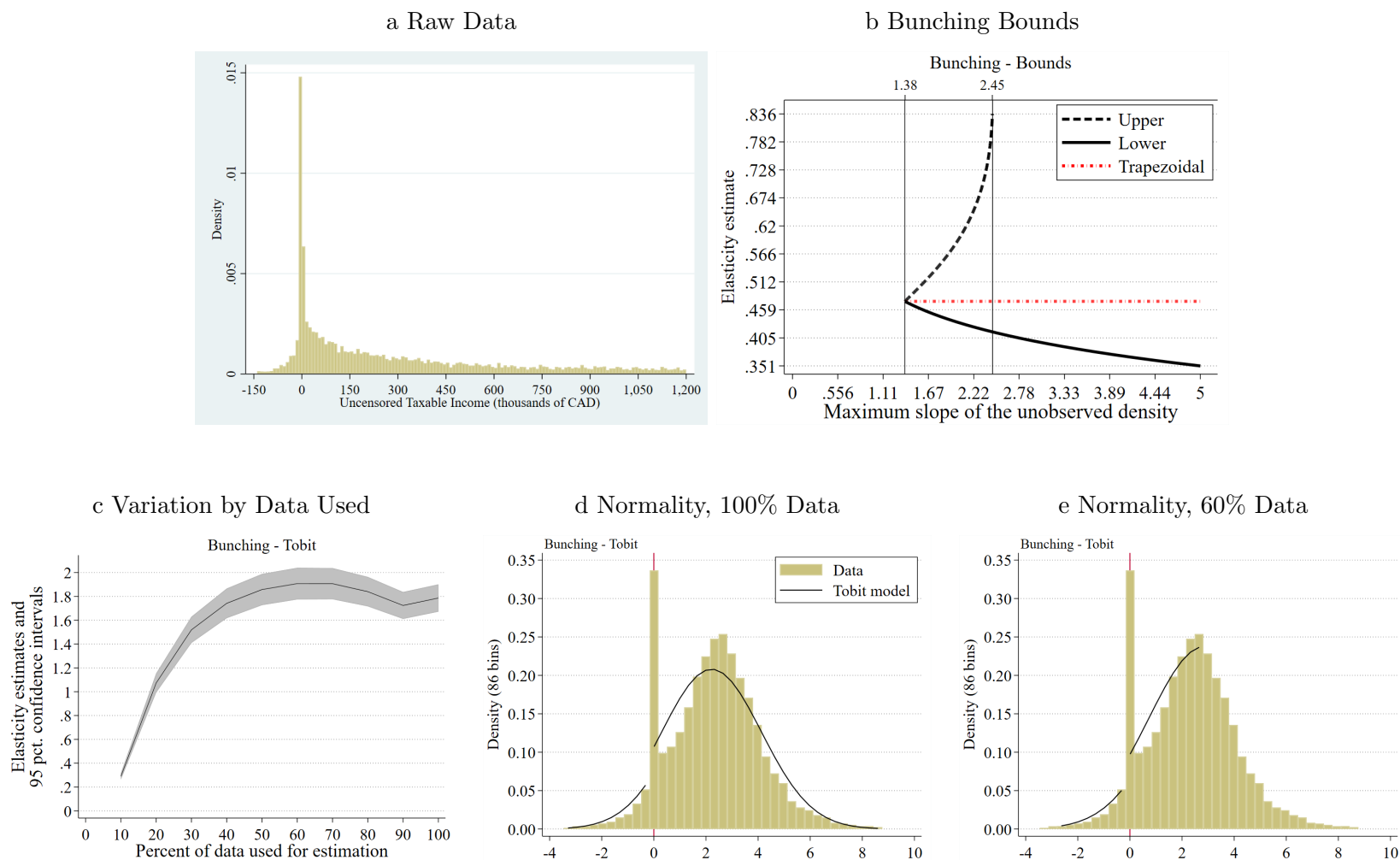
Corporate Tax Context Corporations are liable for federal and provincial corporate income tax on their worldwide income (foreign tax credits are available to offset business income tax paid in other countries). The general corporate tax rate at the federal level is 15% since 2012, and ranges at the provincial level from 8% to 16% for a total tax burden of 26.5% on average in our sample. Expenditures in scientific research and experimental development (SRED) are eligible for a 15% non-refundable tax credit.

Canadian-controlled private corporations (CCPC) with total assets below \$10 million and passive investment income below \$50,000 are facing the small business tax rate on their first \$500,000 of active business income, which is 9% at the federal level and between 0% and 3.2% at the provincial level, for a total tax burden of 11.8% on average. Active business income in excess of \$500,000 is taxed at the general rate of 15%. The SRED credit rate is 35% on the first \$3 million of eligible expenditures for small CCPCs and is refundable. As our sample is limited to public corporations and their subsidiaries, CCPCs are excluded from the analysis.

The main differences between book income and taxable income are that only half of capital gains are included in taxable income (taxed at realization, not on an accrual basis),

generally more generous tax depreciation rules, some expenses are not deductible for tax purposes (for example only half of meals and entertainment expenses are allowed), and dividends received are generally excluded from taxable income (to avoid double taxation). Non-capital losses can be carried-back to the three previous tax years or carried-forward for 20 years (after which they expire). Unused non-refundable SRED credits can also be carried-back to the three previous tax years or carried-forward for 20 years. Capital losses (50% inclusion rate applying to capital gains also applies to capital losses) can only be applied against capital gains. They can be carried-back to the three previous tax years or carried-forward indefinitely.

Figure D2: Corporate Elasticity of Taxable Income: Canada, 2018



Notes: This figure plots diagnostic graphs related to the estimation of the corporate elasticity of taxable income using administrative data from Canada. Panel (a) plots the raw distribution of accounting profit and highlights bunching behavior. Panel (b) plots diagnostic tools for the traditional bunching estimate. Panel (c) plots variation in the estimated elasticity of taxable income based on the amount of data used in the estimation. Panel (d) plots the fit of the normal distribution in the tobit model for 100% of the data. Panel (e) plots the fit of the normal distribution in the tobit model using 60% of the data, and reflects our preferred specification.

D.3 Chile

Tax Data The estimations are based on administrative data provided by the Chilean IRS. We use the universe of income tax returns of firms filing taxes in 2018. The data consists of the full income tax form (F22) for each firm, which contain economic sector, tax regime, sales, VAT debits and credits, wages, operating expenses, financial expenses, tax credits (for fixed assets purchases, Research and Development expenses, training expenses, donations, property taxes paid), depreciation, amortization, taxable income, corporate income tax base, and corporate income taxes paid.

Corporate Tax Context There are two general tax regimes for corporate taxes in Chile, both with a flat rate:

1. Partial-Integration: under this regime the corporate tax rate is 27%, the tax base for personal income taxes is distributed profits and 65% of corporate taxes paid are credited against personal income taxes

2. Full Integration: under this regime the corporate tax rate is 25%, the tax base for personal income is accrued profits, and corporate taxes paid are fully credited against personal income taxes

All firms must choose one tax regime and cannot switch to the other one for 5 years

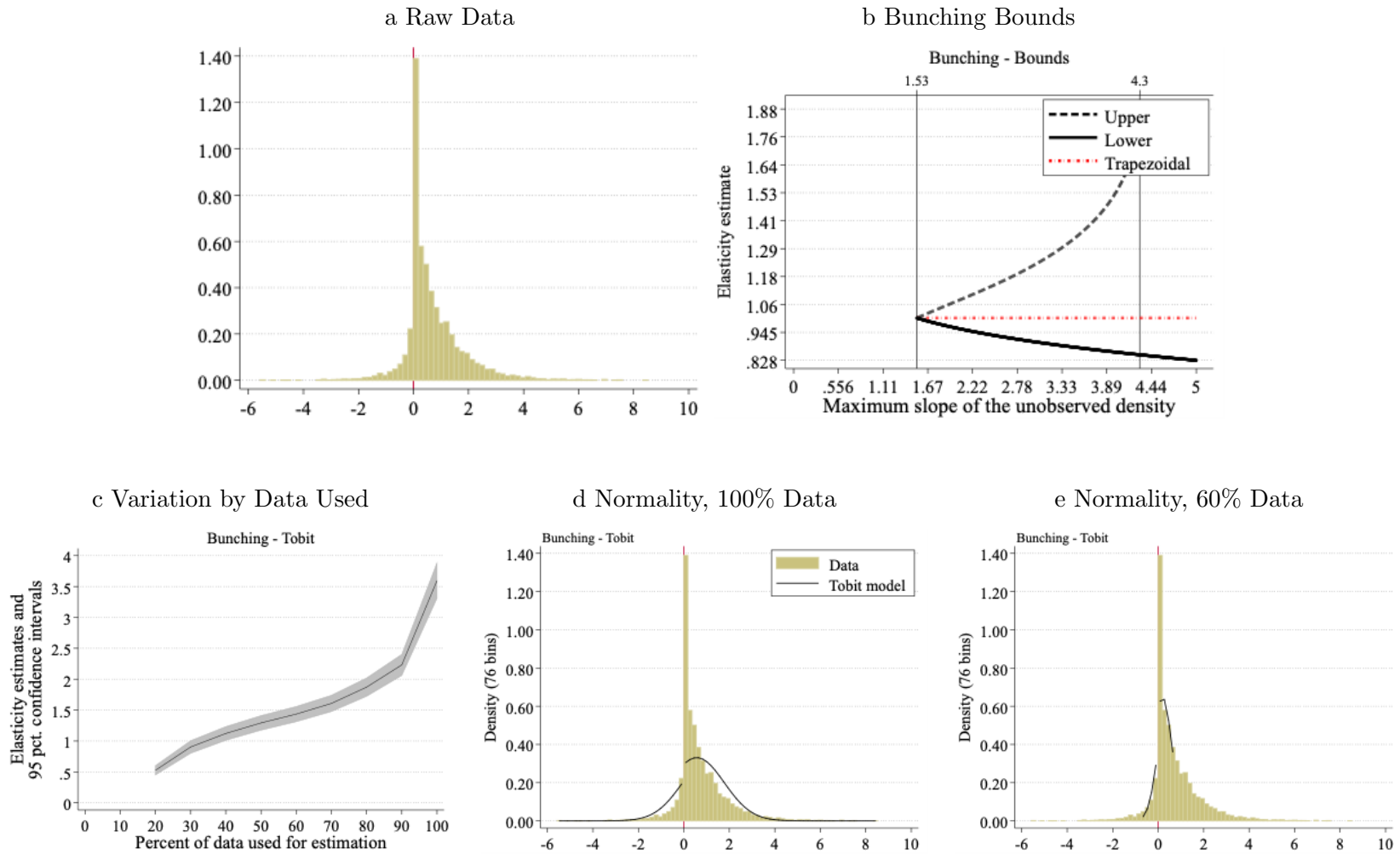
In addition, there is a unique tax regime for small businesses. Under this regime they pay no corporate taxes, only personal income taxes on distributed profits. There is a cap on annual sales of around USD4 million to be considered small business for tax purposes.

Corporate taxes are paid in an annual tax return filled in April (Form 22, which is the source of the data used), but monthly provisional payments have to be made based on an average of previous years' tax payments. Only loss carryforwards are allowed; no loss carrybacks

There exist two types of investment incentives. The first one, applies to: all firms under full accounting and sales up to USD4 million, which can deduct from the corporate income tax base up to USD200,000. The second one, benefits taxpayers who declare CIT on effective income determined according to full accounting records that acquire, finish build or take in leasing fixed assets, and that register a maximum of average annual sales. The benefit consists of a tax credit, whose magnitude depends on the size of the firm. Firms with average annual sales less than or equal to UF 25,000 (around USD 840,000) receive a tax credit of 6% of the value of the fixed assets, acquired new, finished build during the fiscal year or taken under leasing. Firms with average annual sales of more than UF 25,000 but less than or equal to

UF 100,000 (around USD 3,365,000), receive a tax credit of either $6\% \times [(100,000 - \text{annual income}) / 75,000]$ or 4%, whichever is larger. Finally, firms with average annual sales of more than UF 100,000 receive a tax credit of 4%.

Figure D3: Corporate Elasticity of Taxable Income: Chile, 2018



Notes: This figure plots diagnostic graphs related to the estimation of the corporate elasticity of taxable income using administrative data from Chile. Panel (a) plots the raw distribution of accounting profit and highlights bunching behavior. Panel (b) plots diagnostic tools for the traditional bunching estimate. Panel (c) plots variation in the estimated elasticity of taxable income based on the amount of data used in the estimation. Panel (d) plots the fit of the normal distribution in the tobit model for 100% of the data. Panel (e) plots the fit of the normal distribution in the tobit model using 60% of the data, and reflects our preferred specification.

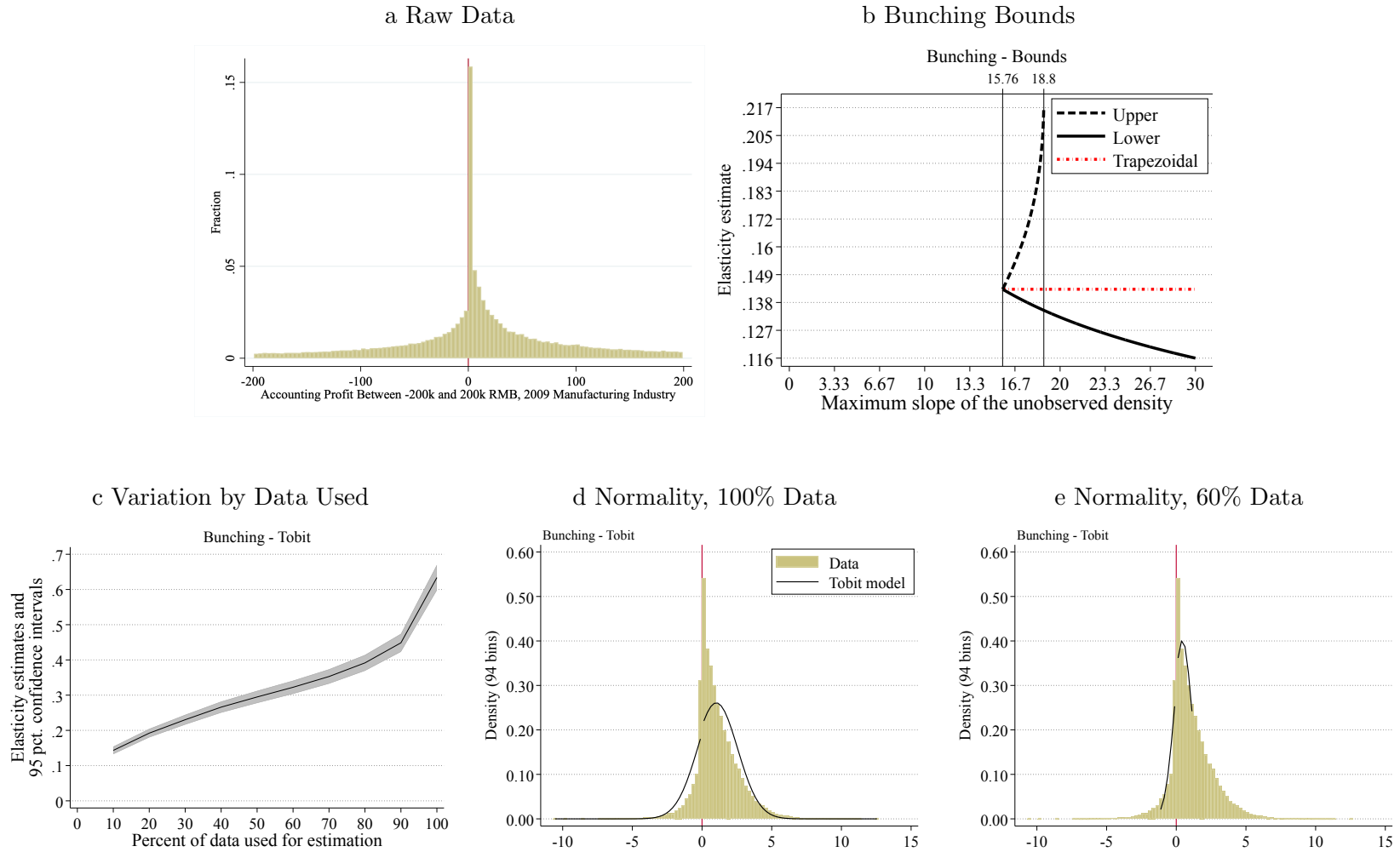
D.4 China

Tax Data Our estimates are based on businesses in the manufacturing industry in China in 2009. Data is drawn from the 2009 China Tax Survey, conducted by the State Taxation Administration (STA) in China — the counterpart to the IRS in the United States. These data provide firm-level information of the components of tax payment in addition to limited financial information for 269,225 firms. Specifically, we observe accounting profit, net profit after tax adjustment, taxable profit, tax adjustments, operating revenue, operating costs, depreciation expenses, total wage and bonus expenses, interest payments, intangible assets purchased, fixed assets held at year end, R&D expenditures, and loss carryforwards from prior tax years.

Corporate Tax Context In 2009 corporations were subject to 17 different taxes within the Chinese business tax system. Most important among these are the Value-Added Tax (VAT), the Corporate Income Tax, the Business Tax, the VAT and Excise Tax on imports, and the domestic Excise tax. In total, these five taxes account for 80% of total corporate tax revenue. VAT and excise taxes are consumption taxes that are levied on goods. Specifically, the VAT taxes the value-added at each stage of goods production and sales, and the baseline VAT rate is 17%. 2009 saw a VAT reform in which firms were allowed to deduct the VAT paid on investment in fixed assets. Excise taxes are levied on a selective list of goods, and business taxes apply to the provision of services, intangible assets, and real estate.

Here, we study how firms respond to the corporate income tax. Prior to 2008, domestic enterprises paid a higher corporate income tax rate than foreign-invested enterprises (33% compared with either 15% or 24%). In 2008, China consolidated corporate income tax rates to a flat 25%, regardless of foreign vs domestic distinctions. Those businesses that faced a preferential rate prior to 2008 were granted a phased-in increase in the corporate income tax rate from 18% – 25% over 5 years. After 2012, all businesses faced the flat 25% rate. Generally, firms are permitted to carry losses from prior tax years forward up to five years to offset current-year positive taxable income.

Figure D4: Corporate Elasticity of Taxable Income: China, 2009



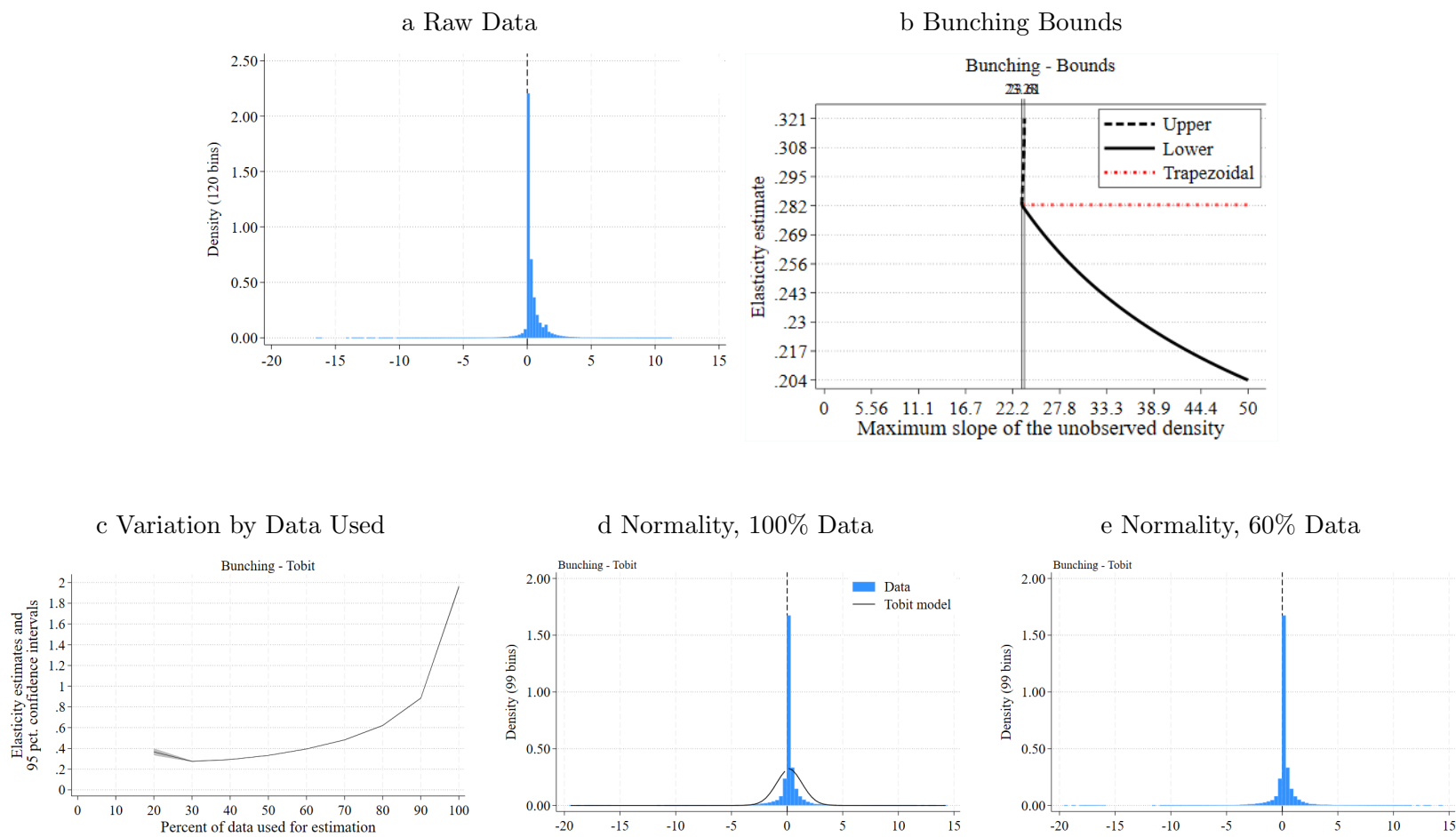
Notes: This figure plots diagnostic graphs related to the estimation of the corporate elasticity of taxable income using administrative data from Chinese manufacturing firms in 2009. Panel (a) plots the raw distribution of accounting profit and highlights bunching behavior. Panel (b) plots diagnostic tools for the traditional bunching estimate. Panel (c) plots variation in the estimated elasticity of taxable income based on the amount of data used in the estimation. Panel (d) plots the fit of the normal distribution in the tobit model for 100% of the data. Panel (e) plots the fit of the normal distribution in the tobit model using 60% of the data, and reflects our preferred specification.

D.5 Costa Rica

Tax Data The sample includes firms identified as legal entities. We rely on an entity type indicator to classify firms as either legal or natural persons. Cases with missing values in this indicator are dropped to avoid accidentally including natural persons.

Tax Context The highest statutory corporate tax rate was 30 percent during the years covered by our data (2006–2019). The tax system includes two other tax brackets for smaller firms with rates at 10 and 20 percent. Tax rates apply to profits, but tax brackets are based on firms' revenue, with thresholds adjusted for inflation annually.

Figure D5: Corporate Elasticity of Taxable Income: Costa Rica, 2006-2020



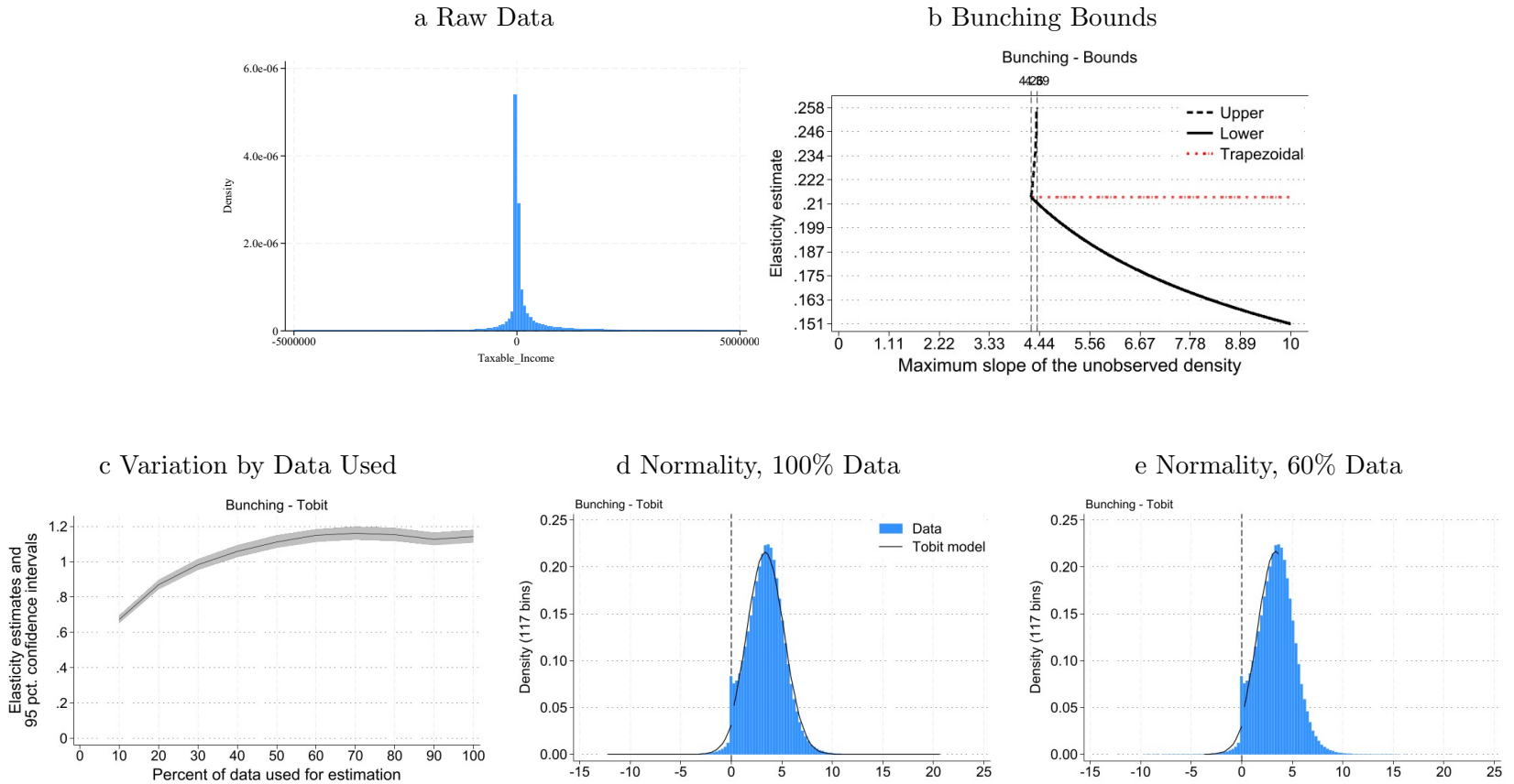
Notes: This figure plots diagnostic graphs related to the estimation of the corporate elasticity of taxable income using administrative data. Panel (a) plots the raw distribution of accounting profit and highlights bunching behavior. Panel (b) plots diagnostic tools for the traditional bunching estimate. Panel (c) plots variation in the estimated elasticity of taxable income based on the amount of data used in the estimation. Panel (d) plots the fit of the normal distribution in the Tobit model for 100% of the data. Panel (e) plots the fit of the normal distribution in the Tobit model using 60% of the data, and reflects our preferred specification.

D.6 Czechia

Tax Data We use administrative tax data from the Czechia provided by the General Financial Directorate. The dataset covers all corporate income tax returns filed by legal entities for the tax year 2021. The tax data are linked with accounting information from firms' balance sheets and profit and loss statements, enabling a comprehensive analysis of firm-level financial characteristics. The final sample includes over 300,000 firms and comprises hundreds of variables, including the tax base, revenues, depreciation, and various categories of costs. For the purposes of our analysis, we focus on three key variables: the tax base, turnover, and material and energy inputs. We exclude firms with negative reported material and energy costs and those with turnover below CZK 250 (approximately EUR 10), in order to restrict the sample to firms with a minimum threshold of genuine economic activity. Firms operating in the financial and insurance sectors are excluded due to their distinct tax treatment.

Tax Context In 2021, the standard statutory corporate income tax rate in Czechia was 19%, applicable to most firms, including branches of foreign companies. Investment funds were taxed at a reduced rate of 5%, and pension funds were fully exempt (0%). Resident firms are taxed on their worldwide income, while non-resident firms are taxed only on their Czechia-source income. The value-added tax (VAT) rate is 21% for most goods and services, with a reduced rate of 12% applied to items such as food and pharmaceuticals. Firms may carry forward tax losses for up to five years; since 2020, losses may also be carried back for up to two years. Consolidation of tax losses across affiliated firms is not permitted—each entity is taxed on a standalone basis.

Figure D6: Corporate Elasticity of Taxable Income: Czechia, 2021



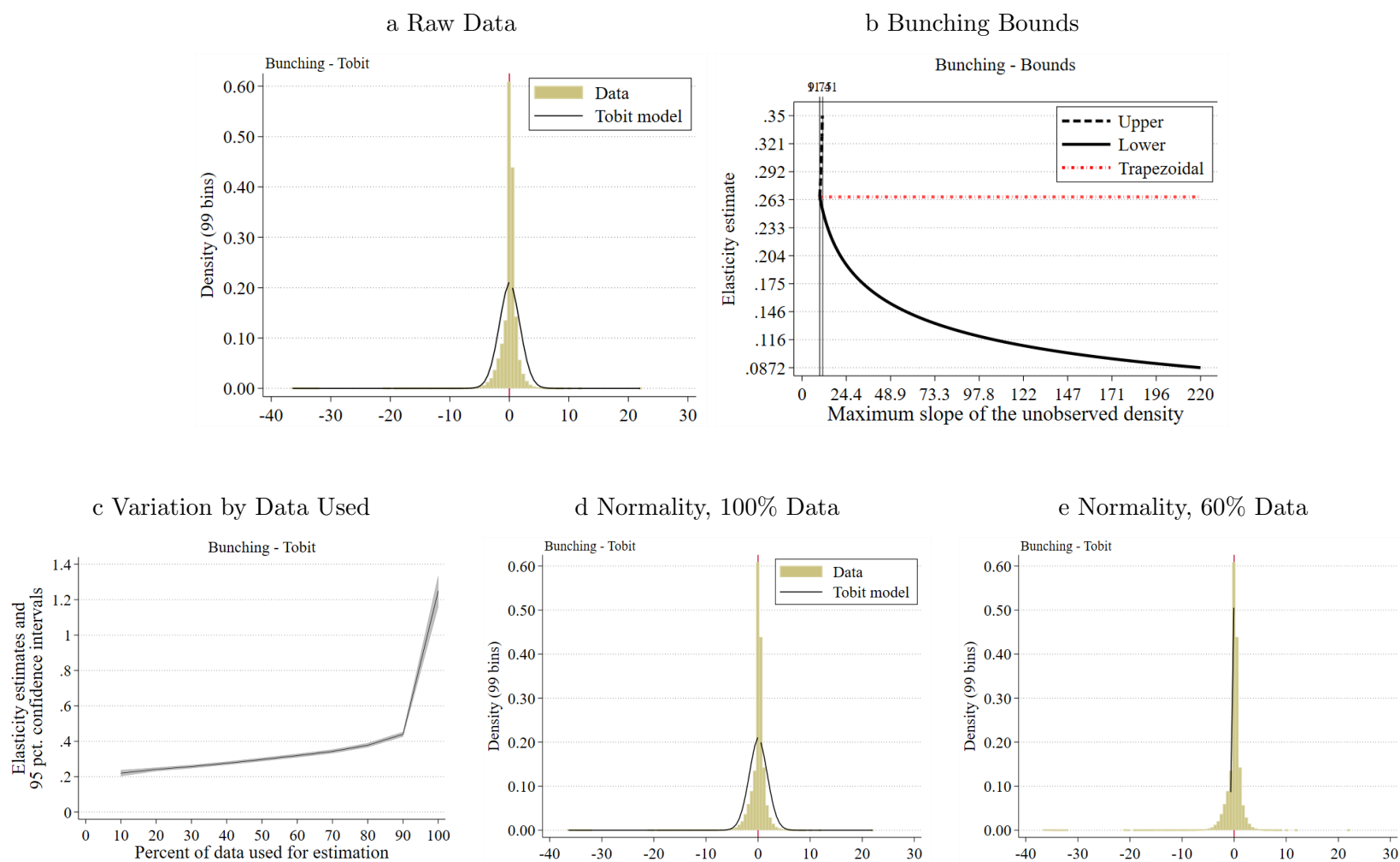
Notes: This figure plots diagnostic graphs related to the estimation of the corporate elasticity of taxable income using administrative data. Panel (a) plots the raw distribution of accounting profit and highlights bunching behavior. Panel (b) plots diagnostic tools for the traditional bunching estimate. Panel (c) plots variation in the estimated elasticity of taxable income based on the amount of data used in the estimation. Panel (d) plots the fit of the normal distribution in the Tobit model for 100% of the data. Panel (e) plots the fit of the normal distribution in the Tobit model using 60% of the data, and reflects our preferred specification.

D.7 Dominican Republic

Tax Data

Tax Context

Figure D7: Corporate Elasticity of Taxable Income: Dominican Republic, 2007-2010



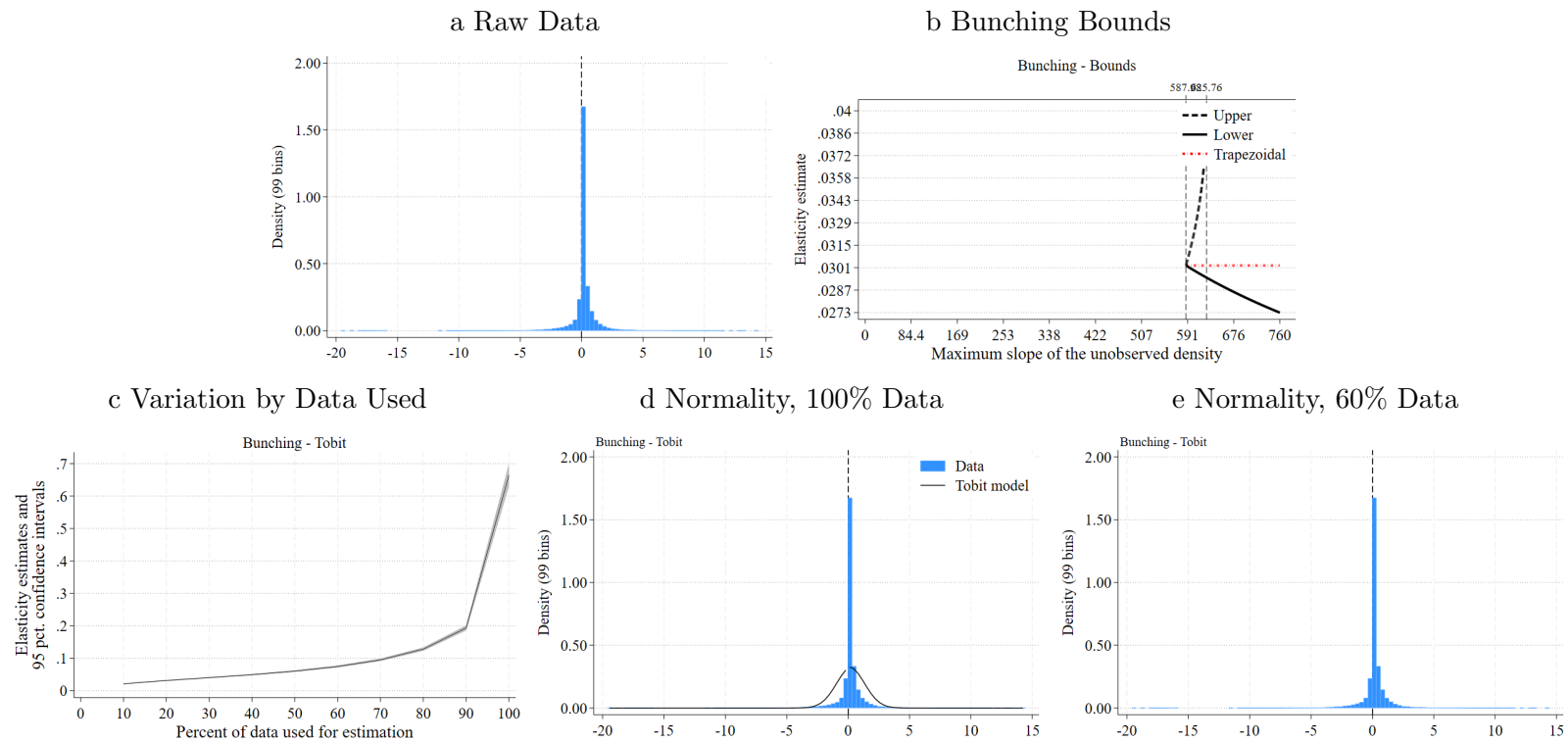
Notes: This figure plots diagnostic graphs related to the estimation of the corporate elasticity of taxable income using administrative data. Panel (a) plots the raw distribution of accounting profit and highlights bunching behavior. Panel (b) plots diagnostic tools for the traditional bunching estimate. Panel (c) plots variation in the estimated elasticity of taxable income based on the amount of data used in the estimation. Panel (d) plots the fit of the normal distribution in the Tobit model for 100% of the data. Panel (e) plots the fit of the normal distribution in the Tobit model using 60% of the data, and reflects our preferred specification.

D.8 Ecuador

Tax Data The data consist of firms' balance sheets provided by the "Superintendencia de Compañías, Valores y Seguros". This information is unrestricted and available under this link .

Tax Context The STR was 22 percent from 2013 to 2017 and increased to 25 percent in 2018. Micro-firms with revenue below 1,000,000 LCU in 2018 remained subject to the 22 percent rate, along with firms in the mining and extractive industries. Depending on the shareholder structure and disclosure compliance, the STR can also be 28 percent.

Figure D8: Corporate Elasticity of Taxable Income: Ecuador



Notes: This figure plots diagnostic graphs related to the estimation of the corporate elasticity of taxable income using administrative data. Panel (a) plots the raw distribution of accounting profit and highlights bunching behavior. Panel (b) plots diagnostic tools for the traditional bunching estimate. Panel (c) plots variation in the estimated elasticity of taxable income based on the amount of data used in the estimation. Panel (d) plots the fit of the normal distribution in the Tobit model for 100% of the data. Panel (e) plots the fit of the normal distribution in the Tobit model using 60% of the data, and reflects our preferred specification.

D.9 France

Tax Data The estimations are based on administrative data provided by the Direction Générale des Finances Publiques (DGFIP), the French equivalent of the IRS. These data are remotely accessible to researchers via the Centre d’Accès Sécurisé aux Données (CASD). We use the universe of corporate income tax returns of firms filing taxes over the period 2010-2016, a period over which exhaustive data are available and there were at the same time no significant reforms of the corporate tax regime. The data consist of the full income tax form (2033 and 2050) for each firm, which contain detailed balance sheets and profits and loss accounts, economic sector, tax regime, tax deductions and credits, depreciation, taxable income, and corporate income taxes paid. Business groups as defined per the French tax code must file, in addition to the former, information on the group-level taxable income.

Tax Context Broadly speaking, there are three distinct business income tax regimes in France :

1. Pass-through regime (BIC-IR): this is the equivalent of the American partnership and S-corp regimes; there is no tax paid at the level of the corporate entity, and all accrued profits are taxed at the shareholder level, following the (progressive) personal income tax schedule (up to 60% if one includes all forms of personal income tax) if the shareholder is an individual and the corporate tax schedule if the shareholder is a corporation subject to the default corporate tax system.
2. Default corporate tax regime (BIC-IS): this is the equivalent of the American C-corp regime; accrued profits are taxed at the level of each unconsolidated corporation at a marginal rate of 34.3% (before 2017) for the part above 38k euros, and 15% for the part below. Dividends are then taxed only if they are distributed to individuals (under the progressive schedule, after a deduction of 40%) or minority corporate owners (under the corporate tax schedule, without deduction).
3. Business group tax regime (IS Groupe fiscal): this regime follows similar rules as the default corporate tax regime for the most part, except that tax losses recognized by some members of the business group contemporaneously offset taxable income recognized in other parts of the business group, leading to the definition of a group-level taxable income against which the default corporate tax schedule is applied.

The pass-through regime is the default regime for partnerships and can be chosen by

some limited liability corporations. Because the tax schedule is specific to each shareholder under this tax regime, we do not include these firms in our analysis.

The business group regime can be chosen by limited liability corporations who share ownership links greater than 95% between each other. Under this regime, the schedule is really only applied to the group-level taxable income. Unfortunately, it is not possible to thinly decompose group-level taxable income into group-level turnover and group-level costs. This is why in our baseline, we leave all business group members out of the analysis. However, we provide in the Appendix an analysis including these business group members.

This means the bulk of the analysis on French data is run only on firms choosing the default corporate tax regime. Under this regime, corporate taxes are paid in an annual tax return filled in May (forms 2033 and 2033, which are the sources of the data used in the estimation), but quarterly provisional payments have to be made based on the previous year's tax payments. Losses can be carried forward without any limitation of time, but past losses cannot offset more than 50% of the taxable income before carryforwards; loss carrybacks are allowed only up to a small amount in absolute terms.

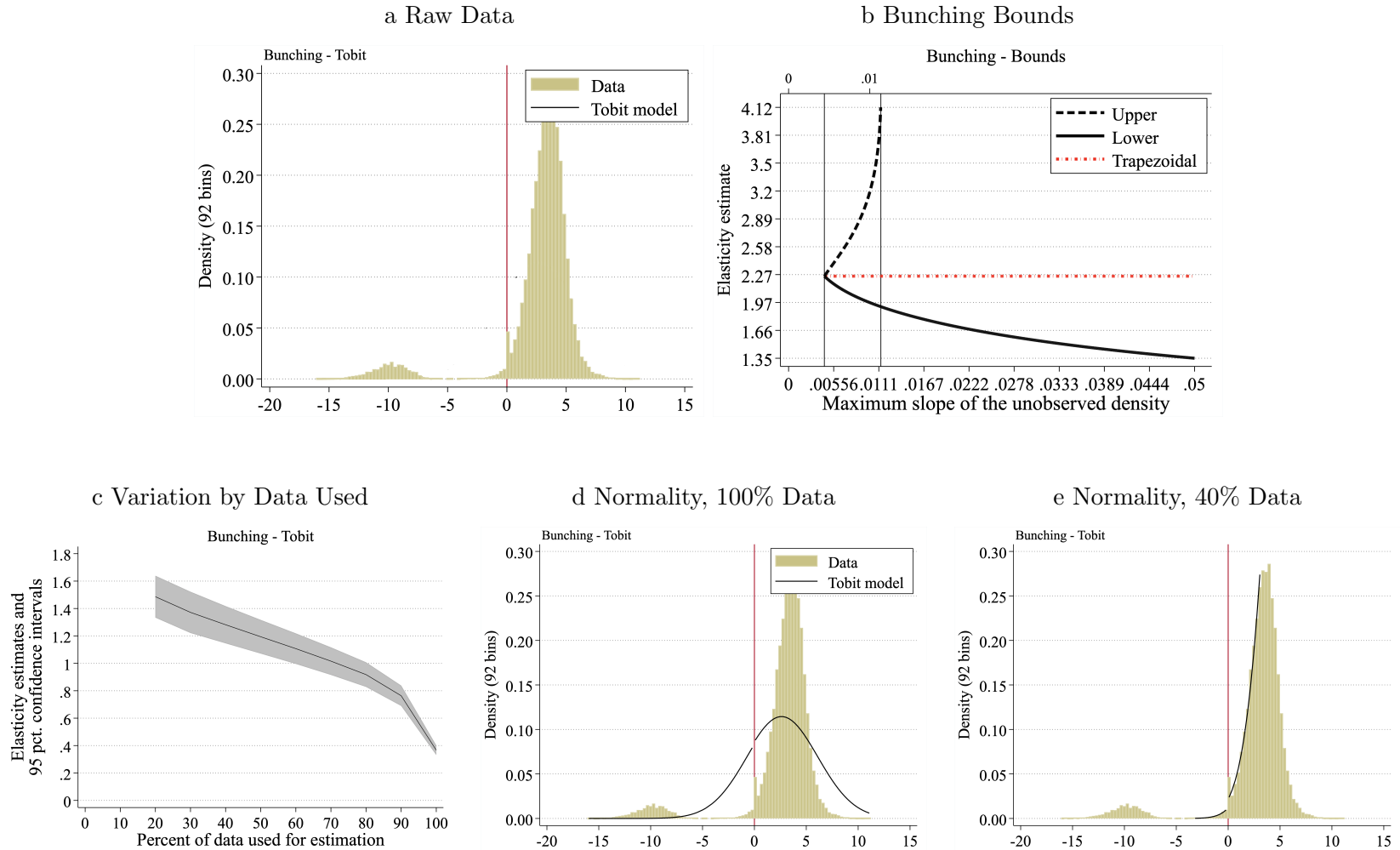
There are substantial tax deductions targeted at specific industries and locations, but no widely-applied investment deductions. R&D subsidies are corporate tax credits and hence do not affect bunching incentives.

D.10 Greece

Tax Data The Greek sample consists of the population of firms established as corporations (mainly Societe Anonyme, Limited Liabilities Companies, Private Capital Companies) for the period 2002-2004. The dataset has been compiled by using two different sources: tax returns available through the Tax Administration of the Ministry of Finance and financial variables from ICAP, the leading provider in Greece. There are more than 50 variables available from the tax form and more than 100 from ICAP records. Financial information from ICAP is limited to larger corporations based on revenue, asset, and employee size thresholds.

Tax Context The corporate tax system in Greece is extremely complicated, characterized by overregulation and low tax collectability. Resident corporations are taxed on their worldwide income. Until 2003, LLCs were taxed differently compared to SAs: half of their profits were taxed in the name of the company and the rest in the name of the partners (natural persons-owners). Beginning in 2003 all corporations, no matter their specific legal type, are taxed in the same way, i.e., all their profits are taxed in the name of the firm. The statutory corporate tax rate demonstrates noticeable volatility over time: the rate has been changed 9 times in the last 20 years. Advanced tax must be also prepaid up to a certain percentage (which is unstable ranging from 55% to 100% during our study period) of the tax obligation in the current year. Businesses are permitted to carry tax losses forward up to five years to reduce taxable profit.

Figure D9: Corporate Elasticity of Taxable Income: Greece, 2002–2004



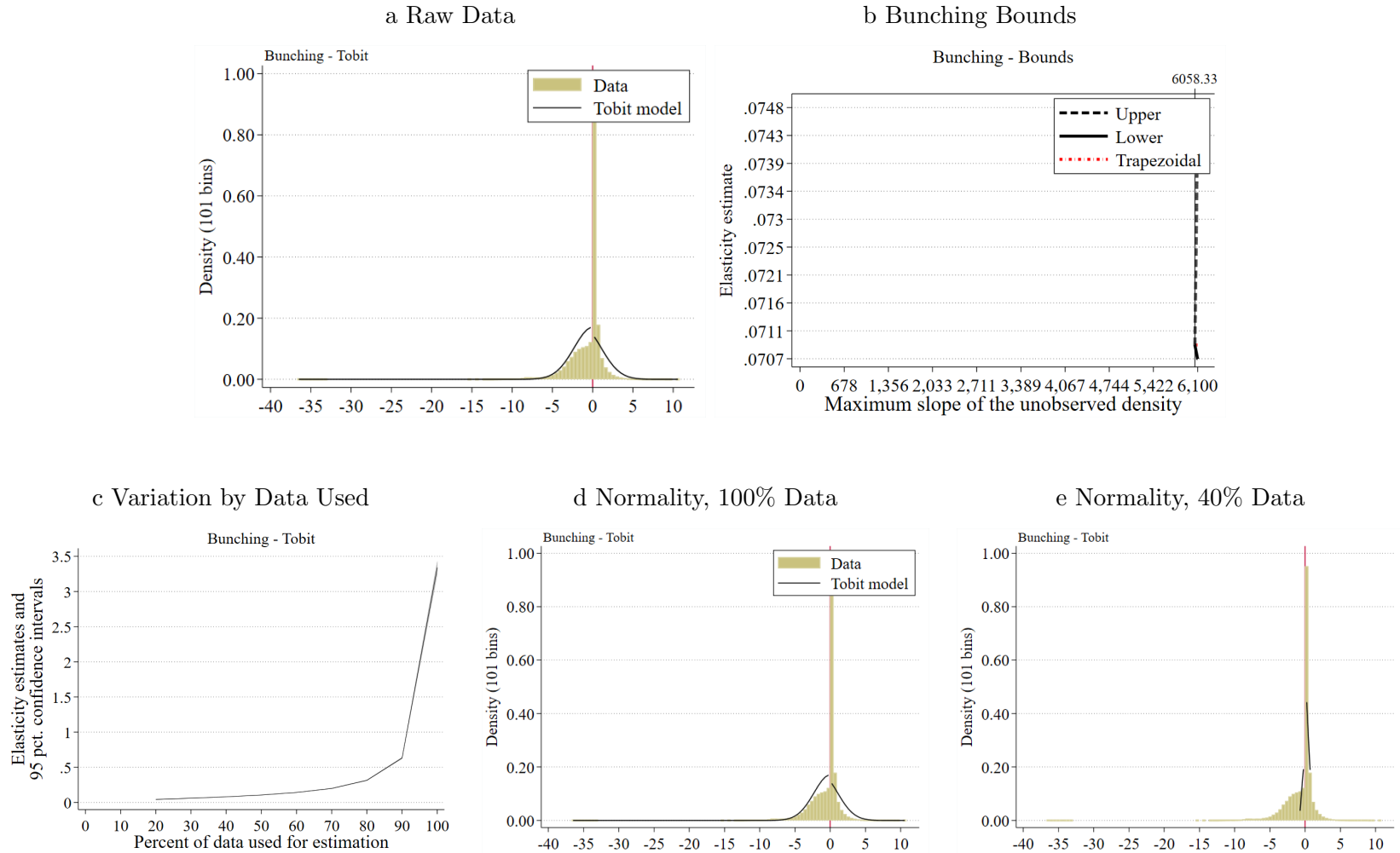
Notes: This figure plots diagnostic graphs related to the estimation of the corporate elasticity of taxable income using administrative data from large Greek firms using data from 2002–2004. Panel (a) plots the raw distribution of accounting profit and highlights bunching behavior. Panel (b) plots diagnostic tools for the traditional bunching estimate. Panel (c) plots variation in the estimated elasticity of taxable income based on the amount of data used in the estimation. Panel (d) plots the fit of the normal distribution in the tobit model for 100% of the data. Panel (e) plots the fit of the normal distribution in the tobit model using 60% of the data, and reflects our preferred specification.

D.11 Montenegro

Tax Data

Tax Context

Figure D10: Corporate Elasticity of Taxable Income: Montenegro, 2011–2020



Notes: This figure plots diagnostic graphs related to the estimation of the corporate elasticity of taxable income using administrative data from large Greek firms using data from 2002–2004. Panel (a) plots the raw distribution of accounting profit and highlights bunching behavior. Panel (b) plots diagnostic tools for the traditional bunching estimate. Panel (c) plots variation in the estimated elasticity of taxable income based on the amount of data used in the estimation. Panel (d) plots the fit of the normal distribution in the tobit model for 100% of the data. Panel (e) plots the fit of the normal distribution in the tobit model using 60% of the data, and reflects our preferred specification.

D.12 Norway

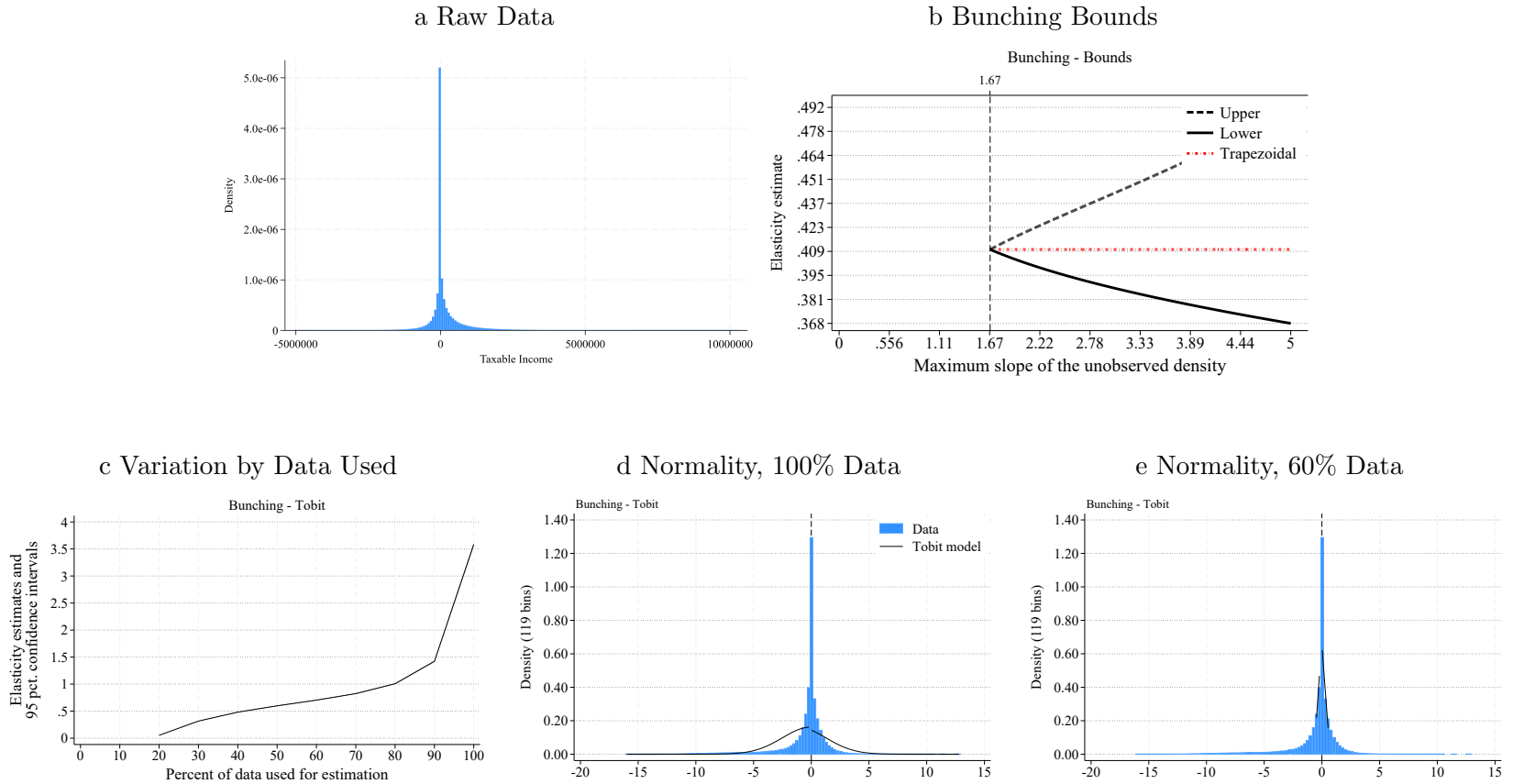
Tax Data We construct our estimation sample from the the universe of Norwegian private and public limited liability firms (AS, *Aksjeselskap* and ASA, *Allmennaksjeselskap*). We draw data from the Norwegian Tax Authority covering the tax years 2006-2013. Information is reported on two mandatory tax forms that must be submitted simultaneously: the actual tax return (form RF-1028) and the income statement (RF-1167). Taxable profit is reported on the tax return and defined as the pre-tax earnings less special deductions. Special deductions include losses from previous years, losses from resource extraction on the Norwegian continental shelf, and group contributions paid to other firms in the corporate group.

Revenue and ordinary deductions are taken from the income statement and computed as the sum of financial and operating income or costs, respectively. For the two-step procedure, we use additional information from the Accounting Register of Norway which collects mandatory balance sheet and profit statement information from all private and public limited liability firms. In addition, we collect information on total intangible fixed assets, depreciation and write-down of fixed assets and long-term liabilities to financial institutions.

Corporate Tax Context During the observation period 2006-2013, Norwegian companies were subject to a flat tax rate of 28% on their corporate profit. This tax base includes the operating and financial profits generated either in Norway or on the Norwegian continental shelf. Income and deductions are assigned to tax years following the realization principle. The tax year is identical to the accounting year and coincides with the calendar year for most firms.

Businesses are permitted to carry tax losses forward to future periods indefinitely to reduce taxable profit. Dividends received by corporate shareholders are exempt from taxation. This also applies to income received from foreign subsidiaries. There is no municipal or local corporate income tax. Finally, businesses face a special tax of 56% on income from offshore production and pipeline transportation of petroleum.

Figure D11: Corporate Elasticity of Taxable Income: Norway, 2006–2013



Notes: This figure plots diagnostic graphs related to the estimation of the corporate elasticity of taxable income using administrative data from Norwegian firms using data from 2006–2013. Panel (a) plots the raw distribution of accounting profit and highlights bunching behavior. Panel (b) plots diagnostic tools for the traditional bunching estimate. Panel (c) plots variation in the estimated elasticity of taxable income based on the amount of data used in the estimation. Panel (d) plots the fit of the normal distribution in the tobit model for 100% of the data. Panel (e) plots the fit of the normal distribution in the tobit model using 60% of the data, and reflects our preferred specification.

D.13 Portugal

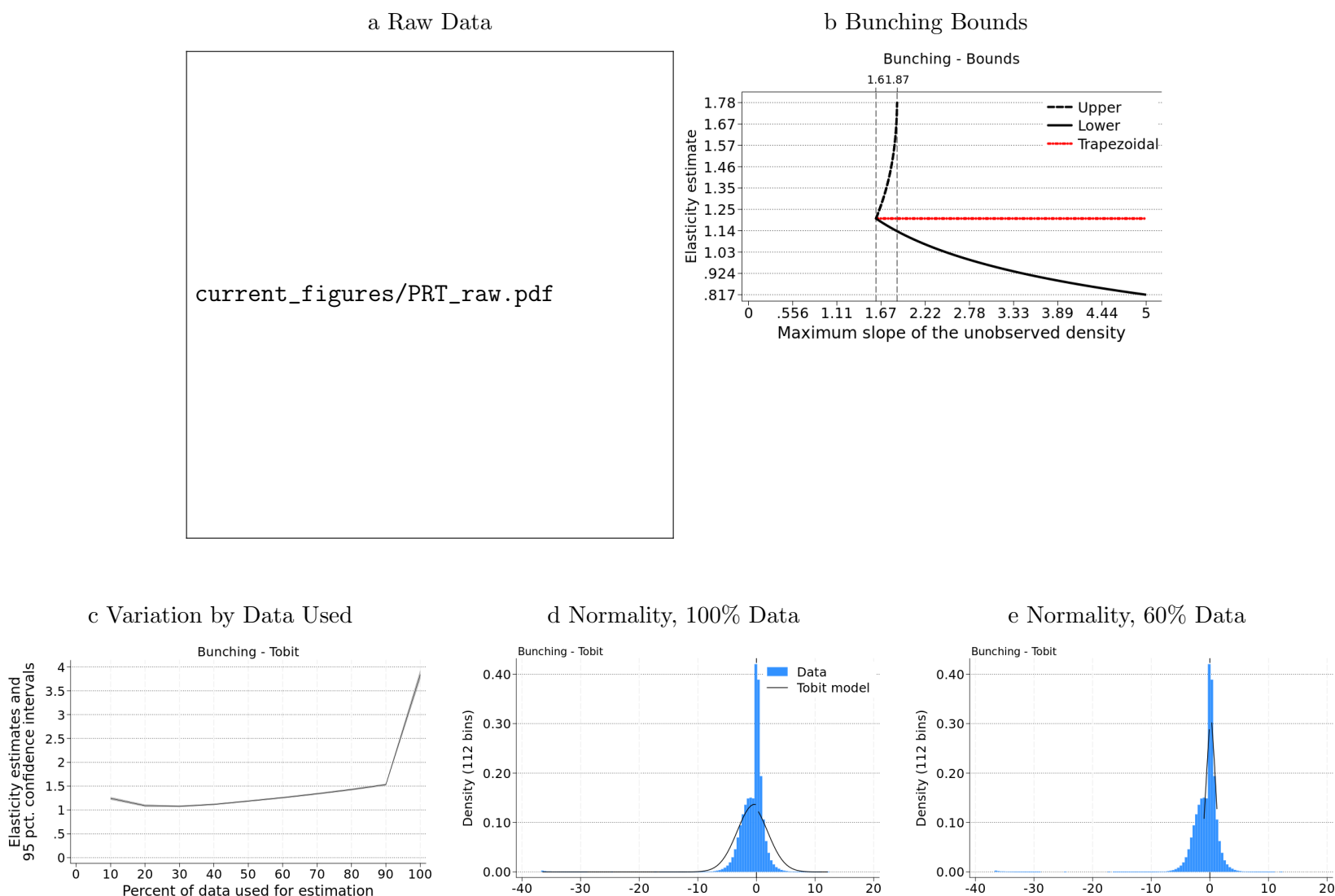
Tax Data The data comes from the Central Balance Sheet Harmonized Panel (CBHP), harmonized and provided by Banco de Portugal through BPLIM. We use the universe of non-financial corporations reporting through the mandatory Informação Empresarial Simplificada survey between 2016 and 2019. The CBHP includes sector of activity, location, and a wealth of variables from balance sheets, profit and loss statements, employment records, trade information, and corporate actions. We exclude public entities and firms in the islands of Azores and Madeira.

For the purposes of our analysis, we focus on three key variables: earnings before tax, turnover, and a proxy for fixed costs estimated using a cubic functional form based on variable costs (costs of goods sold and material consumed, supplies and external services, and employee expenses). We only keep firms with earnings before tax that are not extremely far from the kink point (between -100,000 and 100,000). The final sample includes 1,400,277 firm-year observations.

Corporate Tax Context Between 2016 and 2019, Portuguese corporations were subject to a general corporate income tax (IRC) rate of 21% on taxable profits. In addition to this base rate, companies could face municipal surtaxes (derrama municipal) of up to 1.5%, depending on the municipality, and a progressive state surtax (derrama estadual) based on profit thresholds: 3% for profits over 1.5 million, 5% for profits over 7.5 million, and 7-9% for profits exceeding 35 million. Despite the relatively high tax rates, Portugal offered various incentives, including exemptions for certain public and social entities and deductions for R&D and investment in strategic sectors.

Small and medium-sized enterprises (SMEs) benefited from a reduced IRC rate of 17% on the first 25,000 of taxable income. Additionally, SMEs with annual turnover below 200,000 and a total balance sheet not exceeding 500,000 could opt for a simplified taxation regime. Under this regime, taxable income was calculated using fixed percentages of gross invoicing, depending on the type of activity: 15% for sales, 35% for tourism-related services, and 75% for liberal professions. This method eliminated the need for detailed expense tracking and professional accounting services, making it particularly attractive for micro-enterprises.

Figure D12: Corporate Elasticity of Taxable Income: Portugal, 2016-2019



Notes: This figure plots diagnostic graphs related to the estimation of the corporate elasticity of taxable income using administrative data from Portugal firms using data from 2016-2019. Panel (a) plots the raw distribution of accounting profit and highlights bunching behavior. Panel (b) plots diagnostic tools for the traditional bunching estimate. Panel (c) plots variation in the estimated elasticity of taxable income based on the amount of data used in the estimation. Panel (d) plots the fit of the normal distribution in the tobit model for 100% of the data. Panel (e) plots the fit of the normal distribution in the tobit model using 60% of the data, and reflects our preferred specification.

D.14 Slovakia

Tax Data We construct our estimation sample based on administrative tax data capturing the population of corporate tax returns in 2013. These data are confidential and owned by the Financial Directorate of the Slovak Republic (FDSR).¹⁸ The data includes tax variables which correspond to individual items recorded on tax return forms. We utilize especially the information about corporate taxable income (or loss) before companies carry forward losses from previous fiscal years (row 400).

We merge the tax return data with additional information from corporate balance sheets and profit and loss statements. The information is publicly available from the Slovak Register of Financial Statements, into which companies are required to submit financial data when they file tax returns to the tax office.

Using these data we limit our analysis to companies with positive (non-zero) sales. In addition, we collect information about the depreciation expense for long-term tangible and intangible assets and information about the net value of non-current intangible assets.

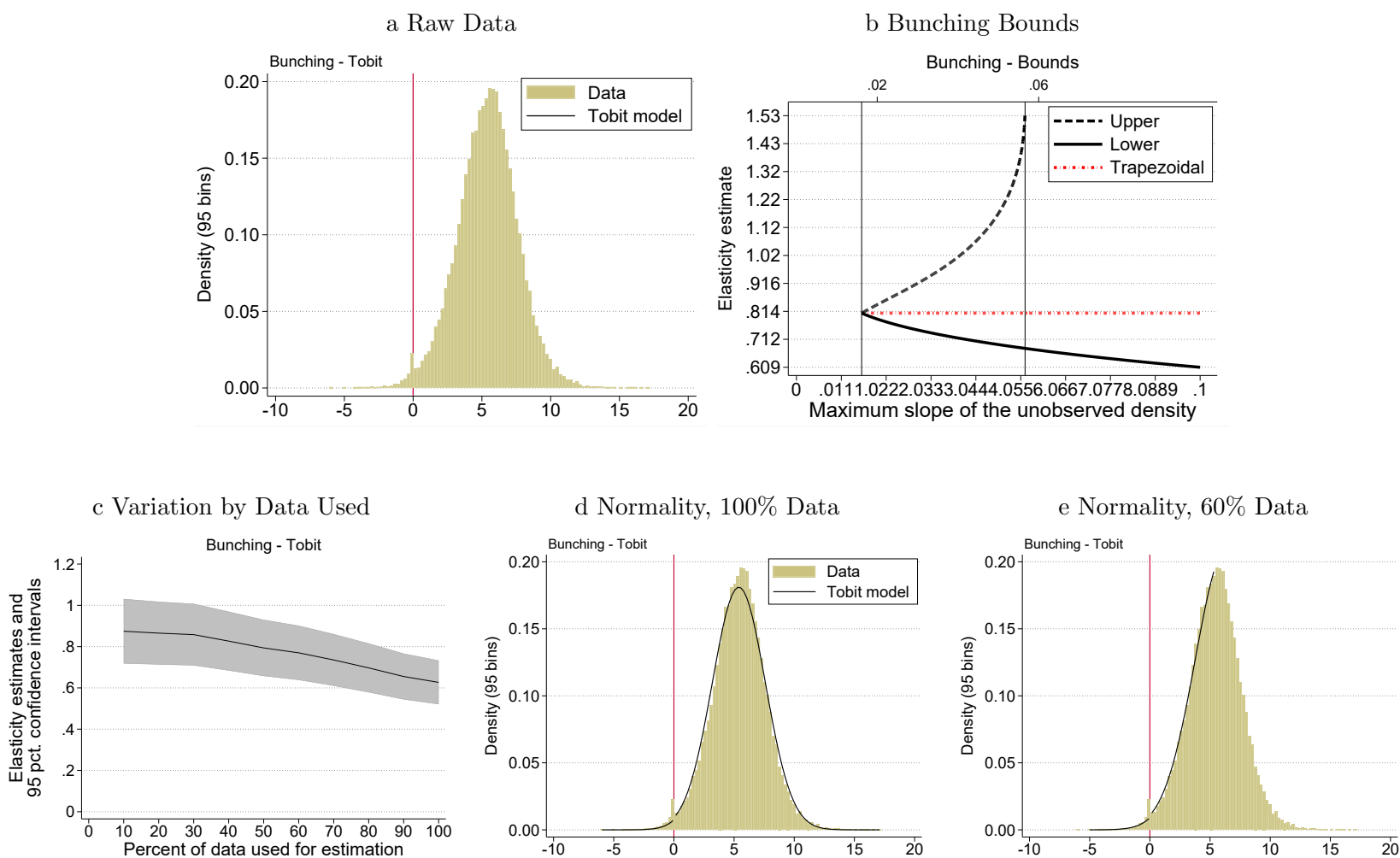
Corporate Tax Context In 2013, governmental tax revenue in Slovakia amounted to 31% of GDP, 11% of which was derived from the corporate income tax. Prior to 2013, incorporated companies were subject to a flat corporate tax rate of 19% on all profits.¹⁹ In 2013, the corporate tax rate increased to 23%. Businesses are permitted to carry tax losses forward to future periods for up to seven years to reduce taxable profit. Loss carrybacks are not permitted.

In addition, companies must register for the VAT once their revenue in the previous 12 months exceeds a fixed threshold specified by the tax law. Furthermore, companies are required to pay quarterly (or monthly) tax advances to the tax office if their tax liability exceeds specific thresholds, also given by the tax law. In 2013, the revenue threshold for mandatory VAT registration was 49,790 euro. The tax liability threshold for quarterly tax advances was 1659.7 euro, while the tax liability threshold for monthly tax advances was 16,597 euro.

¹⁸FDSR provides the data to other state organs of the Slovak Republic following article 11 of the Slovak Tax Code Act no. 563/2009 on tax secrecy. For details, see: <https://www.zakonypreludi.sk/zz/2009-563>

¹⁹In contrast, the profits of unincorporated legal entities, such as sole proprietorships and partnerships, were taxed according to the personal income tax schedule, once profits were attributed to individual partners. Unincorporated companies yet generate only around 4% of tax revenue collected from legal entities.

Figure D13: Corporate Elasticity of Taxable Income: Slovakia, 2013



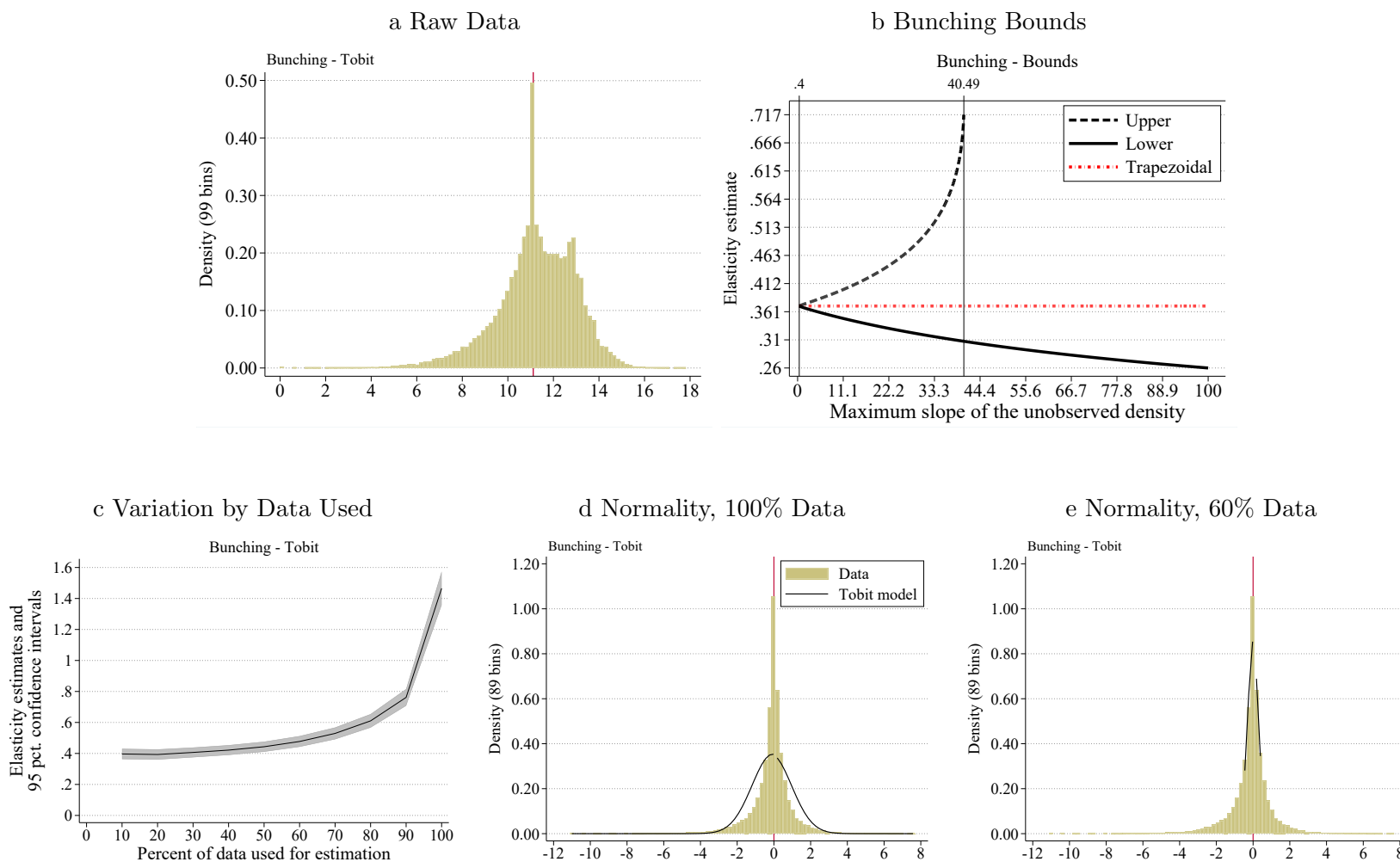
Notes: This figure plots diagnostic graphs related to the estimation of the corporate elasticity of taxable income using administrative data from Slovakian firms using data from 2013. Panel (a) plots the raw distribution of accounting profit and highlights bunching behavior. Panel (b) plots diagnostic tools for the traditional bunching estimate. Panel (c) plots variation in the estimated elasticity of taxable income based on the amount of data used in the estimation. Panel (d) plots the fit of the normal distribution in the tobit model for 100% of the data. Panel (e) plots the fit of the normal distribution in the tobit model using 60% of the data, and reflects our preferred specification.

D.15 South Africa

Tax Data We use administrative corporate tax data from the South African Revenue Service (SARS) which was made available in a joint project with National Treasury (NT) (see Pieterse et al. (2018) for full description of the NT-SARS database). It is an unbalanced panel dataset that includes all kinds of firm’s balance sheet items like total assets, sales, cost of sales as well as deduction items like, for example, directors’ remuneration, donations, travel expenses. The data also includes exact taxable income from the income statement. In total, the bunching analysis uses 2 million firm-year observations for approximately 200,000 firms.

Corporate Tax Context Corporate income taxes are levied by the national government of South Africa under the Income Tax Act 58 of 1962. The tax system is residence-based, and the headline company tax rate is 28%. All tax returns submitted by a tax-registered firm must be completed electronically or at a SARS branch within 12 months of its financial year end (usually at the end of February). Small and Medium Businesses (SBCs) in South Africa benefit from a progressive tax schedule, starting with a tax rate of 0%. To qualify as a SBC, a company must i) not have elected to be classified as a Micro Business for the year of tax assessment and ii) meet specific criteria. These include, among others, gross income not exceeding R20 million (R14 million prior to the 2013 tax year) and limited shareholding. Depending on the tax year considered, there are two to three tax kinks where the marginal tax rate jumps. For the tax year 2015, for example, the marginal tax rate jumps to 7% at the income threshold 70,000 Rand and then increases to 21% and 28% at threshold values of 365,000 and 550,000 Rand. The tax rate of 28% is then applied to all SBCs with taxable income larger than 550,000 Rand. Over time, the thresholds as well as the tax rates have changed over time.

Figure D14: Corporate Elasticity of Taxable Income: South Africa, 2014



Notes: This figure plots diagnostic graphs related to the estimation of the corporate elasticity of taxable income using administrative data from South African firms using data from 2014. Panel (a) plots the raw distribution of accounting profit and highlights bunching behavior. Panel (b) plots diagnostic tools for the traditional bunching estimate. Panel (c) plots variation in the estimated elasticity of taxable income based on the amount of data used in the estimation. Panel (d) plots the fit of the normal distribution in the tobit model for 100% of the data. Panel (e) plots the fit of the normal distribution in the tobit model using 60% of the data, and reflects our preferred specification.

